Mathematics - Probability and Statistics (S2)

Thomas de Jaeger

thomas.de-jaeger@epita.fr

Spring 2026

Class details

- 21h: 2h*9+ 1h mid term +2h final exam

- Grade:

  10% Participation

  35% Mid term (1h)

  55% Final Exam (2h)

- For anonymous feedback:[Feedback - Google Docs]

Class organisation

- Lectures and exercices

Syllabus

- **Session 1:** Review semester 1
- **Session 2:** Confidence Intervals for a Single Sample
- **Session 3:** Confidence Intervals for a two Sample
- **Session 4**: Hypothesis testing
- **Session 5:** Correction mid term+ANOVA
- **Session 6:** Simple Linear Regression and Correlation
- **Session 7:** Multiple Linear Regression
- **Session 8:** Analysis of Categorical Data
- **Session 9**: Bayesian inference

Environment & Requirements

Tools & Libraries to Install

Library
numpy
matplotlib
Pandas
stats

You can run this course on **Google Colab** or **locally**.

Google Colab

What is Google Colab? Google Colab is a free cloud service that supports Jupyter notebooks and provides free access to computing resources including GPUs.

To use it:

- Log in with a Google account
- Open a .ipynb notebook from Drive or GitHub, or create a new one
- To enable GPU: Runtime > Change runtime type > GPU

Why we use it:

- Free and powerful
- No local install needed
- Familiar interface for Jupyter users

Upload files as a drive from your local machine

```
from google.colab import files
input_file = files.upload()
```

Bibliography

- https://online.stat.psu.edu/stat414/
- Probability & Statistics for Machine Learning and Data Science
- Applied Statistics and Probability for Engineers, 5e
- statbook3.pdf
- Statistics and Probability Full Course || Statistics For Data Science – YouTube
- https://hsong1.github.io/teachings/psu-sp21-stat415

S1 summary

Session 1

S1 summary

Session 1

Objectives:

- Summary of Semester 1
- Final exam correction

Sample vs Population

- The **population** is the complete set of individuals or observations we are interested in.

Usually **too large** or **impossible** to observe entirely.

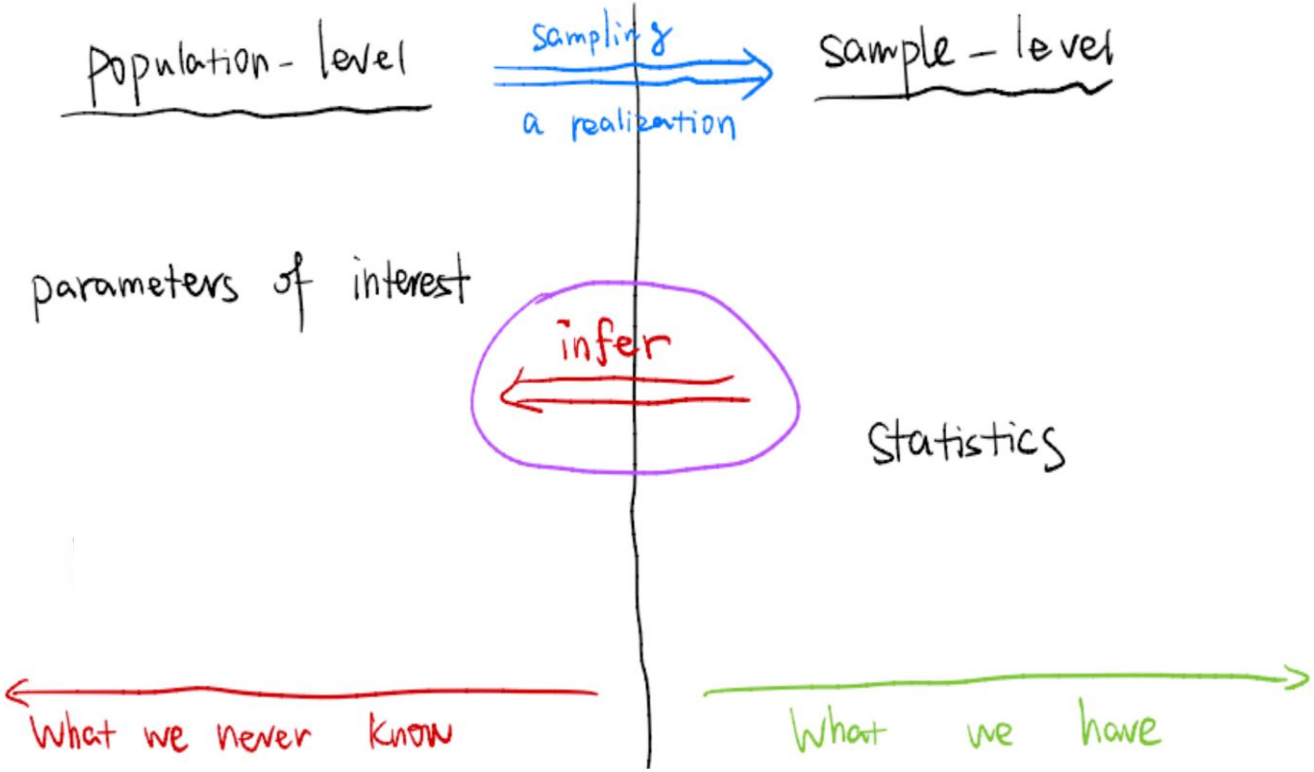
- **The sample** is a subset of the population, usually collected randomly.

$$X_1, X_2, \dots, X_n$$

Each X_i is a **random variable.**

- The sample is **observable**
- The population parameters are **unknown**

Population	Sample
Mean μ	Sample mean $\bar{X}$
Variance σ^2	Sample variance S^2



Goal of statistics: To make inferences on parameters of a population which are assumed to be fixed but unknown.

What is a random variable?

It is one of the most important concepts in probability!!

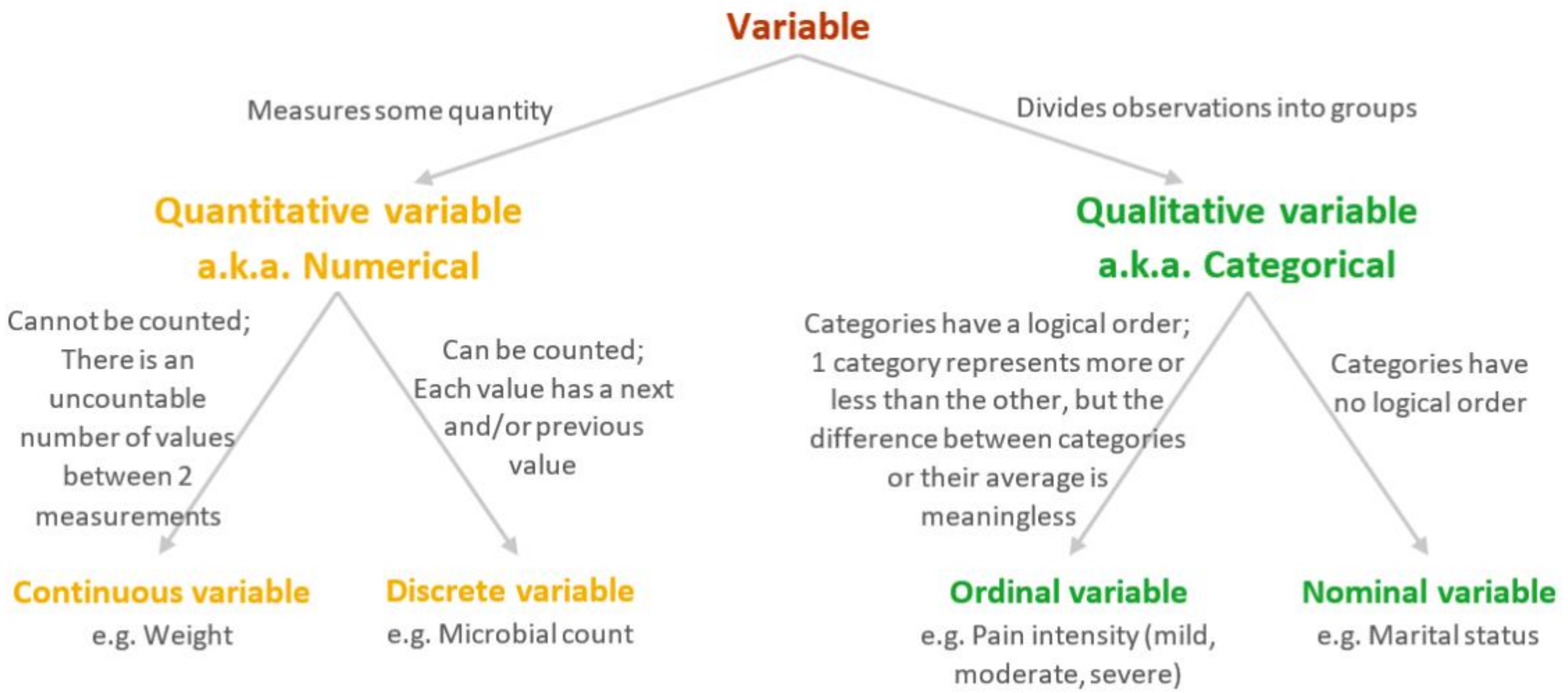
The concept of variable is not new for you: algebra, **variable X=3**

A **random variable** assigns a numerical value to the outcome of a random experiment.

Example:

- Toss a coin → $X = 1$if Head, 0 if Tail
- Draw a card → value depends on rank
- Temperature

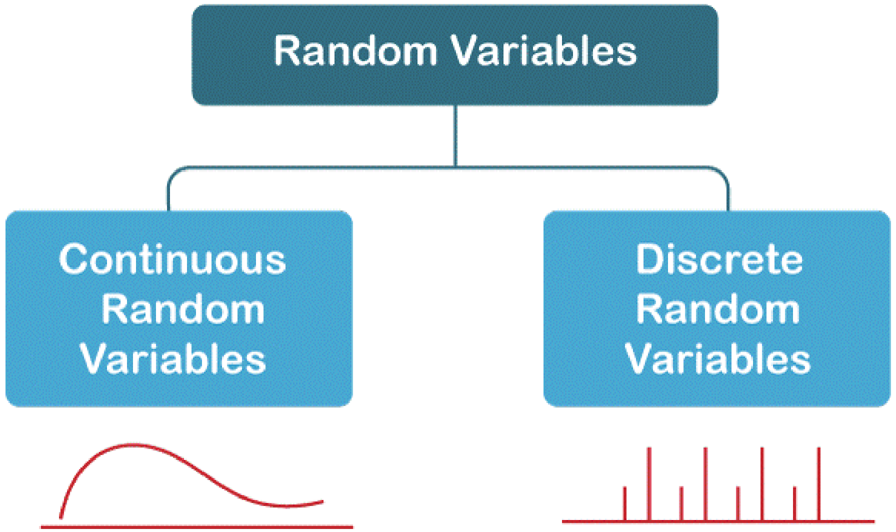
Discrete vs continous



Discrete vs continous

A continuous variable is a type of quantitative variable consisting of numerical **values that can be measured but not counted**, because there are infinitely many values between 1 measurement and another.

A discrete variable is a type of quantitative variable consisting of numerical values **that can be measured and counted**, because these values are separate or distinct.



Discrete: PMF

Definition:

The most important way to characterize a random variable is through the probabilities of the values that it can take. For a discrete random variable X, these are captured by the probability mass function (PMF for short) of X, denoted f_X.

If x is any possible value of X, the probability mass of x, denoted $f_X(x)$, is the probability of the event {X = x} consisting of all outcomes that give rise to a value of X equal to x: $f_X(x) = P(\{X = x\})$

Heads	Flip Outcomes	Probability
0	TTT	$1/8 = 0.125$
1	TTH, THT, HTT	$3/8 = 0.375$
2	HHT, HTH, THH	$3/8 = 0.375$
3	HHH	$1/8 = 0.125$

$$\mathbb{P}[X = x] = \begin{cases} 0.125 & x = 0 \text{ or } 3 \\ 0.375 & x = 1 \text{ or } 2 \\ 0 & \text{Otherwise} \end{cases}$$

Examples:

- Rolling one fair die: $X \in \{1,2,3,4,5,6\}$ with $\mathbf{P(X=k)=1/6}$

Mean and variance

The **mean** or **expected value** of the discrete random variable X, denoted as μ or $E(X)$, is

$$\mu = E(X) = \sum_x x f(x) \tag{3-3}$$

The **variance** of X, denoted as σ^2 or $V(X)$, is

$$\sigma^2 = V(X) = E(X - \mu)^2 = \sum_x (x - \mu)^2 f(x) = \sum_x x^2 f(x) - \mu^2$$

The **standard deviation** of X is $\sigma = \sqrt{\sigma^2}$.

Exam exercice 7

Exercise 7 : Discrete random variables (2 pts)

A card is drawn uniformly at random from a deck of 32 cards, consisting of the ranks $7, 8, 9, 10, J, Q, K, A$ in each of the four suits (spades, hearts, diamonds, clubs).

Define the random variable X as follows:

$$X = \begin{cases} -1 & \text{if the card drawn is a } 7, 8, 9 \text{ or } 10, \\ 1 & \text{if the card drawn is a Jack, Queen, or King,} \\ 2 & \text{if the card drawn is an Ace.} \end{cases}$$

1. Find the probability density function of X (0.5pt).
2. Compute the expected value $E[X]$ and the variance $\mathrm{Var}(X)$ (0.5pt).
3. Compute $P(X \geq 0)$ (0.5pt).
4. Let $Y = 2X - 1$. Compute the expected value $E[Y]$ (0.5pt).

Exam exercice 7

1) First we need to derive the probabilities for each outcomes.
X=-1 for 7,8,9,10 therefore 4*4/32=16/32=1/2
X=1 for J,Q,K therefore 4*3/32=12/32=3/8
X=2 for Ace, 4/32=1/8

$$P(X = x) = \begin{cases} \frac{16}{32} = \frac{1}{2}, & x = -1, \\ \frac{12}{32} = \frac{3}{8}, & x = 1, \\ \frac{4}{32} = \frac{1}{8}, & x = 2, \\ 0, & \text{otherwise.} \end{cases}$$

2) $\mathbb{E}[X] = \sum_x x\, P(X = x)$ $\quad \mathbb{E}[X] = (-1) \cdot \frac{1}{2} + 1 \cdot \frac{3}{8} + 2 \cdot \frac{1}{8} = \frac{1}{8}$

$$\mathrm{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$$

$$\mathbb{E}[X^2] = (-1)^2 \cdot \frac{1}{2} + 1^2 \cdot \frac{3}{8} + 2^2 \cdot \frac{1}{8} = \frac{11}{8}$$

$$\mathrm{Var}(X) = \frac{11}{8} - \left(\frac{1}{8}\right)^2 = \frac{87}{64}$$

3) $P(X \geq 0) = \sum_{x \geq 0} P(X = x) = P(X = 1) + P(X = 2) = \frac{3}{8} + \frac{1}{8} = \frac{1}{2}$

4) $\mathbb{E}[aX + b] = a\mathbb{E}[X] + b$

$$\mathbb{E}[Y] = 2\mathbb{E}[X] - 1 = 2 \cdot \frac{1}{8} - 1 = -\frac{3}{4}$$

Continous

Definition:

For a continuous random variable X, a **probability density function** is a function such that

(1) $f(x) \geq 0$

(2) $\int_{-\infty}^{\infty} f(x)\, dx = 1$

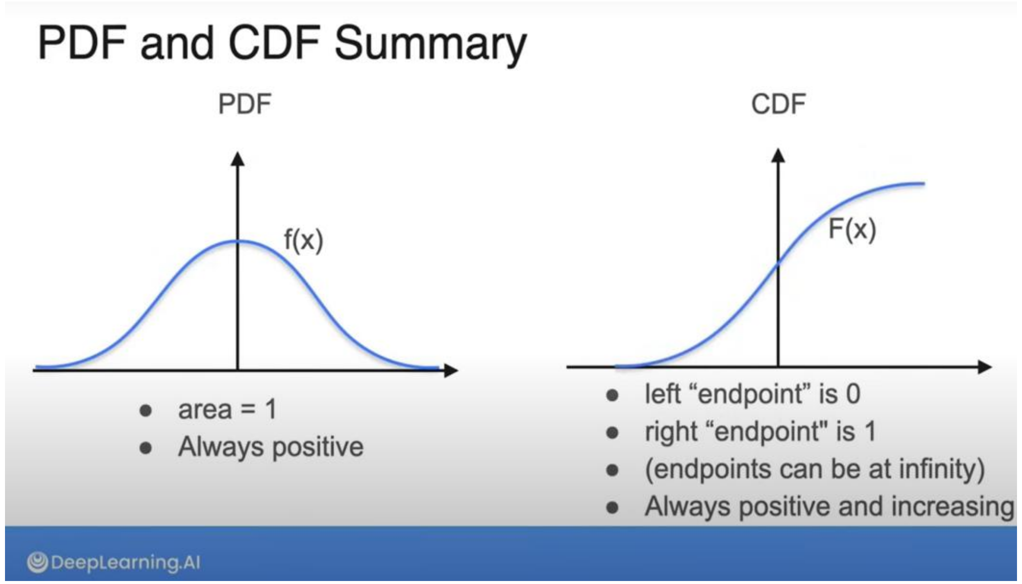
(3) $P(a \leq X \leq b) = \int_{a}^{b} f(x)\, dx = \text{area under } f(x) \text{ from } a \text{ to } b$

for any a and b

The **cumulative distribution function** of a continuous random variable X is

$$F(x) = P(X \leq x) = \int_{-\infty}^{x} f(u)\, du$$

for $-\infty < x < \infty$.



Suppose X is a continuous random variable with probability density function $f(x)$. The **mean** or **expected value** of X, denoted as μ or $E(X)$, is

$$\mu = E(X) = \int_{-\infty}^{\infty} x f(x)\, dx \qquad (4\text{-}4)$$

The **variance** of X, denoted as $V(X)$ or σ^2, is

$$\sigma^2 = V(X) = \int_{-\infty}^{\infty} (x - \mu)^2 f(x)\, dx = \int_{-\infty}^{\infty} x^2 f(x)\, dx - \mu^2$$

The **standard deviation** of X is $\sigma = \sqrt{\sigma^2}$.

Exam exercice 2

Let X be a continuous random variable with probability density function

$$f(x) = \begin{cases} \dfrac{1}{b-a}, & a \leq x \leq b, \\ 0, & \text{otherwise}, \end{cases}$$

where $a < b$ are real numbers.

1. Show that f is a probability density function.
2. Compute the expectation $E[X]$.
3. Compute the variance $\mathrm{Var}(X)$.

Exam exercice 2

1) A function f is a pdf if:

   1. $f(x) \geq 0$ for all x
   2. $\int_{-\infty}^{\infty} f(x)\,dx = 1$

$$\int_{-\infty}^{\infty} f(x)\,dx = \int_a^b \frac{1}{b-a}\,dx = \frac{1}{b-a}(b-a) = 1$$

2) $\mathbb{E}[X] = \int_{-\infty}^{\infty} x f(x)\,dx$

$$\mathbb{E}[X] = \int_a^b x \cdot \frac{1}{b-a}\,dx = \frac{1}{b-a}\int_a^b x\,dx = \frac{1}{b-a}\left[\frac{x^2}{2}\right]_a^b = \frac{b^2-a^2}{2(b-a)} = \frac{a+b}{2}$$

3) $\mathrm{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$

$$\mathbb{E}[X^2] = \int_a^b x^2 \cdot \frac{1}{b-a}\,dx = \frac{1}{b-a}\left[\frac{x^3}{3}\right]_a^b = \frac{b^3-a^3}{3(b-a)}$$

$$\mathrm{Var}(X) = \frac{b^3-a^3}{3(b-a)} - \left(\frac{a+b}{2}\right)^2 = \frac{(b-a)^2}{12}$$

Exam exercice 3

Exercise 3 : continuous random variables (5 pts)

Let X be a continuous random variable supported on $[1, 2]$ with density

$$f_X(x) = \begin{cases} 2k\,x^2, & 1 \leq x \leq 2, \\ 0, & \text{otherwise,} \end{cases}$$

where $k > 0$ is a constant.

1. Determine the value of k so that f_X is a valid probability density function. (1 pt)
2. Find $F_X(x)$, the cdf of X. (0.75 pt)
3. Compute $P(X < 3/2)$ (0.25 pt)
4. Compute $E[X]$ (1 pt)
5. Compute $E[X^2]$ and $\mathrm{Var}(X)$ (1 pt)

Exam exercice 3

1) $$\int_{-\infty}^{\infty} f_X(x)\,dx = 1$$

$$\int_1^2 2kx^2\,dx = 1 \qquad 2k\left[\frac{x^3}{3}\right]_1^2 = 2k\left(\frac{8-1}{3}\right) = \frac{14k}{3} \qquad k = \frac{3}{14}$$

2) CDF: $$F_X(x) = \int_{-\infty}^{x} f_X(t)\,dt$$

$$F_X(x) = \int_{-\infty}^{x} 0\,dt = 0$$

$$F_X(x) = \begin{cases} 0, & x < 1, \\ \dfrac{x^3-1}{7}, & 1 \le x \le 2, \\ 1, & x > 2. \end{cases}$$

$$F_X(x) = \int^{1} 0\,dt + \int^{x} 2kt^2\,dt$$

$$F_X(x) = \int_{-\infty}^{1} 0\,dt + \int_1^2 2kt^2\,dt + \int_2^x 0\,dt$$

3) $$P(X < a) = \int_{-\infty}^{a} f_X(x)\,dx = F_X(a) = \frac{19}{56}$$ = 0.339

4) $$\mathbb{E}[X] = 2k\int_1^2 x^3\,dx = \frac{45}{28}$$

5) $$\mathbb{E}[X^2] = \int_1^2 x^2(2kx^2)\,dx = 2k\int_1^2 x^4\,dx = \frac{93}{35}$$

$$\mathrm{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$$

Var [X] = 93/35 – $(45/28)^2$= 0.075

Joint distribution, covariance and correlation

- Joint distribution describes how two variables behave **together**

The **joint probability mass function** of the discrete random variables X and Y, denoted as $f_{XY}(x, y)$, satisfies

$$(1) \quad f_{XY}(x, y) \geq 0$$

$$(2) \quad \sum_x \sum_y f_{XY}(x, y) = 1$$

$$(3) \quad f_{XY}(x, y) = P(X = x, Y = y) \qquad (5\text{-}1)$$

If X and Y are discrete random variables with joint probability mass function $f_{XY}(x, y)$, then the **marginal probability mass functions** of X and Y are

$$f_X(x) = P(X = x) = \sum_{R_x} f_{XY}(x, y) \quad \text{and} \quad f_Y(y) = P(Y = y) = \sum_{R_y} f_{XY}(x, y) \qquad (5\text{-}2)$$

where R_x denotes the set of all points in the range of (X, Y) for which $X = x$ and R_y denotes the set of all points in the range of (X, Y) for which $Y = y$

Joint distribution, covariance and correlation

The **covariance** between the random variables X and Y, denoted as $\text{cov}(X, Y)$ or σ_{XY}, is

$$\sigma_{XY} = E[(X - \mu_X)(Y - \mu_Y)] = E(XY) - \mu_X \mu_Y \qquad (5\text{-}28)$$

The **correlation** between random variables X and Y, denoted as ρ_{XY}, is

$$\rho_{XY} = \frac{\text{cov}(X, Y)}{\sqrt{V(X)\,V(Y)}} = \frac{\sigma_{XY}}{\sigma_X \sigma_Y} \qquad (5\text{-}29)$$

Correlation = normalized covariance ⇒ correlation takes values between **-1 and 1**

independance

Discrete random variables

Formula

X and Y are independent if and only if, for all x, y,

$$P(X = x,\ Y = y) = P(X = x)\, P(Y = y).$$

Continuous random variables

Formula

Let $f_{X,Y}(x, y)$ be the joint pdf.
X and Y are independent if and only if

$$f_{X,Y}(x, y) = f_X(x)\, f_Y(y) \quad \text{for all } x, y.$$

independance

Discrete random variables

Formula

X and Y are independent if and only if, for all x, y,

$$P(X = x,\ Y = y) = P(X = x)\, P(Y = y).$$

Continuous random variables

Formula

Let $f_{X,Y}(x, y)$ be the joint pdf.
X and Y are independent if and only if

$$f_{X,Y}(x, y) = f_X(x)\, f_Y(y) \quad \text{for all } x, y.$$

independance

Independence ⇒ Covariance = 0

If X and Y are independent, then:

$$\mathrm{Cov}(X,Y)=0$$

But the converse is false

$$\mathrm{Cov}(X,Y)=0 \quad \not\Rightarrow \quad X \text{ and } Y \text{ independent}$$

Zero covariance only means **no linear relationship**, not full independence.

Example: Try X uniform from -1 and 1, and Y=X^2

- Y is not independent of X.
- Cov(X,Y)=E[XY]-E[X]E[Y]=E[X^3]-E[X]E[X^2]
- E[X] =(b+a)/2=0 and E[X^3]=0 (integral from -1 to 1…)

Exam exercise 4

Exercise 4 : Joint Distribution, Covariance, and Correlation (4 pts)

Let U and V be discrete random variables with the following marginal distributions:

$$P(U=2)=P(U=-2)=\frac{1}{2}, \qquad P(V=3)=P(V=-3)=\frac{1}{2}.$$

Define

$$d=P(U=2 \text{ and } V=3).$$

1. Express the **joint distribution** of U and V in terms of d (1 pt).
2. Compute the **covariance** $\text{Cov}(U,V)$ as a function of d (1 pt).
3. Compute the **correlation** $\text{Cor}(U,V)$ as a function of d (1 pt).
4. For which value(s) of d are U and V **independent** (0.5 pt)?
5. For which value(s) of d are U and V **perfectly correlated** (0.5 pt)

Exam exercise 4

1) We know that P(U=2,V=3) =d

P(U=2)=0.5 =P(U=2,V=3)+P(U=2,V=-3)=d+P(U=2,V=-3) therefore P(U=2,V=-3)=0.5-d
P(V=3)=0.5 =P(U=2,V=3)+P(U=-2,V=3)=d+P(U=-2,V=3) therefore P(U=-2,V=3) =0.5-d

P(2,3)+P(2,-3)+P(-2,3)+P(-2,-3)=1 -> 1-d-(0.5-d) -(0.5-d)=P(-2,-3) therefore P(-2,-3)=1-d-0.5+d-0.5+d=d

$U\backslash V$	$V=3$	$V=-3$
$U=2$	d	$\frac{1}{2}-d$
$U=-2$	$\frac{1}{2}-d$	d

Exam exercise 4

$U\backslash V$	$V=3$	$V=-3$
$U=2$	d	$\frac{1}{2}-d$
$U=-2$	$\frac{1}{2}-d$	d

2) $\mathrm{Cov}(U,V)=\mathbb{E}[UV]-\mathbb{E}[U]\mathbb{E}[V]$

$$\mathbb{E}[U]=\sum_{u,v} u\cdot P(U=u,V=v),\quad \mathbb{E}[V]=\sum_{u,v} v\cdot P(U=u,V=v)$$

$$\mathbb{E}[UV]=\sum_{u,v} uv\cdot P(U=u,V=v)$$

$$\mathbb{E}[U]=2\cdot P(U=2)+(-2)\cdot P(U=-2)=2\cdot\frac{1}{2}+(-2)\cdot\frac{1}{2}=0$$

$$\mathbb{E}[UV]=(2\cdot 3)\cdot d+(2\cdot -3)\cdot(\frac{1}{2}-d)+(-2\cdot 3)\cdot(\frac{1}{2}-d)+(-2\cdot -3)\cdot d=24d-6$$

$$\mathrm{Cov}(U,V)=24d-6$$

Exam exercise 4

$U\backslash V$	$V=3$	$V=-3$
$U=2$	d	$\frac{1}{2}-d$
$U=-2$	$\frac{1}{2}-d$	d

3) $$\mathrm{Cor}(U,V)=\frac{\mathrm{Cov}(U,V)}{\sigma_U\sigma_V}$$

$$\sigma_U^2=\mathrm{Var}(U)=\mathbb{E}[U^2]-(\mathbb{E}[U])^2$$

$$\mathbb{E}[U^2]=2^2\cdot\frac{1}{2}+(-2)^2\cdot\frac{1}{2}=4\cdot\frac{1}{2}+4\cdot\frac{1}{2}=4$$

$$\sigma_U=\sqrt{4}=2\qquad\sigma_V=\sqrt{9}=3$$

$$\mathrm{Cor}(U,V)=4d-1$$

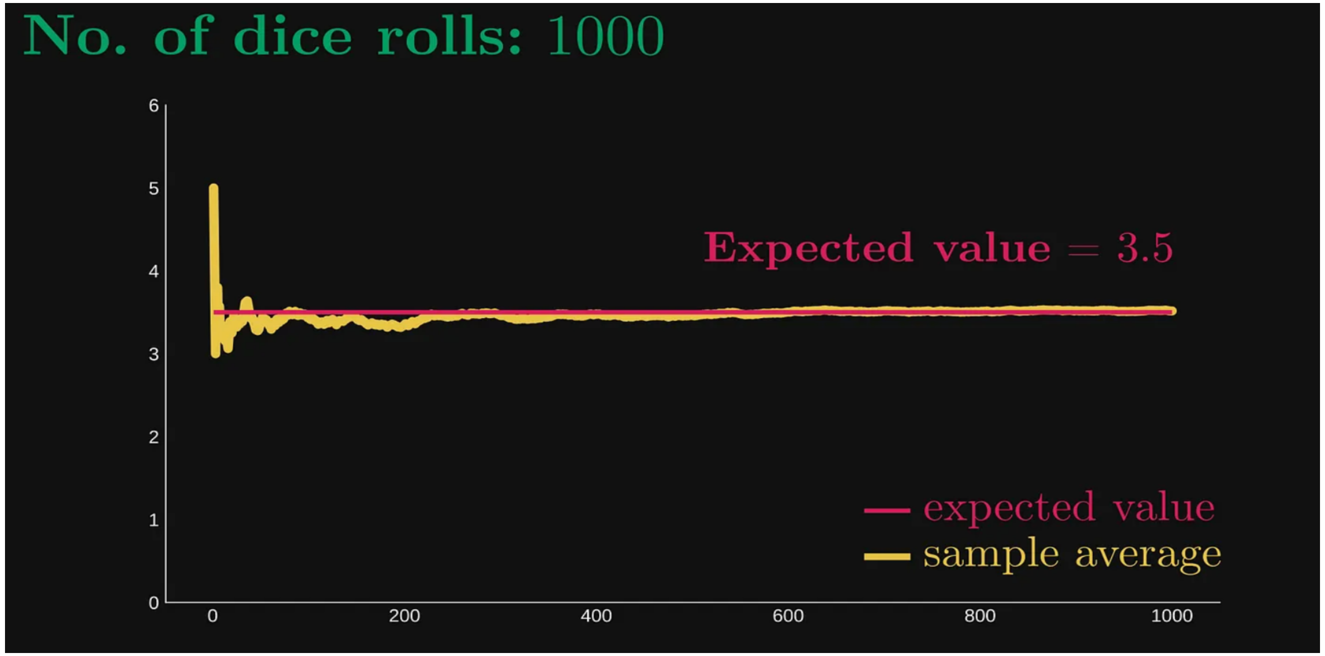
4) If U,V independent Cov(UV)=0 therefore, 4d-1=0 so d=1/4

5) Perfectly correlated if Cor(U,V)=1

4d-1=-1 -> d=0

4d-1=1 -> d=1/2

Law of large number



The law of large numbers states precisely this: the sample average converges to the expected value.

$$\bar{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i \to \mu$$

Central limit theorem

When you average a large number of independent, identically distributed random variables, the distribution of the average becomes approximately normal, regardless of the original distribution.

If $X_1, X_2, \ldots, X_n$ is a random sample of size n taken from a population (either finite or infinite) with mean μ and finite variance σ^2, and if $\overline{X}$ is the sample mean, the limiting form of the distribution of

$$Z = \frac{\overline{X} - \mu}{\sigma/\sqrt{n}} \tag{7-1}$$

as $n \to \infty$, is the standard normal distribution.

How many standard deviations the sample mean $\bar{X}_n$ is away from the true mean μ.

Central limit theorem : formalism

If $X_1, X_2, \ldots, X_n$ is a random sample of size n taken from a population (either finite or infinite) with mean μ and finite variance σ^2, and if $\bar{X}$ is the sample mean, the limiting form of the distribution of

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \tag{7-1}$$

as $n \to \infty$, is the standard normal distribution.

How many standard deviations the sample mean $\bar{X}_n$ is away from the true mean μ.

Exam exercise 5

Exercise 5 : inferential statistics (5 pts)

Carbon monoxide (CO) emissions (in gm/mi) for a certain kind of car vary with mean

$$\mu = 3.0$$

and standard deviation

$$\sigma = 0.45.$$

A company owns 64 cars of this type, acquired from various (i.e., random) sources. Assume that the CO emissions of different cars are independent and identically distributed.

1. What is the probability that a randomly selected car from the fleet has CO emissions greater than 3.2 gm/mi? (0.75 pt)
2. What is the probability that the average CO emissions of the 64 cars is greater than 3.2 gm/mi? (0.75 pt)
3. There is only a 1% chance that the fleet's mean CO emissions exceeds what value? (0.75 pt)
4. If the distribution of individual car emissions is not normal, explain why the approximation used in Question 2 remains valid. (0.5 pt)

Exam exercise 5

1) For a normal distribution:

$$P(X > x) = P\left(Z > \frac{x-\mu}{\sigma}\right)$$

Z=(3.2-3.0)/0.45= 0.2/0.45=0.44

P(Z>0.44)=1-P(Z<0.44)= 1-0.67= 0.33 -> 33%

2) Using CLT

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right) \qquad Z = \frac{\bar{X}-\mu}{\sigma_{\bar{X}}} = \frac{\bar{X}-\mu}{\sigma/\sqrt{n}}$$

$$P(\bar{X} > x) = P\left(Z > \frac{x-\mu}{\sigma/\sqrt{n}}\right)$$

$$P(\bar{X} > 3.2) = P(Z > 3.556)$$

$$P(Z > 3.556) = 1 - P(Z < 3.556) \qquad P(Z > 3.556) = 1 - 0.99981 = 0.00019$$

Exam exercise 5

3) $P(\bar{X} > x) = 0.01$

$$x = \mu + Z\frac{\sigma}{\sqrt{n}}$$

$$x = \mu + z_{0.99} \cdot \frac{\sigma}{\sqrt{n}} = 3.0 + 2.33 \cdot 0.05625 = 3.131$$

4) CLT

Here, $n = 64$ is large, so $\bar{X}$ is approximately normal, **regardless of the original distribution of X**.

Estimator

An **estimator** is a **rule or function** that uses **sample data** to estimate an **unknown population parameter**.

$$\hat{\theta} = \text{estimator of } \theta$$

It is unbiased if: $\mathbb{E}[\hat{\theta}] = \theta$

The basic idea behind this form of the method is to:

1. Equate the first sample moment about the origin $M_1 = \frac{1}{n}\sum_{i=1}^{n} X_i = \bar{X}$ to the first theoretical moment $E(X)$.
2. Equate the second sample moment about the origin $M_2 = \frac{1}{n}\sum_{i=1}^{n} X_i^2$ to the second theoretical moment $E(X^2)$.
3. Continue equating sample moments about the origin, M_k, with the corresponding theoretical moments $E(X^k),\ k = 3, 4, \ldots$ until you have as many equations as you have parameters.
4. Solve for the parameters.

Method of Moment

The basic idea behind this form of the method is to:

1. Equate the first sample moment about the origin $M_1 = \frac{1}{n}\sum_{i=1}^{n} X_i = \bar{X}$ to the first theoretical moment $E(X)$.
2. Equate the second sample moment about the origin $M_2 = \frac{1}{n}\sum_{i=1}^{n} X_i^2$ to the second theoretical moment $E(X^2)$.
3. Continue equating sample moments about the origin, M_k, with the corresponding theoretical moments $E(X^k),\ k = 3, 4, \ldots$ until you have as many equations as you have parameters.
4. Solve for the parameters.

Maximum likelihood estimator

Goal:

Find the parameter value $\hat{\theta}$ that makes the likelihood **as large as possible**.

Steps to compute an MLE

1. **Write the likelihood function**

$$L(\theta) = \prod_{i=1}^{n} f(x_i \mid \theta)$$

2. **Take the logarithm** (optional but almost always easier)

$$\ell(\theta) = \log L(\theta)$$

   Turns products into sums.

3. **Differentiate** the log-likelihood with respect to the parameter

$$\frac{d}{d\theta}\ell(\theta)$$

4. **Set the derivative equal to zero**

$$\frac{d}{d\theta}\ell(\theta) = 0$$

5. **Solve for θ**

   This gives the **candidate** for the MLE.

6. **Check it is a maximum**

   Usually:

   - second derivative < 0
   - or log-likelihood is clearly maximized there (often obvious for simple models)

Result:

$$\boxed{\hat{\theta}_{MLE} = \arg\max_{\theta} L(\theta)}$$

Exam exercise 7

Exercise 6 : MoM and MLE (5 pts)

Let X be a continuous random variable with probability density function

$$f(x;\theta) = \begin{cases} \frac{1}{\theta} e^{-(x-1)/\theta}, & x \geq 1, \\ 0, & \text{otherwise}, \end{cases}$$

where $\theta > 0$ is an unknown parameter.

A random sample of size $n = 10$ is observed:

$$x_1 = 1.42,\ x_2 = 1.35,\ x_3 = 1.58,\ x_4 = 1.26,\ x_5 = 1.47,$$
$$x_6 = 1.31,\ x_7 = 1.62,\ x_8 = 1.39,\ x_9 = 1.51,\ x_{10} = 1.44.$$

Method of Moment (2 pts)

1. Compute the theoretical $E[X]$ as a function of θ using the integral
   Hint: Use the substitution $u = x - 1$ to reduce the integral to a sum of standard exponential integrals.
2. Derive the Method of Moments estimator $\hat{\theta}_{\text{MoM}}$.
3. Compute the numerical value of $\hat{\theta}_{\text{MoM}}$ using the observed data.
4. Is your estimator biased?

Exam exercise 7

1) Theoretical E[X]

$$E[X] = \int_1^\infty x f(x;\theta) dx = \int_1^\infty x \frac{1}{\theta} e^{-(x-1)/\theta} dx$$

u=x-1 so du=dx,

$$\int_0^\infty (u+1) \frac{1}{\theta} e^{-u/\theta} du = \frac{1}{\theta} \int_0^\infty u e^{-u/\theta} du + \frac{1}{\theta} \int_0^\infty e^{-u/\theta} du$$

Exponential distri so
E[X]=1/theta

$$\int_0^\infty e^{-u/\theta} du = \theta$$

$$E[X] = 1 + \theta$$

Exam exercise 7

2) MoM $M_1 = \frac{1}{n}\sum_{i=1}^{n} X_i = \bar{X}$

$$\bar{X} = E[X] = 1 + \theta$$

$$\hat{\theta}_{\text{MoM}} = \bar{X} - 1$$

3) Use data to compute the estimator

$$\hat{\theta}_{\text{MoM}} = \bar{X} - 1 = 1.435 - 1 = 0.435$$

4) $E[\hat{\theta}_{\text{MoM}}] = E[\bar{X} - 1] = E[X] - 1 = \theta$

Unbiased

Exam exercise 7

Exercise 6 : MoM and MLE (5 pts)

Let X be a continuous random variable with probability density function

$$f(x;\theta) = \begin{cases} \frac{1}{\theta} e^{-(x-1)/\theta}, & x \geq 1, \\ 0, & \text{otherwise,} \end{cases}$$

where $\theta > 0$ is an unknown parameter.

A random sample of size $n = 10$ is observed:

$$x_1 = 1.42,\ x_2 = 1.35,\ x_3 = 1.58,\ x_4 = 1.26,\ x_5 = 1.47,$$
$$x_6 = 1.31,\ x_7 = 1.62,\ x_8 = 1.39,\ x_9 = 1.51,\ x_{10} = 1.44.$$

Maximum Likelihood Estimator (2 pts)

1. Write the likelihood function $L(\theta)$ based on the sample.
2. Find the Maximum Likelihood Estimator $\hat{\theta}_{\text{MLE}}$.

   Hint: Use the log-likelihood.
3. Compute its numerical value using the observed sample.
4. Is your estimator biased?

Exam exercise 7

1) $L(\theta) = \prod_{i=1}^{n} f(x_i; \theta) = \prod_{i=1}^{10} \frac{1}{\theta} e^{-(x_i - 1)/\theta} = \theta^{-10} \exp\left(-\frac{1}{\theta} \sum_{i=1}^{10} (x_i - 1)\right)$

2) $\ell(\theta) = \ln L(\theta) = -10 \ln \theta - \frac{1}{\theta} \sum_{i=1}^{10} (x_i - 1)$

$$\frac{d\ell}{d\theta} = -\frac{10}{\theta} + \frac{1}{\theta^2} \sum_{i=1}^{10} (x_i - 1) = 0$$

$$\sum (x_i - 1) = 10\theta$$

$$\hat{\theta}_{\text{MLE}} = \frac{1}{n} \sum_{i=1}^{n} (x_i - 1) = \bar{X} - 1$$

3) $\hat{\theta}_{\text{MLE}} = \bar{X} - 1 = 0.435$

Which one is better?

Both **MoM** and **MLE** estimators give the same value here

Confidence Intervals

Session 2

Confidence Intervals

Session 2

Objectives:

- Understand the meaning of a confidence interval
- Correctly interpret confidence levels
- Construct confidence intervals for:
    - One mean (Z and t)
    - One proportion
    - One variance
- Understand the role of assumptions (normality, σ known/unknown)
- Use bootstrap to build confidence intervals.

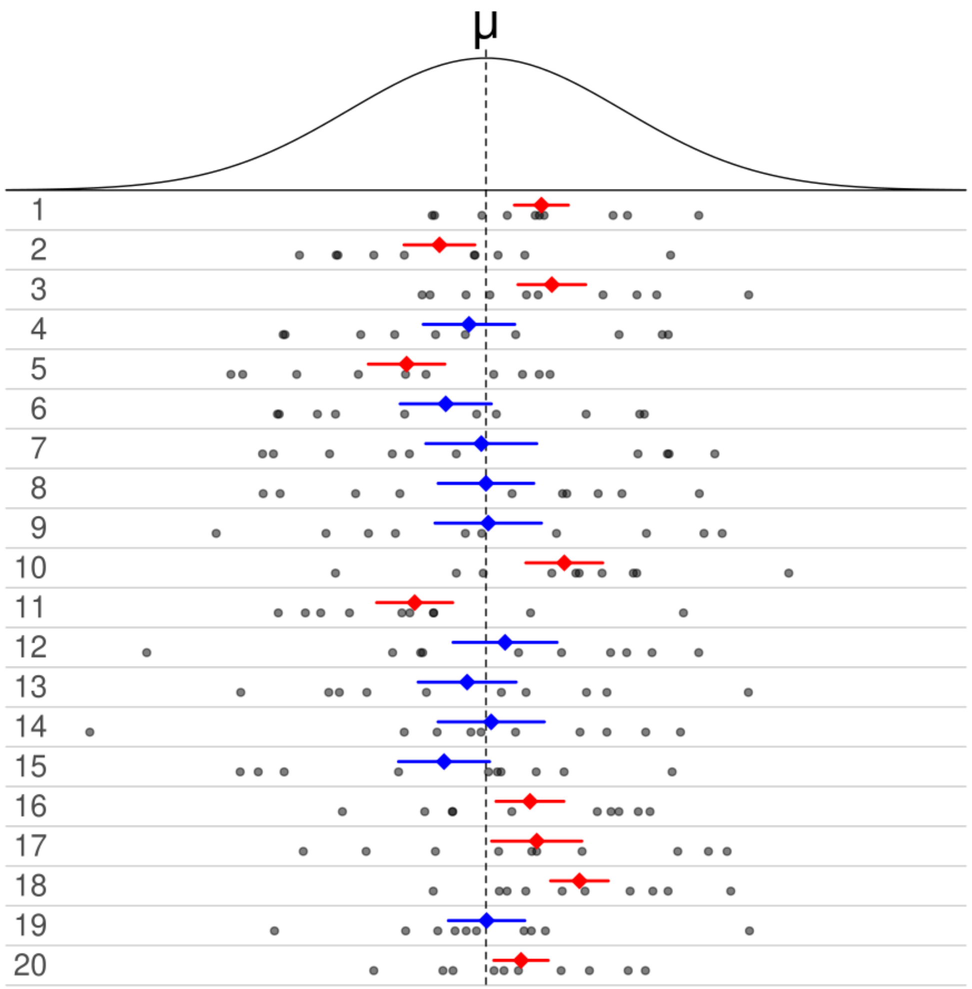
What is it?

Definition: a confidence interval (CI) is a region (interval), constructed under the assumptions of our model, that contains the “true” value (the parameter of interest) with a specified probability.

A confidence interval estimates are intervals within which the parameter is expected to fall, with a certain degree of confidence

Each interval has a **confidence coefficient** (reported as a proportion): **1-α** and a **degree of confidence** (reported as a %) : **(1-α)100%**

Typical confidence coefficients are 0.90, 0.95, and 0.99, with corresponding confidence levels 90%, 95%, and 99%.



Source: Wikipedia
(FRuDIxAFLG)

Interpretation

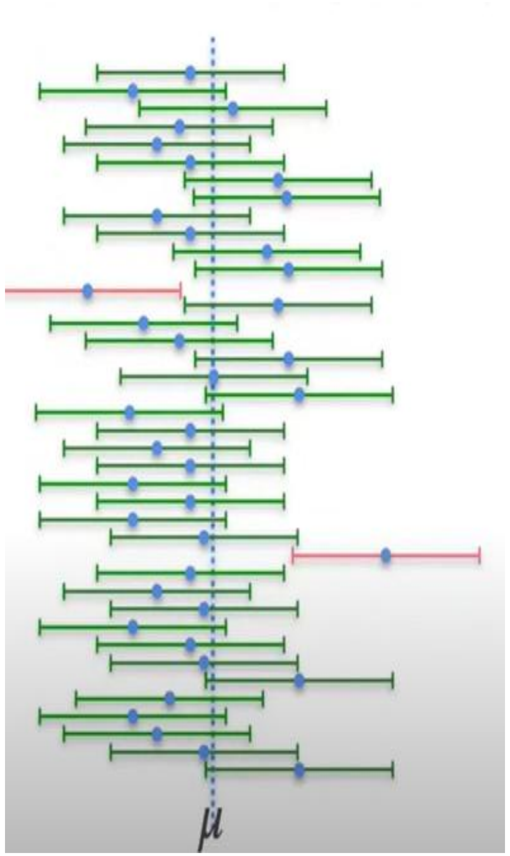
For a 95% confidence interval , is there a 95% probability that the population parameter falls within confidence interval?

Suppose we want to estimate the **average height of students** at Epita.
We take a random sample of **n = 50 students**
The sample mean height is **170 cm**
A 95% confidence interval is calculated as: [167cm; 173cm]

Do you think that there is 95% of probability that the average height of the students at Epita is between 167 and 173cm?

Interpretation

It is a common (and natural) mistake, when confronted with, for example, a 95% confidence interval, to say something like: “There is a 95% probability that the population parameter falls within confidence interval.



Why the statement is wrong:

- **The parameter is fixed** :The true mean of the population is a constant (though unknown). It either lies inside the interval, or it doesn’t
- **The randomness is in the sample** : What varies is the sample we take, and therefore the interval we compute

The correct interpretation, at least for Frequentist confidence intervals, requires us to invert our thinking: instead of focusing on the probability of the parameter being in the interval, we need to focus on the probability of the interval containing the parameter.

If we repeated the sampling process many times and constructed a confidence interval each time, about 95% of those intervals would contain the true average height.

CI estimation: Bootstrap

If we have sample data, then we can use bootstrapping methods to construct **a bootstrap sampling distribution** and construct a **confidence interval**.

Definition:

Bootstrapping is a resampling procedure that uses data from one sample to generate a sampling distribution by repeatedly taking random samples from the known sample.

CI estimation: Bootstrap

Steps:

We have data concerning the heights of individuals in a random sample of n=15. To construct a bootstrap distribution for the mean height we would:

- first randomly select one individual from that sample and record their height.
- Then, with the that individual placed back into the sample, we would randomly select a second individual and record their height.

This is known as "sampling with replacement" because we are putting each case back into the sample after recording their height.

- We would repeat this process until we have selected 15 values. Because we are sampling with replacement, some individuals may appear in the bootstrap sample more than once.
- We would use those 15 selected values to compute a bootstrapped sample mean.

This process is repeated many times. The distribution of many bootstrapped sample means is known as the bootstrap distribution or bootstrap sampling distribution.

CI estimation: Bootstrap

Steps:

Finally, to obtain the CI from the boostrap distribution:

- Order all the bootstrap means from smallest to largest.
- Take the 2.5th percentile (lower bound) and the 97.5th percentile (upper bound).
- That interval is your bootstrap 95% confidence interval for the mean height.

$$\mathrm{CI}_{95\%} = \left[\hat{\mu}^*_{2.5\%},\ \hat{\mu}^*_{97.5\%}\right]$$

CI estimation: Bootstrap (example)

We will use data from the Hubble Space Telescope. This data contains distances and velocities of 24 galaxies containing Cepheid stars, from the Hubble space telescope key project to measure the Hubble Constant
The **Hubble constant** is a fundamental number in cosmology that describes the **rate at which the universe is expanding**.

The data contains three columns:
Galaxy: A factor label identifying the galaxy
y: The galaxy's relative velocity measured in kilometers/second (km/s)
x: The galaxy's distance from Earth measured in Megaparsecs (Mpc)
(Note: 1 Mpc = 3.09e13 km)

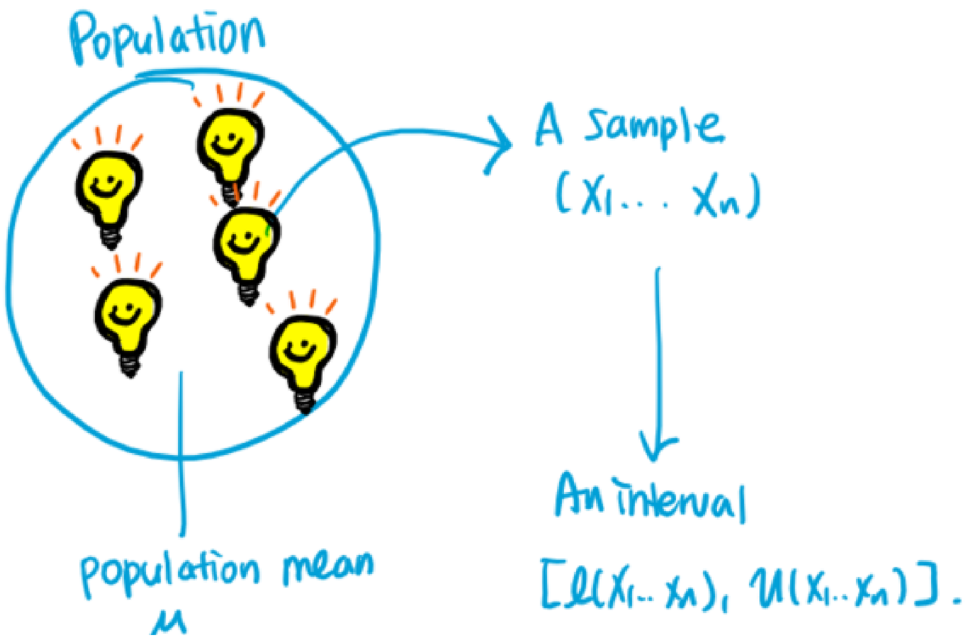
$$v = H_0 d$$

```
Galaxy,"y","x"
NGC0300,133,2
NGC0925,664,9.16
NGC1326A,1794,16.14
NGC1365,1594,17.95
NGC1425,1473,21.88
NGC2403,278,3.22
NGC2541,714,11.22
NGC2090,882,11.75
NGC3031,80,3.63
NGC3198,772,13.8
NGC3351,642,10
NGC3368,768,10.52
NGC3621,609,6.64
NGC4321,1433,15.21
NGC4414,619,17.7
NGC4496A,1424,14.86
NGC4548,1384,16.22
NGC4535,1444,15.78
NGC4536,1423,14.93
NGC4639,1403,21.98
NGC4725,1103,12.36
IC4182,318,4.49
NGC5253,232,3.15
NGC7331,999,14.72
```

Confidence intervals for one mean

Know how to compute a confidence interval (1-α) for when we have:

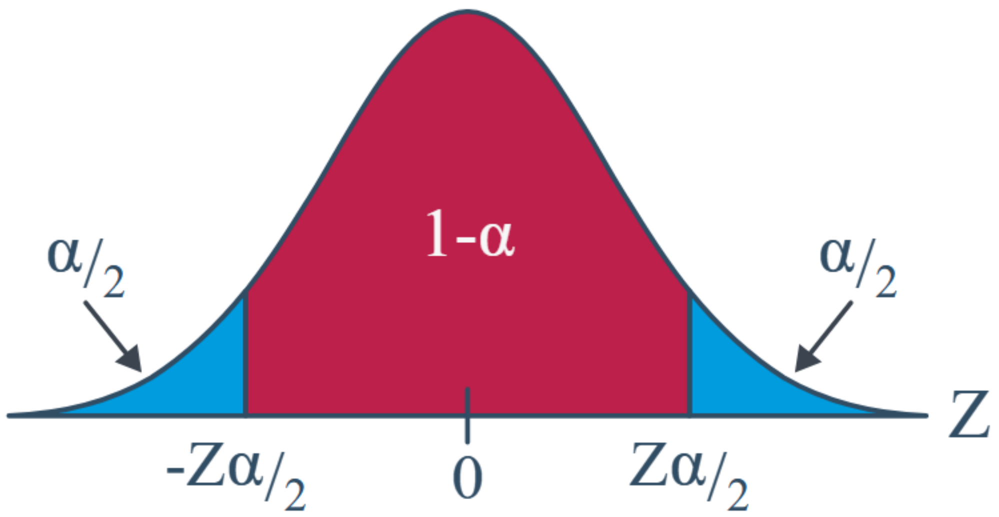
- a random sample ($X_1 \ldots X_n$) from $N(\mu,\sigma^2)$ with known σ^2
- a random sample ($X_1 \ldots X_n$) from $N(\mu,\sigma^2)$ with unknown σ^2
- a random sample from an unknown distribution but when we have a large sample size



Confidence intervals for μ from $N(\mu,\sigma^2)$ with known σ^2: Z interval

Notation: $z_{\alpha/2}$ is the Z value (obtained from a standard normal table) such that the area to the right of it under the standard normal curve is $\alpha/2$.

$$P(Z \geq z_{\alpha/2}) = \alpha/2$$



Confidence intervals for μ from $N(\mu,\sigma^2)$ with known σ^2: Z interval

Definition

1. $X_1, X_2, \ldots, X_n$ is a random sample from a normal population with mean μ and variance σ^2. So that:

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right) \text{ and } Z = \frac{\bar{X}-\mu}{\sigma/\sqrt{n}} \sim N(0,1)$$

2. The population variance $\sigma^2$is known.

Then, a $(1-\alpha)100\%$ **confidence interval for the mean** μ is:

$$\bar{x} \pm z_{\alpha/2}\left(\frac{\sigma}{\sqrt{n}}\right)$$

The interval, because it depends on Z, is often referred to as the **Z-interval for a mean.**

Confidence intervals for μ from $N(\mu,\sigma^2)$ with known σ^2: Z interval

Example: A random sample of **126 persons exposed to contamination** had an average blood lead level concentration **of 29.2 g/dl** . Assume , the blood lead level of a randomly selected person exposed to contamination, is **normally distributed with a standard deviation of 7.5** g/dl.

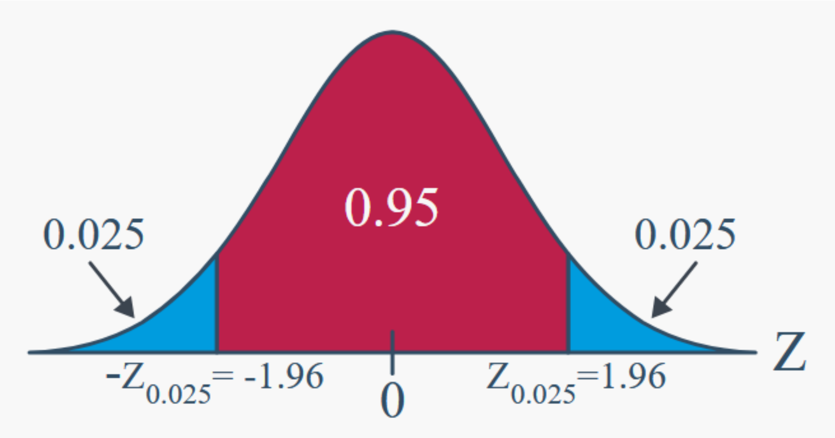
It is known that the **average blood lead level concentration** of humans with **no exposure is 18.2 g/dl** . Is there convincing evidence that the persons exposed to contamination have elevated blood lead level concentrations?

Confidence Level	Value of z_c
80% confidence	1.280
90% confidence	1.645
95% confidence	1.960
99% confidence	2.575

Let derive the 95% confidence interval, therefore $1-\alpha=0.95$ so $\alpha/2=0.025$

From Z-table, $z_{\alpha/2}=1.96$

$$\bar{x} \pm z_{\alpha/2}\left(\frac{\sigma}{\sqrt{n}}\right)$$



$$\left[29.2 - 1.96\left(\frac{7.5}{\sqrt{126}}\right), 29.2 + 1.96\left(\frac{7.5}{\sqrt{126}}\right)\right]$$

$$[27.89, 30.51]$$

Yes, **there is convincing evidence** that persons exposed to contamination have **elevated blood lead level concentrations** compared to humans with no exposure.

Confidence intervals for μ from $N(\mu,\sigma^2)$ with known σ^2: Z interval

Example: A researcher measures the heights of 100 randomly selected adult men in a city. The sample mean height is **175 cm**, and the population standard deviation is known to be **10 cm**. Assume the data is normally distributed

1) Calculate a **95% confidence interval** for the **true mean height** of adult men in this city

2) Imagine that you are not happy with this margin, you want only 0.5cm. What should be the smallest size of your random sample?

Confidence intervals for μ from $N(\mu,\sigma^2)$ with known σ^2: Z interval

Solution: A researcher measures the heights of 100 randomly selected adult men in a city. The sample mean height is **175 cm**, and the population standard deviation is known to be **10 cm**. Assume the data is normally distributed

1) Calculate a **95% confidence interval** for the **true mean height** of adult men in this city

$$\bar{x} \pm z_{\alpha/2}\left(\frac{\sigma}{\sqrt{n}}\right)$$

Let derive the 95% confidence interval, therefore $1-\alpha=0.95$ so $\alpha/2=0.025$

From Z-table, $z_{\alpha/2}=1.96$

CI: [173.04,176.96]

2) 0.5=1.96*10/sqrt(n) -> n=1537

Confidence intervals for μ from $N(\mu,\sigma^2)$ with unknown σ^2: T interval

To use the Z-interval method we need to know the population standard deviation but what happens if YOU DO NOT KNOW IT?

We need another procedure to derive the CI.

Therefore, instead of the Z procedure we will use **the T procedure** and use the **sample standard deviation S instead of the unknow population standard deviation**.

Definition:

Let $X_1, X_2, \ldots, X_n$ be a random sample from a normal distribution with unknown mean μ and unknown variance σ^2. The random variable

$$T = \frac{\overline{X} - \mu}{S/\sqrt{n}}$$

has a t distribution with $n - 1$ degrees of freedom.

A t-Interval for a Mean with unknown σ^2

Definition:

If $\bar{x}$ and s are the mean and standard deviation of a random sample from a normal distribution with unknown variance σ^2, a **$100(1-\alpha)\%$ confidence interval on μ** is given by

$$\bar{x} - t_{\alpha/2,n-1}s/\sqrt{n} \leq \mu \leq \bar{x} + t_{\alpha/2,n-1}s/\sqrt{n} \tag{8-16}$$

where $t_{\alpha/2,n-1}$ is the upper $100\alpha/2$ percentage point of the t distr $n-1$ degrees of freedom.

$$\bar{x} \pm t_{\alpha/2,n-1}\left(\frac{s}{\sqrt{n}}\right)$$

in mind, we say that:

1. $\bar{x}$ is a "**point estimate**" of μ
2. $\bar{x} \pm t_{\alpha/2,n-1}\left(\frac{s}{\sqrt{n}}\right)$ is an "**interval estimate**" of μ
3. $\frac{s}{\sqrt{n}}$ is the "standard error of the mean"
4. $t_{\alpha/2,n-1}\left(\frac{s}{\sqrt{n}}\right)$ is the "margin of error"

A t-Interval for a Mean with unknown σ^2

Example: The sample mean and standard deviation of a sample of n=22 are 13.71, and 3.55 respectively.
We want to find a 95% CI on population mean

Degrees of freedom *(df)*	.2	.15	.1	.05	.025	.01	.005	.001
1	3.078	4.165	6.314	12.706	25.452	63.657	127.321	636.619
2	1.886	2.282	2.920	4.303	6.205	9.925	14.089	31.599
3	1.638	1.924	2.353	3.182	4.177	5.841	7.453	12.924
4	1.533	1.778	2.132	2.776	3.495	4.604	5.598	8.610
5	1.476	1.699	2.015	2.571	3.163	4.032	4.773	6.869
6	1.440	1.650	1.943	2.447	2.969	3.707	4.317	5.959
7	1.415	1.617	1.895	2.365	2.841	3.499	4.029	5.408
8	1.397	1.592	1.860	2.306	2.752	3.355	3.833	5.041
9	1.383	1.574	1.833	2.262	2.685	3.250	3.690	4.781
10	1.372	1.559	1.812	2.228	2.634	3.169	3.581	4.587
11	1.363	1.548	1.796	2.201	2.593	3.106	3.497	4.437
12	1.356	1.538	1.782	2.179	2.560	3.055	3.428	4.318
13	1.350	1.530	1.771	2.160	2.533	3.012	3.372	4.221
14	1.345	1.523	1.761	2.145	2.510	2.977	3.326	4.140
15	1.341	1.517	1.753	2.131	2.490	2.947	3.286	4.073
16	1.337	1.512	1.746	2.120	2.473	2.921	3.252	4.015
17	1.333	1.508	1.740	2.110	2.458	2.898	3.222	3.965
18	1.330	1.504	1.734	2.101	2.445	2.878	3.197	3.922
19	1.328	1.500	1.729	2.093	2.433	2.861	3.174	3.883
20	1.325	1.497	1.725	2.086	2.423	2.845	3.153	3.850
21	1.323	1.494	1.721	2.080	2.414	2.831	3.135	3.819
[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]

A t-Interval for a Mean with unknown σ^2

Solution: The sample mean and standard deviation of a sample of n=22 are 13.71, and 3.55 respectively.
We want to find a 95% CI on population mean

$$\bar{x} - t_{\alpha/2,n-1}s/\sqrt{n} \le \mu \le \bar{x} + t_{\alpha/2,n-1}s/\sqrt{n}$$

n-1=21
$t_{\alpha/2,n-1}$=2,414

[12.14,15.28]

A t-Interval for a Mean with unknown σ^2

Exercice :

Determine the t-percentile that is required to construct each of the following two-sided confidence intervals:

- Confidence level 95%, degrees of freedom 10
- Confidence level 95%, degrees of freedom 20

Determine the t-percentile that is required to construct each of the following one-sided confidence intervals:

- Confidence level 95%, degrees of freedom 10
- Confidence level 95%, degrees of freedom 20

t Table

cum. prob	$t_{.50}$	$t_{.75}$	$t_{.80}$	$t_{.85}$	$t_{.90}$	$t_{.95}$	$t_{.975}$	$t_{.99}$	$t_{.995}$	$t_{.999}$	$t_{.9995}$
one-tail	0.50	0.25	0.20	0.15	0.10	$\boxed{0.05}$	0.025	0.01	0.005	0.001	0.0005
two-tails	1.00	0.50	0.40	0.30	0.20	0.10	0.05	0.02	0.01	0.002	0.001
df											
1	0.000	1.000	1.376	1.963	3.078	6.314	12.71	31.82	63.66	318.31	636.62
2	0.000	0.816	1.061	1.386	1.886	2.920	4.303	6.965	9.925	22.327	31.599
3	0.000	0.765	0.978	1.250	1.638	2.353	3.182	4.541	5.841	10.215	12.924
4	0.000	0.741	0.941	1.190	1.533	2.132	2.776	3.747	4.604	7.173	8.610
5	0.000	0.727	0.920	1.156	1.476	2.015	2.571	3.365	4.032	5.893	6.869
6	0.000	0.718	0.906	1.134	1.440	1.943	2.447	3.143	3.707	5.208	5.959
7	0.000	0.711	0.896	1.119	1.415	1.895	2.365	2.998	3.499	4.785	5.408
8	0.000	0.706	0.889	1.108	1.397	1.860	2.306	2.896	3.355	4.501	5.041
9	0.000	0.703	0.883	1.100	1.383	1.833	2.262	2.821	3.250	4.297	4.781
10	0.000	0.700	0.879	1.093	1.372	1.812	2.228	2.764	3.169	4.144	4.587
11	0.000	0.697	0.876	1.088	1.363	1.796	2.201	2.718	3.106	4.025	4.437
12	0.000	0.695	0.873	1.083	1.356	1.782	2.179	2.681	3.055	3.930	4.318
13	0.000	0.694	0.870	1.079	1.350	1.771	2.160	2.650	3.012	3.852	4.221
14	0.000	0.692	0.868	1.076	1.345	1.761	2.145	2.624	2.977	3.787	4.140
15	0.000	0.691	0.866	1.074	1.341	1.753	2.131	2.602	2.947	3.733	4.073
16	0.000	0.690	0.865	1.071	1.337	1.746	2.120	2.583	2.921	3.686	4.015
17	0.000	0.689	0.863	1.069	1.333	1.740	2.110	2.567	2.898	3.646	3.965
18	0.000	0.688	0.862	1.067	1.330	1.734	2.101	2.552	2.878	3.610	3.922
19	0.000	0.688	0.861	1.066	1.328	1.729	2.093	2.539	2.861	3.579	3.883
20	0.000	0.687	0.860	1.064	1.325	1.725	2.086	2.528	2.845	3.552	3.850

A t-Interval for a Mean with unknown σ^2

Solution : Two sided

- Confidence level 95%, degrees of freedom 10

$1-\alpha=0.95$ so $\alpha/2=0.025$ (two sided)

$t_{\alpha/2,n-1}=t_{0.025,10}=2{,}228$

- Confidence level 95%, degrees of freedom 20

$t_{\alpha/2,n-1}=t_{0.025,20}=2{,}086$

one-sided confidence intervals:

- Confidence level 95%, degrees of freedom 10

$1-\alpha=0.95$ so $\alpha=0.05$ (one sided)

$t_{\alpha,n-1}=t_{0.05,10}=1.812$

- Confidence level 95%, degrees of freedom 20

$t_{\alpha,n-1}=t_{0.05,20}=1.725$

t Table

cum. prob	$t_{.50}$	$t_{.75}$	$t_{.80}$	$t_{.85}$	$t_{.90}$	$t_{.95}$	$t_{.975}$	$t_{.99}$	$t_{.995}$	$t_{.999}$	$t_{.9995}$
one-tail	**0.50**	**0.25**	**0.20**	**0.15**	**0.10**	**0.05**	**0.025**	**0.01**	**0.005**	**0.001**	**0.0005**
two-tails	**1.00**	**0.50**	**0.40**	**0.30**	**0.20**	**0.10**	**0.05**	**0.02**	**0.01**	**0.002**	**0.001**
df											
1	0.000	1.000	1.376	1.963	3.078	6.314	12.71	31.82	63.66	318.31	636.62
2	0.000	0.816	1.061	1.386	1.886	2.920	4.303	6.965	9.925	22.327	31.599
3	0.000	0.765	0.978	1.250	1.638	2.353	3.182	4.541	5.841	10.215	12.924
4	0.000	0.741	0.941	1.190	1.533	2.132	2.776	3.747	4.604	7.173	8.610
5	0.000	0.727	0.920	1.156	1.476	2.015	2.571	3.365	4.032	5.893	6.869
6	0.000	0.718	0.906	1.134	1.440	1.943	2.447	3.143	3.707	5.208	5.959
7	0.000	0.711	0.896	1.119	1.415	1.895	2.365	2.998	3.499	4.785	5.408
8	0.000	0.706	0.889	1.108	1.397	1.860	2.306	2.896	3.355	4.501	5.041
9	0.000	0.703	0.883	1.100	1.383	1.833	2.262	2.821	3.250	4.297	4.781
10	0.000	0.700	0.879	1.093	1.372	1.812	2.228	2.764	3.169	4.144	4.587
11	0.000	0.697	0.876	1.088	1.363	1.796	2.201	2.718	3.106	4.025	4.437
12	0.000	0.695	0.873	1.083	1.356	1.782	2.179	2.681	3.055	3.930	4.318
13	0.000	0.694	0.870	1.079	1.350	1.771	2.160	2.650	3.012	3.852	4.221
14	0.000	0.692	0.868	1.076	1.345	1.761	2.145	2.624	2.977	3.787	4.140
15	0.000	0.691	0.866	1.074	1.341	1.753	2.131	2.602	2.947	3.733	4.073
16	0.000	0.690	0.865	1.071	1.337	1.746	2.120	2.583	2.921	3.686	4.015
17	0.000	0.689	0.863	1.069	1.333	1.740	2.110	2.567	2.898	3.646	3.965
18	0.000	0.688	0.862	1.067	1.330	1.734	2.101	2.552	2.878	3.610	3.922
19	0.000	0.688	0.861	1.066	1.328	1.729	2.093	2.539	2.861	3.579	3.883
20	0.000	0.687	0.860	1.064	1.325	1.725	2.086	2.528	2.845	3.552	3.850

A t-Interval for a Mean with unknown σ^2

Exercice : A machine produces metal rods used in an automobile suspension system. A random sample of 15 rods is selected, and the diameter is measured. The resulting data (in millimeters) are as follows:

8.24 8.25 8.20 8.23 8.24 8.21 8.26 8.26 8.20 8.25 8.23 8.23 8.19 8.28 8.24

(a) Check the assumption of normality for rod diameter.

(b) Calculate a 95% two-sided confidence interval on mean rod diameter.

(c) Calculate a 95% upper confidence bound on the mean. Compare this bound with the upper bound of the two-sided confidence interval and discuss why they are different

***t* Table**

cum. prob	$t_{.50}$	$t_{.75}$	$t_{.80}$	$t_{.85}$	$t_{.90}$	$t_{.95}$	$t_{.975}$	$t_{.99}$	$t_{.995}$	$t_{.999}$	$t_{.9995}$
one-tail	**0.50**	**0.25**	**0.20**	**0.15**	**0.10**	$\boxed{\textbf{0.05}}$	**0.025**	**0.01**	**0.005**	**0.001**	**0.0005**
two-tails	**1.00**	**0.50**	**0.40**	**0.30**	**0.20**	**0.10**	**0.05**	**0.02**	**0.01**	**0.002**	**0.001**
df											
1	0.000	1.000	1.376	1.963	3.078	6.314	12.71	31.82	63.66	318.31	636.62
2	0.000	0.816	1.061	1.386	1.886	2.920	4.303	6.965	9.925	22.327	31.599
3	0.000	0.765	0.978	1.250	1.638	2.353	3.182	4.541	5.841	10.215	12.924
4	0.000	0.741	0.941	1.190	1.533	2.132	2.776	3.747	4.604	7.173	8.610
5	0.000	0.727	0.920	1.156	1.476	2.015	2.571	3.365	4.032	5.893	6.869
6	0.000	0.718	0.906	1.134	1.440	1.943	2.447	3.143	3.707	5.208	5.959
7	0.000	0.711	0.896	1.119	1.415	1.895	2.365	2.998	3.499	4.785	5.408
8	0.000	0.706	0.889	1.108	1.397	1.860	2.306	2.896	3.355	4.501	5.041
9	0.000	0.703	0.883	1.100	1.383	1.833	2.262	2.821	3.250	4.297	4.781
10	0.000	0.700	0.879	1.093	1.372	1.812	2.228	2.764	3.169	4.144	4.587
11	0.000	0.697	0.876	1.088	1.363	1.796	2.201	2.718	3.106	4.025	4.437
12	0.000	0.695	0.873	1.083	1.356	1.782	2.179	2.681	3.055	3.930	4.318
13	0.000	0.694	0.870	1.079	1.350	1.771	2.160	2.650	3.012	3.852	4.221
14	0.000	0.692	0.868	1.076	1.345	1.761	2.145	2.624	2.977	3.787	4.140
15	0.000	0.691	0.866	1.074	1.341	1.753	2.131	2.602	2.947	3.733	4.073
16	0.000	0.690	0.865	1.071	1.337	1.746	2.120	2.583	2.921	3.686	4.015
17	0.000	0.689	0.863	1.069	1.333	1.740	2.110	2.567	2.898	3.646	3.965
18	0.000	0.688	0.862	1.067	1.330	1.734	2.101	2.552	2.878	3.610	3.922
19	0.000	0.688	0.861	1.066	1.328	1.729	2.093	2.539	2.861	3.579	3.883
20	0.000	0.687	0.860	1.064	1.325	1.725	2.086	2.528	2.845	3.552	3.850

A t-Interval for a Mean with unknown σ^2

Solution :

8.24 8.25 8.20 8.23 8.24 8.21 8.26 8.26 8.20 8.25
8.23 8.23 8.19 8.28 8.24

(a) Check the assumption of normality for rod diameter.

Mean=8.234
Median=8.23
Mode=8.23
Std=s=0.0252

(b) Calculate a 95% two-sided confidence interval on mean rod diameter.

$$\bar{x} \pm t_{0.025,14} \frac{s}{\sqrt{n}}$$

1-α=0.95 so α=0.05 (two sided)

$t_{\alpha/2,n-1}=t_{0.025,14}=2{,}145$

[8.220, 8.248]

(c) Calculate a 95% upper confidence bound on the mean. Compare this bound with the upper bound of the two-sided confidence interval and discuss why they are different

$$\bar{x} + t_{0.05,14} \frac{s}{\sqrt{n}}$$

1-α=0.95 so α=0.05 (one sided)

$t_{\alpha,n-1}=t_{0.05,14}=1{,}761$

8.246

In the two-sided case, we split α=0.05 across two tails (α/2=0.025 each), making the critical value larger.

In the one-sided case, the entire α=0.05 is in one tail, so the critical value is smaller

Confidence intervals for μ from $N(\mu,\sigma^2)$ with unknown σ^2: T interval

Example: You have a population of 6000 Adults and you want to know the average weight. You select randomly 49 persons and you find a mean of 170cm and a standard deviation of 25 cm.

Calculate the 95% CI of the average height of the adults

Confidence intervals for μ from $N(\mu,\sigma^2)$ with unknown σ^2: T interval

Solution: You have a population of 6000 Adults and you want to know the average weight. You select randomly 49 persons and you find a mean of 170cm and a standard deviation of 25 cm.

Calculate the 95% CI of the average height of the adults

Let derive the 95% confidence interval, therefore $1-\alpha=0.95$ so $\alpha/2=0.025$
dof=n-1=48
From t-table, $t_{\alpha/2,n-1}=2.011$

$$SE = \frac{s}{\sqrt{n}} = \frac{25}{\sqrt{49}} = \frac{25}{7}$$

Therefore the 95% CI is: [162.83,177.17]

What happens if you use the Z-distribution even if the dispersion is unknow?

Confidence intervals for μ from $N(\mu,\sigma^2)$ with unknown σ^2: T interval

Solution: You have a population of 6000 Adults and you want to know the average weight. You select randomly 49 persons and you find a mean of 170cm and a standard deviation of 25 cm.

Calculate the 95% CI of the average height of the adults

Let derive the 95% confidence interval, therefore $1-\alpha=0.95$ so $\alpha/2=0.025$
dof=n-1=48
From t-table, $t_{\alpha/2,n-1}=2.011$

$$SE = \frac{s}{\sqrt{n}} = \frac{25}{\sqrt{49}} = \frac{25}{7}$$

Therefore the 95% CI is: [162.83,177.17]

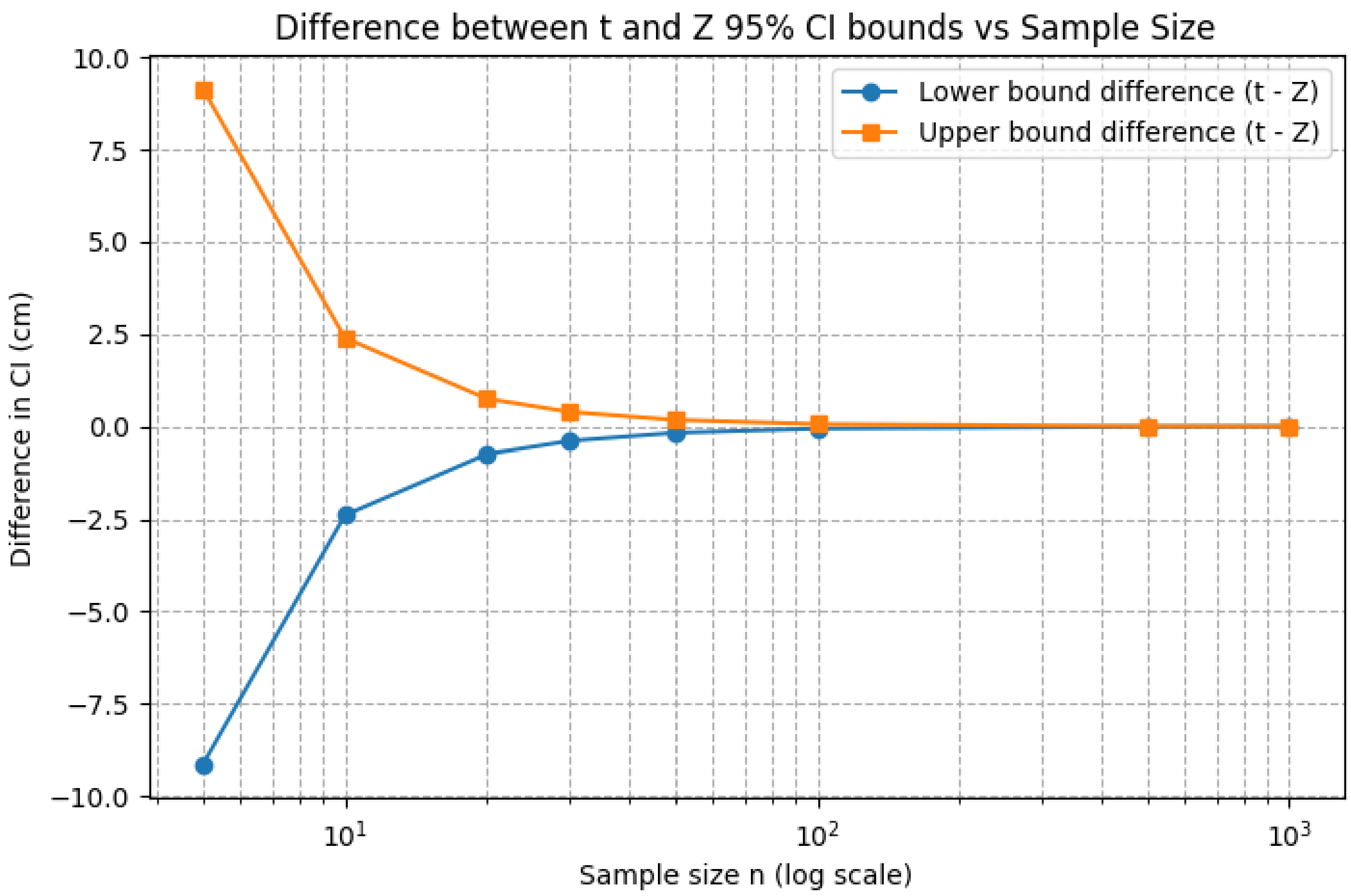
What happens if you use the Z-distribution even if the dispersion is unknow?

From the z-table , $z_{\alpha/2}=1.96$

Therefore the 95% CI is: [163.01, 176.99]

T-distribution -> Z-distribution

As n increases, **s → σ** by the **Central Limit Theorem**, so the t-distribution **approaches the Z-distribution**.



What to do with non normal data?

What happens if our data are skewed, and therefore clearly not normally distributed?

In practice!

1. Use $\bar{x} \pm t_{\alpha/2,n-1}\left(\frac{s}{\sqrt{n}}\right)$ if the data are normally distributed.
2. If you have reason to believe that the data are not normally distributed, then make sure you have a large enough sample ($n \geq 30$ generally suffices, but recall that it depends on the skewness of the distribution.) Then:

$$\bar{x} \pm t_{\alpha/2,n-1}\left(\frac{s}{\sqrt{n}}\right) \text{ and } \bar{x} \pm z_{\alpha/2}\left(\frac{s}{\sqrt{n}}\right)$$

   will give similar results.
3. If the data are not normally distributed *and* you have a small sample, use:

$$\bar{x} \pm t_{\alpha/2,n-1}\left(\frac{s}{\sqrt{n}}\right)$$

   with extreme caution and/or use a nonparametric confidence interval for the median (which we'll learn about later in this course).

Confidence interval : one proportion

It is often necessary to construct confidence intervals on a population proportion.

For example, suppose that a random sample of **size *n*** has been taken from a large population and that *X* observations in this sample belong to a class of interest. Then p_hat=X/n
is a point estimator of the proportion of the population *p* that belongs to this class.

n and p are the parameters of a binomial distribution.

We know that the sampling distribution of is approximately normal with mean np and variance p(p-1)/n

If n is large, the distribution of

$$Z = \frac{X - np}{\sqrt{np(1-p)}} = \frac{\hat{P} - p}{\sqrt{\dfrac{p(1-p)}{n}}}$$

is approximately standard normal.

Confidence interval : one proportion

On 418 surveyed, 280 French blamed rising insurance rates on large court settlements against doctors. That is, the sample proportion is $\hat{p}$=280/418

Use this sample proportion to estimate, with 95% confidence, the parameter p , that is, the proportion of *French* who blame rising insurance rates

Theorem

For large random samples, a $(1-\alpha)100\%$ **confidence interval** for a **population proportion** p is:

$$\hat{p} - z_{\alpha/2}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}} \leq p \leq \hat{p} + z_{\alpha/2}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

Confidence interval for variances: one proportion

Solution $\hat{p}$=280/418 .Use this sample proportion to estimate, with 95% confidence, the parameter p , that is, the proportion of *French* who blame rising insurance rates

α=0.05
$\hat{p}$=280/418=0.67
$z_{\alpha/2}$=1.96

Theorem

For large random samples, a $(1-\alpha)100\%$ **confidence interval** for a **population proportion** p is:

$$\hat{p} - z_{\alpha/2}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}} \leq p \leq \hat{p} + z_{\alpha/2}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

[0.625,0.715]

We can be 95% confident that between 62.5% and 71.5% of all French blame rising insurance rate

Confidence interval for variances: one proportion

Exercice : In a random sample of 85 automobile engine crankshaft bearings,10 have a surface finish that is rougher than the specifications allow. A 95% two-sided confidence interval for *p*?

Confidence interval for variances: one proportion

Solution In a random sample of 85 automobile engine crankshaft bearings,10 have a surface finish that is rougher than the specifications allow. A 95% two-sided confidence interval for *p*?

α=0.05
$\hat{p}$=10/85=0.12
$z_{\alpha/2}$=1.96

Theorem

For large random samples, a $(1-\alpha)100\%$ **confidence interval** for a **population proportion** p is:

$$\hat{p} - z_{\alpha/2}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}} \le p \le \hat{p} + z_{\alpha/2}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

[0.05,0.19]

We can be 95% confident that between 5% and 19% of all engine crankshaft bearings have a surface finish that is rougher than the specifications

Confidence interval for variances: one variance

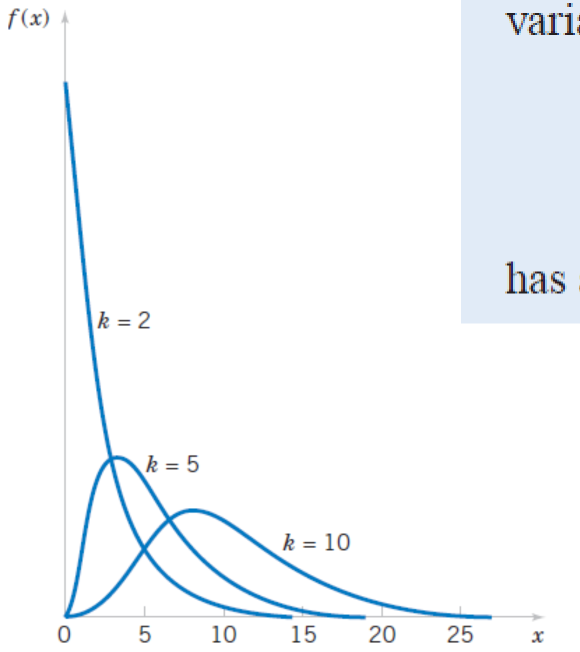
Sometimes confidence intervals on the population variance or standard deviation are needed

Definition :

Let $X_1, X_2, \ldots, X_n$ be a random sample from a normal distribution with mean μ and variance σ^2, and let S^2 be the sample variance. Then the random variable

$$X^2 = \frac{(n-1)\,S^2}{\sigma^2} \qquad (8\text{-}17)$$

has a chi-square (χ^2) distribution with $n-1$ degrees of freedom.



The probability density function of a χ^2 random variable is

$$f(x) = \frac{1}{2^{k/2}\Gamma(k/2)} x^{(k/2)-1} e^{-x/2} \qquad x > 0$$

Confidence interval for variances: one variance

Definition

If s^2 is the sample variance from a random sample of n observations from a normal distribution with unknown variance σ^2, then **a 100(1 − α)% confidence interval on σ^2 is**

$$\frac{(n-1)s^2}{\chi^2_{\alpha/2,n-1}} \leq \sigma^2 \leq \frac{(n-1)s^2}{\chi^2_{1-\alpha/2,n-1}} \tag{8-19}$$

where $\chi^2_{\alpha/2,n-1}$ and $\chi^2_{1-\alpha/2,n-1}$ are the upper and lower $100\alpha/2$ percentage points of the chi-square distribution with $n-1$ degrees of freedom, respectively. A **confidence interval for σ** has lower and upper limits that are the square roots of the corresponding limits in Equation 8-19.

Confidence interval for variances: one variance

Exemple: A large candy manufacturer produces, packages and sells packs of candy targeted to weigh 52 grams. To estimate σ , we took a random sample of n = 10 packs off of the factory line. The random sample yielded a sample variance of 4.2 grams. Use the random sample to derive a 95% confidence interval for σ.

$$\frac{(n-1)s^2}{\chi^2_{\alpha/2,n-1}} \leq \sigma^2 \leq \frac{(n-1)s^2}{\chi^2_{1-\alpha/2,n-1}}$$

Degrees of freedom (*df*)	Significance level (α)							
	.99	.975	.95	.9	.1	.05	.025	.01
1	-------	0.001	0.004	0.016	2.706	3.841	5.024	6.635
2	0.020	0.051	0.103	0.211	4.605	5.991	7.378	9.210
3	0.115	0.216	0.352	0.584	6.251	7.815	9.348	11.345
4	0.297	0.484	0.711	1.064	7.779	9.488	11.143	13.277
5	0.554	0.831	1.145	1.610	9.236	11.070	12.833	15.086
6	0.872	1.237	1.635	2.204	10.645	12.592	14.449	16.812
7	1.239	1.690	2.167	2.833	12.017	14.067	16.013	18.475
8	1.646	2.180	2.733	3.490	13.362	15.507	17.535	20.090
9	2.088	2.700	3.325	4.168	14.684	16.919	19.023	21.666
10	2.558	3.247	3.940	4.865	15.987	18.307	20.483	23.209

Confidence interval for variances: one variance

Solution: A large candy manufacturer produces, packages and sells packs of candy targeted to weigh 52 grams. To estimate σ , we took a random sample of n = 10 packs off of the factory line. The random sample yielded a sample variance of 4.2 grams. Use the random sample to derive a 95% confidence interval for σ.

n=10 -> n-1=9 sqrt(n-1)=3
s=4.2
α =0.05
a=2.70
b=19.023

[1.41,3.74]

$$\left(\frac{\sqrt{(n-1)}}{\sqrt{b}}s \leq \sigma \leq \frac{\sqrt{(n-1)}}{\sqrt{a}}s\right)$$

$a = \chi^2_{1-\alpha/2,n-1}$ and $b = \chi^2_{\alpha/2,n-1}$

Confidence interval for variances: one variance

Exercice: An article in *Urban Ecosystems*, "Urbanization and Warming of Phoenix (Arizona, USA): Impacts, Feedbacks and Mitigation" (2002, Vol. 6, pp. 183–203), mentions that Phoenix is ideal to study the effects of an urban heat island because it has grown from a population of 300,000 to approximately 3 million over the last 50 years and this is a period with a continuous, detailed climate record. The 50-year averages of the mean annual temperatures at eight sites in Phoenix are shown below.

Construct a 95% confidence interval for the standard deviation over the sites of the mean annual temperatures.

$$\frac{(n-1)s^2}{\chi^2_{\alpha/2,n-1}} \leq \sigma^2 \leq \frac{(n-1)s^2}{\chi^2_{1-\alpha/2,n-1}}$$

Site	Average Mean Temperature (°C)
Sky Harbor Airport	23.3
Phoenix Greenway	21.7
Phoenix Encanto	21.6
Waddell	21.7
Litchfield	21.3
Laveen	20.7
Maricopa	20.9
Harlquahala	20.1

Degrees of freedom (df)	Significance level (α) .99	.975	.95	.9	.1	.05	.025	.01
1	--------	0.001	0.004	0.016	2.706	3.841	5.024	6.635
2	0.020	0.051	0.103	0.211	4.605	5.991	7.378	9.210
3	0.115	0.216	0.352	0.584	6.251	7.815	9.348	11.345
4	0.297	0.484	0.711	1.064	7.779	9.488	11.143	13.277
5	0.554	0.831	1.145	1.610	9.236	11.070	12.833	15.086
6	0.872	1.237	1.635	2.204	10.645	12.592	14.449	16.812
7	1.239	1.690	2.167	2.833	12.017	14.067	16.013	18.475
8	1.646	2.180	2.733	3.490	13.362	15.507	17.535	20.090
9	2.088	2.700	3.325	4.168	14.684	16.919	19.023	21.666
10	2.558	3.247	3.940	4.865	15.987	18.307	20.483	23.209

Confidence interval for variances: one variance

Solution:
n=8 -> n-1=7 sqrt(n-1)=2.64
s=0.95 (mean is 21.41, variance 0.896)
α =0.05
a=1.690
b=16.13

[0.63,1.93]

$$\left(\frac{\sqrt{(n-1)}}{\sqrt{b}}s \leq \sigma \leq \frac{\sqrt{(n-1)}}{\sqrt{a}}s\right)$$

$a = \chi^2_{1-\alpha/2,n-1}$ and $b = \chi^2_{\alpha/2,n-1}$

Site	Average Mean Temperature (°C)
Sky Harbor Airport	23.3
Phoenix Greenway	21.7
Phoenix Encanto	21.6
Waddell	21.7
Litchfield	21.3
Laveen	20.7
Maricopa	20.9
Harlquahala	20.1

Summary

Population parameter	Distribution assumption	Symbol	Conditions / Other parameters	Test / Distribution used	Confidence Interval formula
Mean	Normal distribution	μ	σ known	**Z-test** (Normal)	$\bar{x} \pm z_{\alpha/2}\frac{\sigma}{\sqrt{n}}$
Mean	Arbitrary distribution, large n	μ	CLT applies, σ ≈ known	**Z-test** (Normal approx.)	$\bar{x} \pm z_{\alpha/2}\frac{s}{\sqrt{n}}$
Mean	Normal distribution	μ	σ unknown, estimated	**t-test** (Student t)	$\bar{x} \pm t_{\alpha/2,n-1}\frac{s}{\sqrt{n}}$
Variance / Std. dev.	Normal distribution	$\sigma^2,\ \sigma$	Mean unknown	**χ^2 distribution**	$\frac{(n-1)s^2}{\chi^2_{\alpha/2,n-1}} \leq \sigma^2 \leq \frac{(n-1)s^2}{\chi^2_{1-\alpha/2,n-1}}$
Population proportion	Bernoulli / Binomial	p	$n\hat{p} \geq 5,\ n(1-\hat{p}) \geq 5$	**Z-test** (Normal approx.)	$\hat{p} \pm z_{\alpha/2}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$

Confidence Intervals for a two Sample

Session 3

Confidence Intervals for a two Sample

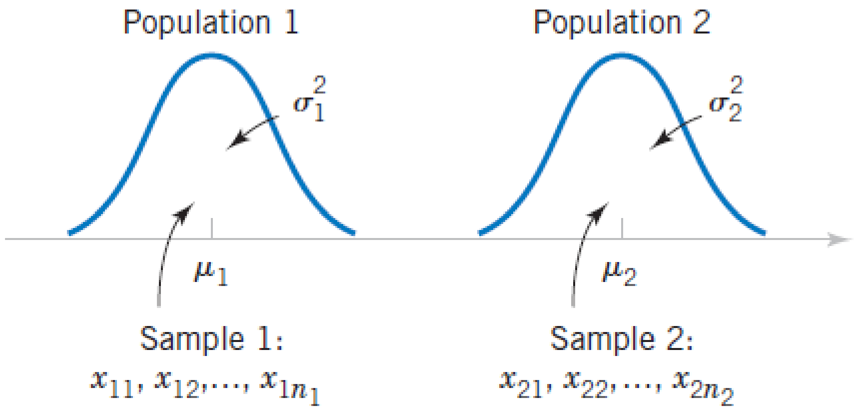
Session 3

Objectives:

- Construct confidence intervals for:
    - The difference of two mean
    - The difference of proportion
    - two variance
- Understand the role of assumptions (normality, σ known/unknown)

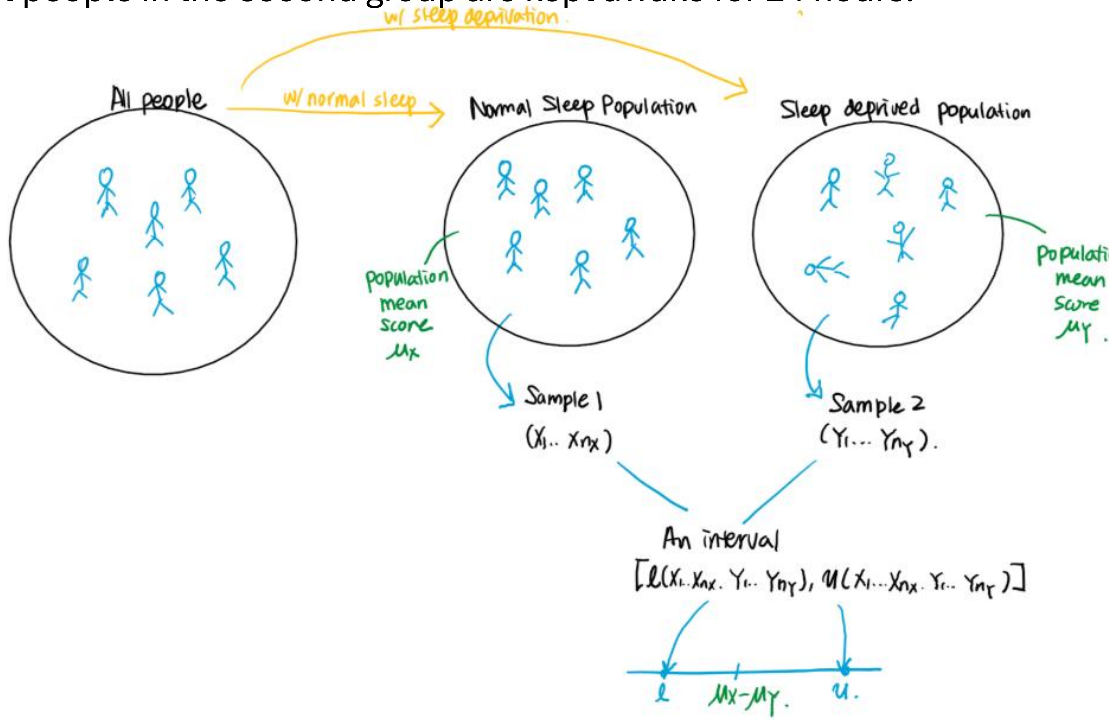
Two independent random samples vs a paired random sample

The previous session presented confidence intervals for a single population parameter (the mean , the variance , or a proportion *p*). This session extends those results to the case of two independent populations.

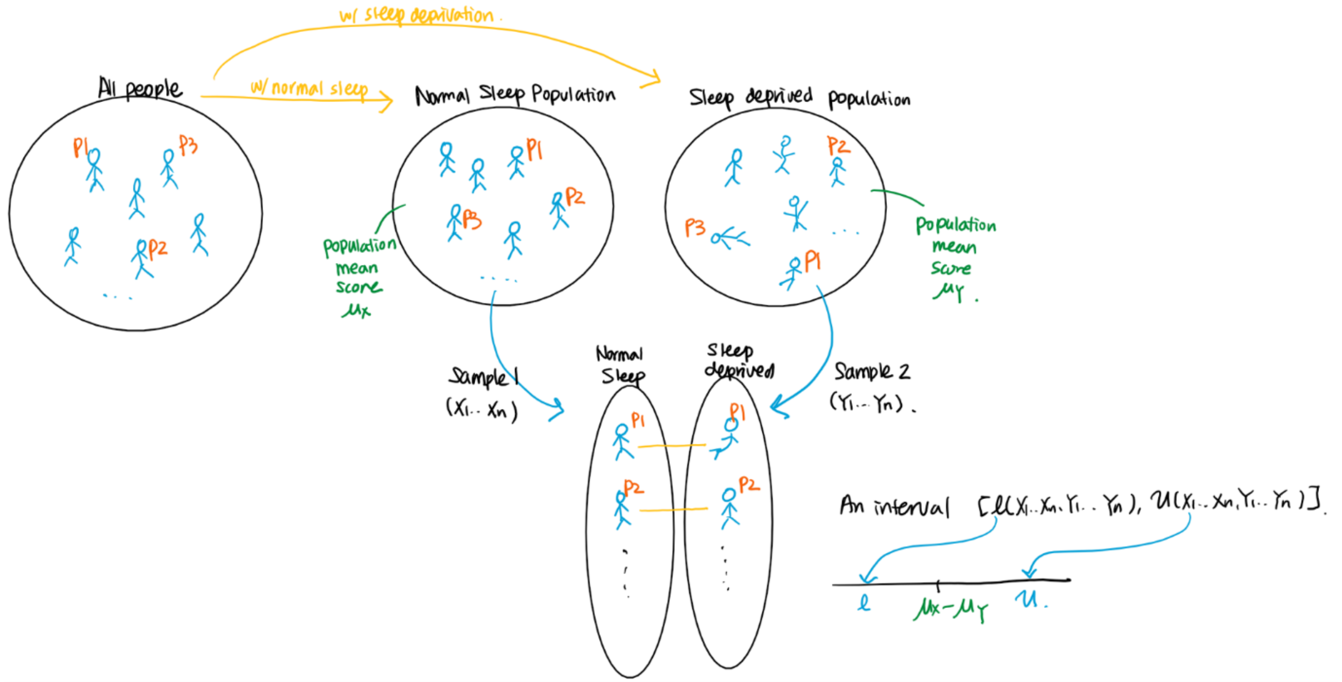


Two independent random samples vs a paired random sample

20 person / into 2 groups. 1st group are allowed to have a normal sleep, but people in the second group are kept awake for 24 hours.



The researcher recruited 10 participants. Each participant is asked to take the tests twice: one after a normal sleep and the other after being kept awake for 24 hours



In the first protocol, there is no association between ith subject from the first and second group, and therefore X_i and Y_i are independent. On the other hand, in the second protocol, X_i and Y_i are dependent, since the test results are from the same ith person. In fact, we have a *paired* random sample, since each X_i and Y_i should be paired. Therefore,

- Following the first protocol, we would have two independent random samples $(X_1, \dots, X_{10})$ and $(Y_1, \dots, Y_{10})$.
- Following the second protocol, we would have a paired random sample $((X_1, Y_1), \dots, (X_{10}, Y_{10}))$.

Confidence interval for two means with the same known variance

Definition

Confidence Interval on the Difference in Means, Variances Known

If $\overline{x}_1$ and $\overline{x}_2$ are the means of independent random samples of sizes n_1 and n_2 from two independent normal populations with known variances σ_1^2 and σ_2^2, respectively, a **$100(1-\alpha)\%$ confidence interval for $\mu_1 - \mu_2$** is

$$\overline{x}_1 - \overline{x}_2 - z_{\alpha/2}\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}} \le \mu_1 - \mu_2 \le \overline{x}_1 - \overline{x}_2 + z_{\alpha/2}\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}} \qquad (10\text{-}7)$$

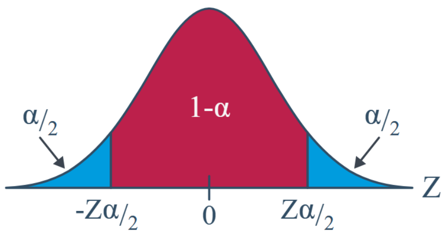
where $z_{\alpha/2}$ is the upper $\alpha/2$ percentage point of the standard normal distribution.

Confidence interval for two means with the same known variance

Dem

Notation: z $_{\alpha/2}$ is the Z value (obtained from a standard normal table) such that the area to the right of it under the standard normal curve is α/2.

$$P(Z \geq z_{\alpha/2}) = \alpha/2$$



$$\bar{X}_1 \sim N\left(\mu_1, \frac{\sigma_1^2}{n_1}\right)$$

$$\bar{X}_2 \sim N\left(\mu_2, \frac{\sigma_2^2}{n_2}\right)$$

$$\mathrm{Var}(\bar{X}_1 - \bar{X}_2) = \mathrm{Var}(\bar{X}_1) + \mathrm{Var}(\bar{X}_2)$$

$$\bar{X}_1 - \bar{X}_2 \sim N\left(\mu_1 - \mu_2, \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}\right)$$

standard normal variable:

$$Z = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

$$P\left(-z_{\alpha/2} \leq Z \leq z_{\alpha/2}\right) = 1 - \alpha$$

$$P\left(-z_{\alpha/2} \leq \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} \leq z_{\alpha/2}\right) = 1 - \alpha$$

$$P\left((\bar{X}_1 - \bar{X}_2) - z_{\alpha/2}\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}} \leq \mu_1 - \mu_2 \leq (\bar{X}_1 - \bar{X}_2) + z_{\alpha/2}\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}\right) = 1 - \alpha$$

Confidence interval for two means with the same known variance

Exercice Tensile strength tests were performed on two different grades of aluminum spars used in manufacturing the wing of a commercial transport aircraft. From past experience with the spar manufacturing process and the testing procedure, the standard deviations of tensile strengths are assumed to be known. The data obtained are as follows: $n1$ =10 x_bar1= 87.6 ,σ1=1 and $n2$ =12 x_bar2= 74.5 ,σ2=1.5

Find a 90% confidence interval on the difference in mean

Confidence Level	Value of z_c
80% confidence	1.280
90% confidence	1.645
95% confidence	1.960
99% confidence	2.575

Confidence interval for two means with the same known variance

Solution $n1$ =10 μ1= 87.6 ,σ1=1 and $n2$ =12 μ2= 74.5 ,σ2=1

Confidence Interval on the Difference in Means, Variances Known

If $\bar{x}_1$ and $\bar{x}_2$ are the means of independent random samples of sizes n_1 and n_2 from two independent normal populations with known variances σ_1^2 and σ_2^2, respectively, a **100(1 − α)% confidence interval for $\mu_1 - \mu_2$** is

$$\bar{x}_1 - \bar{x}_2 - z_{\alpha/2}\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}} \le \mu_1 - \mu_2 \le \bar{x}_1 - \bar{x}_2 + z_{\alpha/2}\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}} \quad (10\text{-}7)$$

where $z_{\alpha/2}$ is the upper $\alpha/2$ percentage point of the standard normal distribution.

$$87.6 - 74.5 - 1.645\sqrt{\frac{(1)^2}{10} + \frac{(1.5)^2}{12}} \le \mu_1 - \mu_2 \le 87.6 - 74.5 + 1.645\sqrt{\frac{(1^2)}{10} + \frac{(1.5)^2}{12}}$$

$$12.22 \le \mu_1 - \mu_2 \le 13.98$$

Confidence interval for two means with the same known variance

Exercice Two machines are used for filling plastic bottles withva net volume of 16.0 ounces. The fill volume can be assumed normal, with standard σ1= 0.020 and σ2 =0.025 ounces. A member of the quality engineering staff suspects that both machines fill to the same mean net volume, whether or not this volume is 16.0 ounces. A random sample of 10 bottles is taken from the output of each machine.

Calculate a 95% confidence interval on the difference in means. Provide a practical interpretation of this interval.

Machine 1		Machine 2	
16.03	16.01	16.02	16.03
16.04	15.96	15.97	16.04
16.05	15.98	15.96	16.02
16.05	16.02	16.01	16.01
16.02	15.99	15.99	16.00

Confidence interval for two means with the same known variance

Solution Two machines are used for filling plastic bottles withva net volume of 16.0 ounces. The fill volume can be assumed normal, with standard σ1= 0.020 and σ2 =0.025 ounces. A member of the quality engineering staff suspects that both machines fill to the same mean net volume, whether or not this volume is 16.0 ounces. A random sample of 10 bottles is taken from the output of each machine.

$$\bar{x}_1 = 16.015, \quad \sigma_1 = 0.020, \quad n_1 = 10$$

$$\bar{x}_2 = 16.005, \quad \sigma_2 = 0.025, \quad n_2 = 10$$

$$z_{\alpha/2} = 1.96$$

$$SE = \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$$

$$CI = (\bar{x}_1 - \bar{x}_2) \pm z_{\alpha/2} \cdot SE$$

$$-0.0098, \ 0.0298)$$

Machine 1		Machine 2	
16.03	16.01	16.02	16.03
16.04	15.96	15.97	16.04
16.05	15.98	15.96	16.02
16.05	16.02	16.01	16.01
16.02	15.99	15.99	16.00

Confidence Interval on the Difference in Means, Variances Known

If $\bar{x}_1$ and $\bar{x}_2$ are the means of independent random samples of sizes n_1 and n_2 from two independent normal populations with known variances σ_1^2 and σ_2^2, respectively, a **100(1 − α)% confidence interval for $\mu_1 - \mu_2$** is

$$\bar{x}_1 - \bar{x}_2 - z_{\alpha/2}\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}} \le \mu_1 - \mu_2 \le \bar{x}_1 - \bar{x}_2 + z_{\alpha/2}\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}} \qquad (10\text{-}7)$$

where $z_{\alpha/2}$ is the upper $\alpha/2$ percentage point of the standard normal distribution.

Confidence interval for two means with the same unknown variance

Definition

Case 1: Confidence Interval on the Difference in Means, Variances Unknowns and Equal

If $\bar{x}_1$, $\bar{x}_2$, s_1^2, and s_2^2 are the sample means and variances of two random samples of sizes n_1 and n_2, respectively, from two independent normal populations with unknown but equal variances, then a **100(1 − α)% confidence interval on the difference in means $\mu_1 - \mu_2$** is

$$\bar{x}_1 - \bar{x}_2 - t_{\alpha/2, n_1+n_2-2}\, s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$$

$$\leq \mu_1 - \mu_2 \leq \bar{x}_1 - \bar{x}_2 + t_{\alpha/2, n_1+n_2-2}\, s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}} \qquad (10\text{-}19)$$

where $s_p = \sqrt{[(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2]/(n_1 + n_2 - 2)}$ is the pooled estimate of the common population standard deviation, and $t_{\alpha/2, n_1+n_2-2}$ is the upper $\alpha/2$ percentage point of the t distribution with $n_1 + n_2 - 2$ degrees of freedom.

- The measurements (X_i and Y_i) are **independent**.
- The measurements in each population are **normally distributed**.
- The measurements in each population have the **same variance** σ^2. $\sigma_1^2 = \sigma_2^2 = \sigma^2$

Confidence interval for two means with the same unknown variance

Example:

Specie 1

sample 1	sample 2	sample 3	sample 4	sample 5	sample 6	sample 7	sample 8	sample 9	sample 10
10.2	6.9	10.9	11.0	10.1	5.3	7.5	10.3	9.2	8.8

Specie 2

sample 1	sample 2	sample 3	sample 4	sample 5	sample 6	sample 7	sample 8	sample 9	sample 10
12.9	10.2	7.4	7.0	10.5	11.9	7.1	9.9	14.4	11.3

What is the difference, if any, in the mean size of the prey (of the entire populations) of the two species? We assume that the population variances are the same

95%

t Table

cum. prob	$t_{.50}$	$t_{.75}$	$t_{.80}$	$t_{.85}$	$t_{.90}$	$t_{.95}$	$t_{.975}$	$t_{.99}$	$t_{.995}$	$t_{.999}$	$t_{.9995}$
one-tail	**0.50**	**0.25**	**0.20**	**0.15**	**0.10**	**0.05**	**0.025**	**0.01**	**0.005**	**0.001**	**0.0005**
two-tails	**1.00**	**0.50**	**0.40**	**0.30**	**0.20**	**0.10**	**0.05**	**0.02**	**0.01**	**0.002**	**0.001**
df											
1	0.000	1.000	1.376	1.963	3.078	6.314	12.71	31.82	63.66	318.31	636.62
2	0.000	0.816	1.061	1.386	1.886	2.920	4.303	6.965	9.925	22.327	31.599
3	0.000	0.765	0.978	1.250	1.638	2.353	3.182	4.541	5.841	10.215	12.924
4	0.000	0.741	0.941	1.190	1.533	2.132	2.776	3.747	4.604	7.173	8.610
5	0.000	0.727	0.920	1.156	1.476	2.015	2.571	3.365	4.032	5.893	6.869
6	0.000	0.718	0.906	1.134	1.440	1.943	2.447	3.143	3.707	5.208	5.959
7	0.000	0.711	0.896	1.119	1.415	1.895	2.365	2.998	3.499	4.785	5.408
8	0.000	0.706	0.889	1.108	1.397	1.860	2.306	2.896	3.355	4.501	5.041
9	0.000	0.703	0.883	1.100	1.383	1.833	2.262	2.821	3.250	4.297	4.781
10	0.000	0.700	0.879	1.093	1.372	1.812	2.228	2.764	3.169	4.144	4.587
11	0.000	0.697	0.876	1.088	1.363	1.796	2.201	2.718	3.106	4.025	4.437
12	0.000	0.695	0.873	1.083	1.356	1.782	2.179	2.681	3.055	3.930	4.318
13	0.000	0.694	0.870	1.079	1.350	1.771	2.160	2.650	3.012	3.852	4.221
14	0.000	0.692	0.868	1.076	1.345	1.761	2.145	2.624	2.977	3.787	4.140
15	0.000	0.691	0.866	1.074	1.341	1.753	2.131	2.602	2.947	3.733	4.073
16	0.000	0.690	0.865	1.071	1.337	1.746	2.120	2.583	2.921	3.686	4.015
17	0.000	0.689	0.863	1.069	1.333	1.740	2.110	2.567	2.898	3.646	3.965
18	0.000	0.688	0.862	1.067	1.330	1.734	2.101	2.552	2.878	3.610	3.922

Confidence interval for two means with the same unknown variance

Solution:

10.2	6.9	10.9	11.0	10.1	5.3	7.5	10.3	9.2	8.8
12.9	10.2	7.4	7.0	10.5	11.9	7.1	9.9	14.4	11.3

What is the difference, if any, in the mean size of the prey (of the entire populations) of the two species?

mean_x=9.02 , std_x=1.9
mean_y=10.26, std_y=2.51
n=m=10
Sp^2=4.955 so, Sp=2.225

$$(\bar{X}-\bar{Y}) \pm (t_{\alpha/2,n+m-2}) S_p \sqrt{\frac{1}{n}+\frac{1}{m}}$$

$$S_p^2 = \frac{(n-1)S_X^2 + (m-1)S_Y^2}{n+m-2}$$

95% confident level.
$t_{\alpha/2,n+m-2}=t_{0.025,18}=2,101$

Therefore, the interval is: **[-0.852,3.332]**

No difference because 0 is included in the interval.

Confidence interval for two means with the different unknown variances

Definition: If the population X and Y do not have the same variance we should use the Welch's t-Interval

Welch's t-interval for $\mu_X - \mu_Y$. If:

the data are normally distributed (or if not, the underlying distributions are not too badly skewed, and n and m are large enough), and the population variances σ_X^2 and σ_Y^2 can't be assumed to be equal,
then, a $(1-\alpha)100\%$ **confidence interval for** $\mu_X - \mu_Y$, the difference in the population means is:

$$\bar{X} - \bar{Y} \pm t_{\alpha/2,r}\sqrt{\frac{s_X^2}{n} + \frac{s_Y^2}{m}}$$

where the r degrees of freedom are approximated by:

$$r = \frac{\left(\frac{s_X^2}{n} + \frac{s_Y^2}{m}\right)^2}{\frac{(s_X^2/n)^2}{n-1} + \frac{(s_Y^2/m)^2}{m-1}}$$

Confidence interval for two means with the different unknown variances

Exercice: A botanist wants to compare the effect of two fertilizers (A and B) on the growth of tomato plants. The heights (in cm) of plants after 6 weeks are measured. Due to variability in soil and watering, the botanist suspects that the variances may not be equal

Fertilizer A (n_1 = 8): 21.4, 23.1, 19.8, 22.5, 20.9, 21.0, 24.3, 22.1
Fertilizer B (n_2 = 10): 18.7, 19.5, 20.3, 17.8, 18.2, 19.9, 21.0, 20.5, 18.8, 19.1

Compute a **95% confidence interval** for the difference in mean growth

Confidence interval for two means with the different unknown variances

Solution:
Fertilizer A (n_1 = 8): 21.4, 23.1, 19.8, 22.5, 20.9, 21.0, 24.3, 22.1
Fertilizer B (n_2 = 10): 18.7, 19.5, 20.3, 17.8, 18.2, 19.9, 21.0, 20.5, 18.8, 19.1

$$\bar{X} - \bar{Y} \pm t_{\alpha/2,r}\sqrt{\frac{s_X^2}{n} + \frac{s_Y^2}{m}}$$

n are approximated by:

$$r = \frac{\left(\frac{s_X^2}{n} + \frac{s_Y^2}{m}\right)^2}{\frac{(s_X^2/n)^2}{n-1} + \frac{(s_Y^2/m)^2}{m-1}}$$

Compute a **95% confidence interval** for the difference in mean growth

$$\bar{x}_A = \frac{175.1}{8} \quad \bar{x}_B = \frac{193.8}{10}$$

$$s_A^2 = \frac{13.071}{n_1 - 1} = \frac{13.071}{7} \approx 1.867 \Rightarrow s_A \approx \sqrt{1.867} \approx 1.366$$

$$s_B^2 = \frac{9.772}{n_2 - 1} = \frac{9.772}{9} \approx 1.0869 \Rightarrow s_B \approx \sqrt{1.0869} \approx 1.043$$

$$\frac{(s_A^2/n_1 + s_B^2/n_2)^2}{\frac{(s_A^2/n_1)^2}{n_1-1} + \frac{(s_B^2/n_2)^2}{n_2-1}} \approx 12.75$$

Two-tailed, α = 0.05, df ≈ 13 → $t_{0.025,13} \approx 2.160$

$$CI \approx (1.247, 3.773)$$

Confidence interval for variances: ratio

Definition: The confidence interval for the ratio of two variances requires the use of the probability distribution known as the ***F*-distribution.**

Distribution of the Ratio of Sample Variances from Two Normal Distributions

Let $X_{11}, X_{12}, \ldots, X_{1n_1}$ be a random sample from a normal population with mean μ_1 and variance σ_1^2, and let $X_{21}, X_{22}, \ldots, X_{2n_2}$ be a random sample from a second normal population with mean μ_2 and variance σ_2^2. Assume that both normal populations are independent. Let S_1^2 and S_2^2 be the sample variances. Then the ratio

$$F = \frac{S_1^2/\sigma_1^2}{S_2^2/\sigma_2^2}$$

has an F distribution with $n_1 - 1$ numerator degrees of freedom and $n_2 - 1$ denominator degrees of freedom.

Confidence interval for variances: ratio

Definition:

Confidence Interval on the Ratio of Variances from Two Normal Distributions

If s_1^2 and s_2^2 are the sample variances of random samples of sizes n_1 and n_2, respectively, from two independent normal populations with unknown variances σ_1^2 and σ_2^2, then a **100(1 − α)% confidence interval on the ratio σ_1^2/σ_2^2** is

$$\frac{s_1^2}{s_2^2} f_{1-\alpha/2, n_2-1, n_1-1} \le \frac{\sigma_1^2}{\sigma_2^2} \le \frac{s_1^2}{s_2^2} f_{\alpha/2, n_2-1, n_1-1} \tag{10-33}$$

where $f_{\alpha/2, n_2-1, n_1-1}$ and $f_{1-\alpha/2, n_2-1, n_1-1}$ are the upper and lower $\alpha/2$ percentage points of the F distribution with $n_2 - 1$ numerator and $n_1 - 1$ denominator degrees of freedom, respectively. A confidence interval on the ratio of the standard deviations can be obtained by taking square roots in Equation 10-33.

Confidence interval for variances: ratio

Examples:

Adult DEINOPIS	Adult MENNEUS
n = 10	m = 10
$\bar{x}$ = 10.26 mm	$\bar{y}$ = 9.02 mm
$s_X^2 = (2.51)^2$	$s_Y^2 = (1.90)^2$

Estimate, with 95% confidence, the ratio of the two population variances.

	$\alpha = 0.025$																		
	d.f.N.																		
d.f.D.	***1***	***2***	***3***	***4***	***5***	***6***	***7***	***8***	***9***	***10***	***12***	***15***	***20***	***24***	***30***	***40***	***60***	***120***	**∞**
1	647.8	799.5	864.2	899.6	921.8	937.1	948.2	956.7	963.3	968.6	976.7	984.9	993.1	997.2	1001	1006	1010	1014	1018
2	38.51	39.00	39.17	39.25	39.30	39.33	39.36	39.37	39.39	39.40	39.41	39.43	39.45	39.46	39.46	39.47	39.48	39.49	39.50
3	17.44	16.04	15.44	15.10	14.88	14.73	14.62	14.54	14.47	14.42	14.34	14.25	14.17	14.12	14.08	14.04	13.99	13.95	13.90
4	12.22	10.65	9.98	9.60	9.36	9.20	9.07	8.98	8.90	8.84	8.75	8.66	8.56	8.51	8.46	8.41	8.36	8.31	8.26
5	10.01	8.43	7.76	7.39	7.15	6.98	6.85	6.76	6.68	6.62	6.52	6.43	6.33	6.28	6.23	6.18	6.12	6.07	6.02
6	8.81	7.26	6.60	6.23	5.99	5.82	5.70	5.60	5.52	5.46	5.37	5.27	5.17	5.12	5.07	5.01	4.96	4.90	4.85
7	8.07	6.54	5.89	5.52	5.29	5.12	4.99	4.90	4.82	4.76	4.67	4.57	4.47	4.42	4.36	4.31	4.25	4.20	4.14
8	7.57	6.06	5.42	5.05	4.82	4.65	4.53	4.43	4.36	4.30	4.20	4.10	4.00	3.95	3.89	3.84	3.78	3.73	3.67
9	7.21	5.71	5.08	4.72	4.48	4.32	4.20	4.10	4.03	3.96	3.87	3.77	3.67	3.61	3.56	3.51	3.45	3.39	3.33
10	6.94	5.46	4.83	4.47	4.24	4.07	3.95	3.85	3.78	3.72	3.62	3.52	3.42	3.37	3.31	3.26	3.20	3.14	3.08

Confidence interval for variances: ratio

Solution:

Adult DEINOPIS	Adult MENNEUS
n = 10	m = 10
$\bar{x}$ = 10.26 mm	$\bar{y}$ = 9.02 mm
$s_X^2 = (2.51)^2$	$s_Y^2 = (1.90)^2$

$$\left(\frac{1}{F_{\alpha/2}(n-1,m-1)}\frac{s_X^2}{s_Y^2} \leq \frac{\sigma_X^2}{\sigma_Y^2} \leq F_{\alpha/2}(m-1,n-1)\frac{s_X^2}{s_Y^2}\right)$$

Estimate, with 95% confidence, the ratio of the two population variances.

$F_{\alpha/2(n-1,m-1)} = F_{0.025(9,9)} = 4.03$

$1/F_{\alpha/2(n-1,m-1)} = 1/4.03 = 0.248$

[0.433,7.033]

1 is included therefore, we cannot conclude that the population variances differs.

Confidence interval for variances: ratio

Exercice: A company manufactures impellers for use in jet-turbine engines. One of the operations involves grinding a particular surface finish on a titanium alloy component. Two different grinding processes can be used, and both processes can produce parts at identical mean surface roughness. The manufacturing engineer would like to select the process having the least variability in surface roughness. A random sample of $n1$ =11 parts from the first process results in a sample standard deviation $s1$ =5.1 microinches, and a random sample of $n2$ =16 parts
from the second process results in a sample standard deviation of $s2$=4.7 microinches.

Find a 90% confidence interval on
the ratio of the two standard deviations

$$\left(\frac{1}{F_{\alpha/2}(n-1,m-1)} \frac{s_X^2}{s_Y^2} \leq \frac{\sigma_X^2}{\sigma_Y^2} \leq F_{\alpha/2}(m-1,n-1) \frac{s_X^2}{s_Y^2} \right)$$

Confidence interval for variances: ratio

Solution:

$$\left(\frac{1}{F_{\alpha/2}(n-1,m-1)}\frac{s_X^2}{s_Y^2} \leq \frac{\sigma_X^2}{\sigma_Y^2} \leq F_{\alpha/2}(m-1,n-1)\frac{s_X^2}{s_Y^2}\right)$$

$$\frac{s_1^2}{s_2^2} f_{0.95,15,10} \leq \frac{\sigma_1^2}{\sigma_2^2} \leq \frac{s_1^2}{s_2^2} f_{0.05,15,10}$$

$$F_{0.05,10,15} = 0.35149, \qquad F_{0.95,10,15} = 2.54372$$

$$\frac{s_1^2}{s_2^2} = \frac{5.1^2}{4.7^2} = 1.17746$$

$$(0.680,\ 1.830)$$

Confidence interval : two proportions

Approximate Confidence Interval on the Difference in Population Proportions

If $\hat{p}_1$ and $\hat{p}_2$ are the sample proportions of observations in two independent random samples of sizes n_1 and n_2 that belong to a class of interest, an **approximate two-sided $100(1-\alpha)\%$ confidence interval on the difference in the true proportions $p_1 - p_2$** is

$$\hat{p}_1 - \hat{p}_2 - z_{\alpha/2}\sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}} \le p_1 - p_2 \le \hat{p}_1 - \hat{p}_2 + z_{\alpha/2}\sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}} \qquad (10\text{-}41)$$

where $z_{\alpha/2}$ is the upper $\alpha/2$ percentage point of the standard normal distribution.

Confidence interval : two proportions

Example

What is the prevalence of anemia in developing countries?

	African Women	Women from Americas
Sample size	2100	1900
Number with anemia	840	323
Sample proportion	$\frac{840}{2100} = 0.40$	$\frac{323}{1900} = 0.17$

Find a 95% confidence interval for the difference in proportions of all African women with anemia and all women from the Americas with anemia.

Confidence interval : two proportions

Solution

$\hat{p}1=0.40$
$\hat{p}2=0.17$

$\alpha=0.05$
$z_{\alpha/2}=1.96$

[20.3%,25.7%]

$$(\hat{p}_1 - \hat{p}_2) \pm z_{\alpha/2}\sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$$

What is the prevalence of anemia in developing countries?

	African Women	Women from Americas
Sample size	2100	1900
Number with anemia	840	323
Sample proportion	$\frac{840}{2100}=0.40$	$\frac{323}{1900}=0.17$

Find a 95% confidence interval for the difference in proportions of all African women with anemia and all women from the Americas with anemia.

We can be 95% confident that there are between 20.3% and 25.7% more African women with anemia than women from the Americas with anemia

Hypothesis Testing – Basics

Session 4

Hypothesis Testing – Basics

Session 4

Objectives:

- State and formalize null (H_0) and alternative (H_1) hypotheses; choose one- vs two-sided.
- Explain Type I / Type II errors, significance level α, power (1−β), p-value.
- Execute hypothesis tests

Hypothesis Testing : What is it?

In the previous chapter, we demonstrated how to build a confidence interval estimate of a parameter using sample data. However, in many engineering problems, the task is not just to estimate a parameter but to determine which of two competing statements about it is true. These statements are called ***hypotheses***, and the process of making such decisions is known **as *hypothesis testing***.

Examples: drug efficacy, manufacturing tolerances, A/B tests

Hypothesis Testing : Theory

Every time we perform a hypothesis test, this is the basic procedure that we will follow:

- State the research hypothesis and the null hypothesis

Null hypothesis (H0) is a statistical assumption suggesting that any observed patterns or differences in a dataset are due to chance and not the result of a meaningful effect.

The other hypothesis is called the alternative hypothesis (H1). Both are mutually exclusive.

- The null hypothesis is generally an equality, and the alternative hypothesis might be either one-sided or two-sided, depending on the conclusion to be drawn if H0 is rejected.

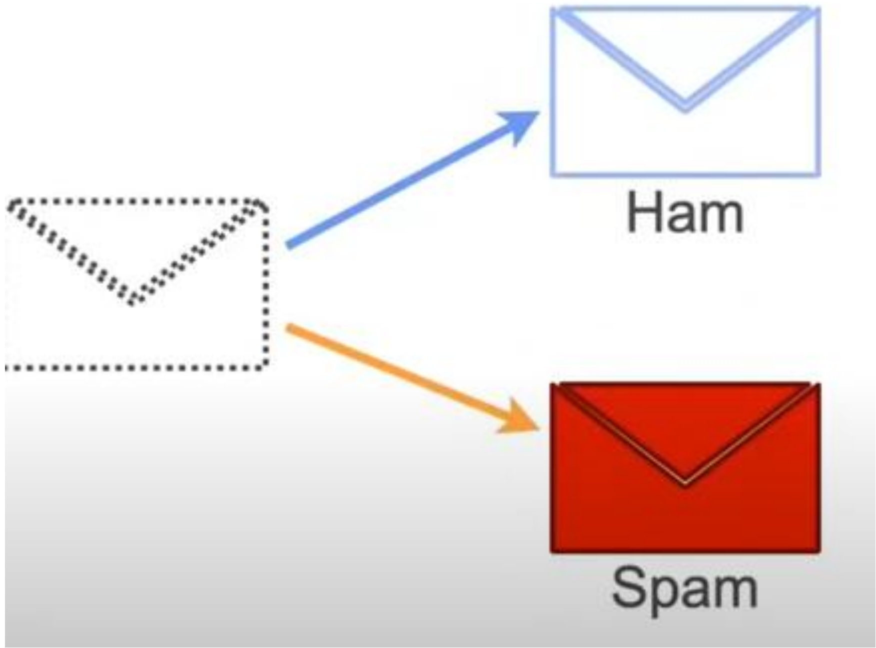
- We'll make an initial assumption about the population parameter.

- We'll collect evidence or else use somebody else's evidence (data).

- Based on the available evidence (data), we'll decide whether to "**reject**" or "**not reject**" our initial assumption.

Hypothesis Testing : example

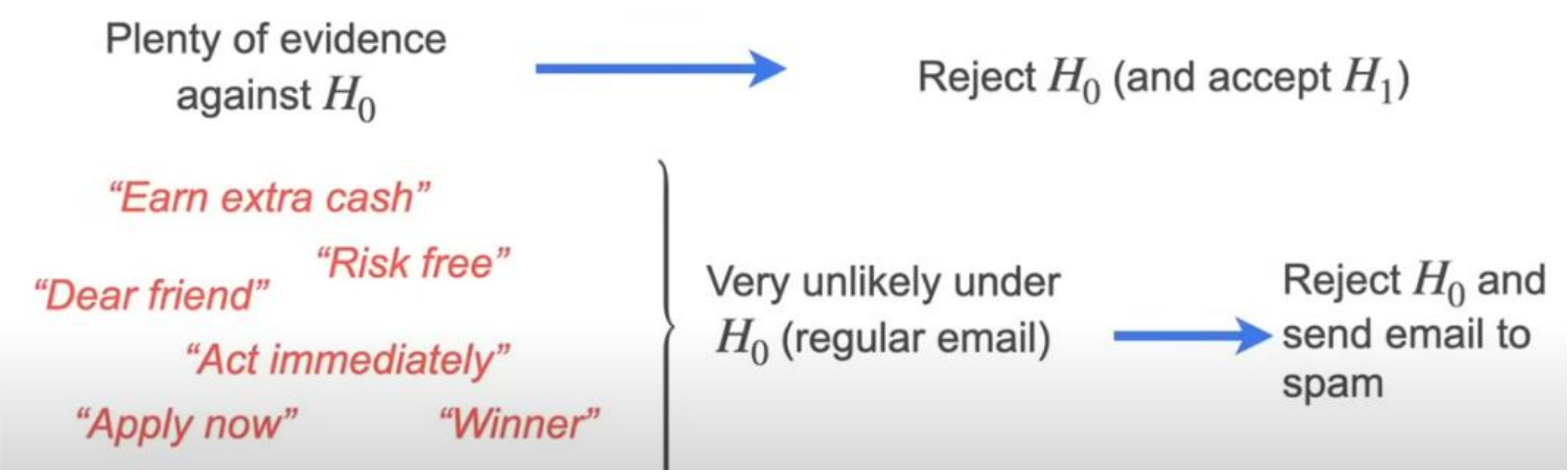
Example: You have a spam email detector, and its role is to say if the email is ham or spam



Hypothesis Testing : example

Example: By default, you assumed that all emails are ham because it is worst to delete a good email than to have a spam!!

Therefore, our base assumption is that an email is ham (H0), meaning that nothing is happening.



Hypothesis Testing : errors

Decision	Reality	
	H_0 True (Not spam)	H_0 False (Spam)
Reject H_0 (Decide spam)	Type I error	Correct
Don't reject H_0 (Decide not spam)	Correct	Type II error

Rejecting the null hypothesis H_0 when it is true is defined as a **type I error.**

Failing to reject the null hypothesis when it is false is defined as a **type II error.**

significance level, or the α-error: $\alpha = P(\text{type I error}) = P(\text{reject } H_0 \text{ when } H_0 \text{ is true})$

the β-error: $\beta = P(\text{type II error}) = P(\text{fail to reject } H_0 \text{ when } H_0 \text{ is false})$

Hypothesis Testing : Making a decision

There are two approaches to making the decision:

- one is called the "**critical value**" approach
- and the other is called the “**p-value**" approach

Hypothesis Testing : critical value

1. **Parameter of interest:** From the problem context, identify the parameter of interest.
2. **Null hypothesis, H_0:** State the null hypothesis, H_0.
3. **Alternative hypothesis, H_1:** Specify an appropriate alternative hypothesis, H_1.
4. **Test statistic:** Determine an appropriate test statistic.
5. **Reject H_0 if:** State the rejection criteria for the null hypothesis.
6. **Computations:** Compute any necessary sample quantities, substitute these into the equation for the test statistic, and compute that value.
7. **Draw conclusions:** Decide whether or not H_0 should be rejected and report that in the problem context.

1. Specify the test statistic to be used (such as Z_0).
2. Specify the location of the critical region (two-tailed, upper-tailed, or lower-tailed).
3. Specify the criteria for rejection (typically, the value of α, or the P-value at which rejection should occur).

Hypothesis Testing : critical value

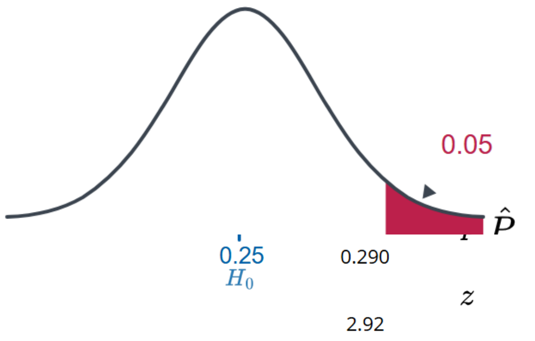
Example: A four-sided die is tossed 1000 times, and 290 fours are observed. Is there evidence to conclude that the die is biased, that is, say, that more fours than expected are observed?

1.) H0: Die is not biaised, therefore the probability to have a 4 is 0.25

H0: p=0.25

2.) The die was tossed 1000 times, and 290 fours were observed. So from this sample we have
p̂=290/1000=0.29

3.) $$Z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}}$$ with n=1000 you find Z=2.92



4.) Critical value approach tells us to define a threshold value, called a "**critical value**“
α=0.05 therefore, z_crit =$z_{0.95}$=1.645. If Z>z_crit, we reject H0

5.) Make a decision: Z=2.92>>z_crit=1.645
we reject the null hypothesis at the 5% significance level and conclude with at least 95% confidence that the die is biased

Hypothesis Testing : critical value

Example: Let *p* equal the proportion of drivers who use a seat belt in a country that does not have a mandatory seat belt law. It was claimed that $p=0.14$. An advertising campaign was conducted to increase this proportion. Two months after the campaign, $y=104$ out of a random sample of $n=590$ drivers were wearing seat belts.

Was the campaign successful? (use a significance level of 0.01)

Hypothesis Testing : critical value

Solution:

1.) H0: p0=0.14

2.) y=104 over n=590. From this sample we have $\hat{p}$=104/590=0.176

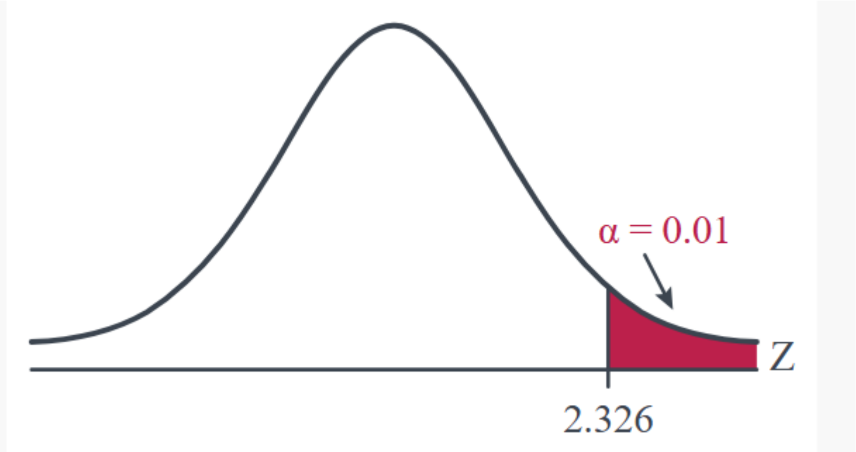
H1: p>P0=0.14

3.) Z=2.52

4.) With alpha=0.01, you have z_crit= $z_{0.99}$=2.326

5.) Z>z_crit,

we reject the null hypothesis at the 1% significance level and conclude with at least 99% confidence that the campaign was successful (p>0.14)



Hypothesis Testing : critical value

Example: 47% of the 1052 U.S. adults surveyed classified themselves as "very happy" when given the choices of:
"very happy"
"fairly happy"
"not too happy"
Suppose that a journalist who is a pessimist took advantage of this poll to write a headline titled "Poll finds that U.S. adults who are very happy are in the minority." Is the pessimistic journalist's headline warranted? (use a significance level of 0.05)

Hypothesis Testing : critical value

Solution:

1.) H0: p0=0.5
2.) p̂=0.47

$$Z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}}$$

3.) Z=-1.946

4.) With alpha=0.05, you have z_crit= $z_{0.05}$=-1.645

5.) Z<z_crit,
we cannot reject the null hypothesis at the 5% significance level
and conclude with at least 95% confidence that
adults who are very happy are in the minority

Hypothesis Testing : p-value

Definition: The **p-value is the smallest significance level α at which you would reject** H_0.

- In other word, the p-value measures how confidence we are about the assumption given the observed data. It ranges from 0 to 1. The smaller the *p*-value, the less likely the results occurred by random chance, and the stronger the evidence that you should reject the null hypothesis

- A p-value of 0.05 is commonly referred to as the level of significance (α).

If the p-value > 0.05: It indicates that the null hypothesis should be accepted.
If the p-value < 0.05: It indicates that the null hypothesis should be rejected.

A p-value of 0.05 means there is only a 5 % chance of obtaining a result at least as extreme as the one observed, assuming the null hypothesis is correct.

Hypothesis Testing : p-value

α : Significance Level

- Chosen before performing the statistical test
- Common values: 0.05, 0.01
- Represents the maximum probability of making a Type I error
- Type I error = rejecting a true null hypothesis $\boldsymbol{H_0}$

P-value

- the smallest significance level α at which you would reject H_0.
- Calculated from the sample data
- Probability of observing a result at least as extreme as the observed result, assuming H_0 is true
- Obtained using the test statistic (e.g., z-score, t-score)

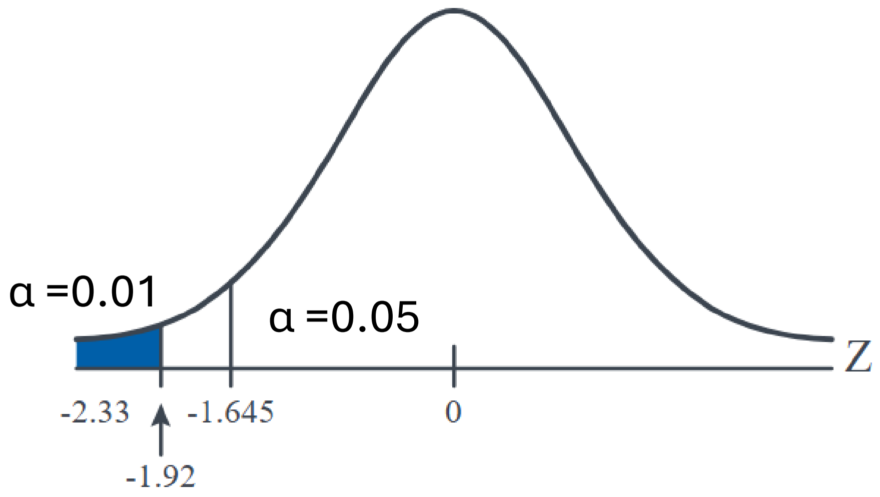
Hypothesis Testing : p-value

Example: Among patients with lung cancer, usually, 90% or more die within three years. As a result of new forms of treatment, it is felt that this rate has been reduced. In a recent study of *n* = 150 lung cancer patients, *y* = 128 died within three years.

- H0: p=0.90, H1:p<0.90
- Statistic test:

$$Z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}} = \frac{0.853 - 0.90}{\sqrt{\frac{0.90(0.10)}{150}}} = -1.92$$

- Using Z table, we found a **p-value of 0.0274**



There is only a **2.74%** chance of seeing a sample this extreme (or more) if the true death rate were **90%**, therefore there is **statistically significant evidence that the true three-year death rate for lung cancer patients is less than 90%**, suggesting that the **new treatments may have reduced the death rate.**

Hypothesis Testing : one population

Summary of Tests on the Mean, Variance Known

Testing Hypotheses on the Mean, Variance Known (*Z*-Tests)

Null hypothesis: $H_0: \mu = \mu_0$

Test statistic: $Z_0 = \dfrac{\overline{X} - \mu_0}{\sigma/\sqrt{n}}$

Alternative Hypotheses	*P*-Value	Rejection Criterion for Fixed-Level Tests
$H_1: \mu \neq \mu_0$	Probability above $\lvert z_0 \rvert$ and probability below $-\lvert z_0 \rvert$, $P = 2[1 - \Phi(\lvert z_0 \rvert)]$	$z_0 > z_{\alpha/2}$ or $z_0 < -z_{\alpha/2}$
$H_1: \mu > \mu_0$	Probability above z_0, $P = 1 - \Phi(z_0)$	$z_0 > z_\alpha$
$H_1: \mu < \mu_0$	Probability below z_0, $P = \Phi(z_0)$	$z_0 < -z_\alpha$

$\Phi(z)$ cumulative distribution function, $\Phi(z)=P(Z\leq z)$

Hypothesis Testing : one population

Summary for the One-Sample t-Test

Testing Hypotheses on the Mean of a Normal Distribution, Variance Unknown

Null hypothesis: $H_0\colon \mu = \mu_0$

Test statistic: $T_0 = \dfrac{\overline{X} - \mu_0}{S/\sqrt{n}}$

Alternative Hypotheses	P-Value	Rejection Criterion for Fixed-Level Tests
$H_1\colon \mu \neq \mu_0$	Probability above $\lvert t_0 \rvert$ and probability below $-\lvert t_0 \rvert$	$t_0 > t_{\alpha/2,n-1}$ or $t_0 < -t_{\alpha/2,n-1}$
$H_1\colon \mu > \mu_0$	Probability above t_0	$t_0 > t_{\alpha,n-1}$
$H_1\colon \mu < \mu_0$	Probability below t_0	$t_0 < -t_{\alpha,n-1}$

The calculations of the P-values and the locations of the critical regions for these situations are shown in Figs. 9-10 and 9-12, respectively.

Hypothesis Testing : Equality of Two Means

When Population Variances Are Equal (pooled t-test)

$$T = \frac{(\bar{X} - \bar{Y}) - (\mu_X - \mu_Y)}{S_p \sqrt{\frac{1}{n} + \frac{1}{m}}}$$

follows a t_{n+m-2} distribution where S_p^2, the pooled sample variance:

$$S_p^2 = \frac{(n-1)S_X^2 + (m-1)S_Y^2}{n+m-2}$$

When Population Variances Are Not Equal (Welch's t-test)

Let's again start with the good news that we've already done the dirty theoretical work here. Recall that if you have two **independent samples** from two **normal distributions** with **unequal variances** $\sigma_X^2 \neq \sigma_Y^2$, then:

$$T = \frac{(\bar{X} - \bar{Y}) - (\mu_X - \mu_Y)}{\sqrt{\frac{S_X^2}{n} + \frac{S_Y^2}{m}}}$$

follows, at least approximately, a t_r distribution where r, the adjusted degrees of freedom is determined by the equation:

$$r = \frac{\left(\frac{s_X^2}{n} + \frac{s_Y^2}{m}\right)^2}{\frac{(s_X^2/n)^2}{n-1} + \frac{(s_Y^2/m)^2}{m-1}}$$

If r doesn't equal an integer, as it usually doesn't, then we take the integer portion of r. That is, we use $\lfloor r \rfloor$ if necessary.

Hypothesis Testing : Equality of Two Means

Example: Is the mean fastest speed driven by male college students different than the mean fastest speed driven by female college students?

Males (*X*)	Females (*Y*)
$n = 34$ $\bar{x} = 105.5$ $s_x = 20.1$	$m = 29$ $\bar{y} = 90.9$ $s_y = 12.2$

Hypothesis Testing : Equality of Two Means

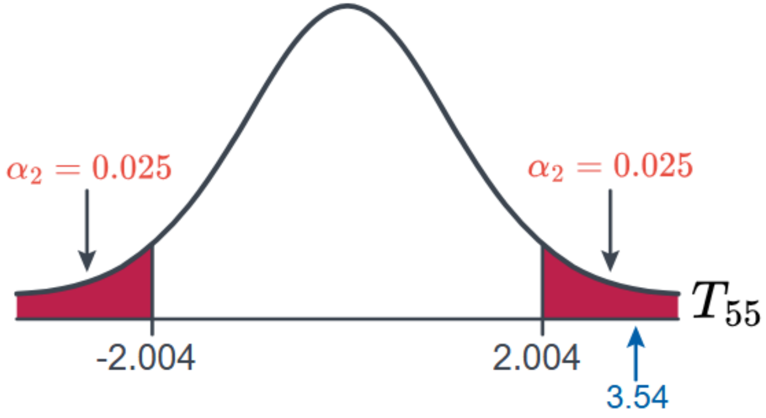
Solution: Is the mean fastest speed driven by male college students different than the mean fastest speed driven by female college students?

Males *(X)*	Females *(Y)*
$n = 34$	$m = 29$
$\bar{x} = 105.5$	$\bar{y} = 90.9$
$s_x = 20.1$	$s_y = 12.2$

Two-sided: $H_0 : \mu_M - \mu_F = 0$

$$t = \frac{(105.5 - 90.9) - 0}{\sqrt{\frac{20.1^2}{34} + \frac{12.2^2}{29}}} = 3.54$$

$$r = \frac{\left(\frac{12.2^2}{29} + \frac{20.1^2}{34}\right)^2}{\left(\frac{1}{28}\right)\left(\frac{12.2^2}{29}\right)^2 + \left(\frac{1}{33}\right)\left(\frac{20.1^2}{34}\right)^2} = 55.5$$



Then, the **critical value approach** tells us to reject the null hypothesis in favor of the alternative hypothesis if:

$$t > t_{0.025,55} = 2.004$$

And again, the decision is the same using the p-value approach. The p-value is 0.0008:

$$P = 2 \times P(T_{55} > 3.54) = 2(0.0004) = 0.0008$$

Therefore, because $p = 0.008 \leq \alpha = 0.05$, we reject the null hypothesis in favor of the alternative hypothesis. Again, we conclude that there is sufficient evidence at the $\alpha = 0.05$ level to conclude that the average fastest speed driven by the population of male college students differs from the average fastest speed driven by the population of female college students.

Hypothesis Testing : Comparing Two Proportions

The test statistic for testing the difference in two population proportions, that is, for testing the null hypothesis $H_0 : p_1 - p_2 = 0$ is:

$$Z = \frac{(\hat{p}_1 - \hat{p}_2) - 0}{\sqrt{\hat{p}(1-\hat{p})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

where:

$$\hat{p} = \frac{Y_1 + Y_2}{n_1 + n_2}$$

the proportion of "successes" in the two samples combined.

Hypothesis Testing : Comparing Two Proportions

Exercice: Is there sufficient evidence at α=0.05 , to conclude that the two populations — smokers and non-smokers — differ significantly with respect to their opinions?

Non- Smokers	Smokers
$n_1 = 605$ $y_1 = 351$ said "yes" $\hat{p}_1 = \frac{351}{605} = 0.58$	$n_2 = 195$ $y_2 = 41$ said "yes" $\hat{p}_2 = \frac{41}{195} = 0.21$

Hypothesis Testing : Comparing Two Proportions

Solution: p1=the proportion of the non-smoker population who reply "yes" and p2= the proportion of the smoker population who reply "yes," then we are interested in testing the null hypothesis:

Non- Smokers	Smokers
$n_1 = 605$ $y_1 = 351$ said "yes" $\hat{p}_1 = \frac{351}{605} = 0.58$	$n_2 = 195$ $y_2 = 41$ said "yes" $\hat{p}_2 = \frac{41}{195} = 0.21$

- Two-sided: H0:p1= p2 and H1:p1≠p2

- Statistic test

$$\hat{p} = \frac{41 + 351}{195 + 605} = \frac{392}{800} = 0.49$$

$$Z = \frac{(0.58 - 0.21) - 0}{\sqrt{0.49(0.51)\left(\frac{1}{195} + \frac{1}{605}\right)}} = 8.99$$

$$Z = \frac{(\hat{p}_1 - \hat{p}_2) - 0}{\sqrt{\hat{p}(1-\hat{p})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

$$\hat{p} = \frac{Y_1 + Y_2}{n_1 + n_2}$$

- Z is >>>>1.96 (alpha=0.05) so we can reject the H0 hypothesis. There is sufficient evidence at the 0.05 level to conclude that the two populations differ with respect to their opinions.

Hypothesis Testing

Exercices: Boys of a certain age are known to have a mean (μ) weight of 85 pounds. A complaint is made that the boys living in a municipal children's home are underfed. As one bit of evidence, $n=25$ boys (of the same age) are weighed and found to have a mean weight of <x> = 80.94 pounds. It is known that the population standard deviation is 11.6 pounds.

A) Based on the available data, what should be concluded concerning the complaint?

B) What is the Type II error probability if the true mean weight of the boys is $\mu = 90$ pounds?

Hypothesis Testing

Solutions: Boys of a certain age are known to have a mean (μ) weight of 85 pounds. A complaint is made that the boys living in a municipal children's home are underfed. As one bit of evidence, n=25 boys (of the same age) are weighed and found to have a mean weight of <x> = 80.94 pounds. It is known that the population standard deviation is 11.6 pounds. Based on the available data, what should be concluded concerning the complaint?

H0: μ=85, alternative H1: μ<85

The variance population is know, therefore, we can apply Z test.

Z=(80.94-85)/[11.6/sqrt(25)]=-1.75

For alpha=0.05, z=-1.65 and one left tailed test

Z<z, we can reject H0

Using p-value=0.0401

There is **statistically significant evidence** that the mean weight of boys in the municipal children's home is **less than 85 pounds**.

1. $X_1, X_2, \ldots, X_n$ is a random sample from a normal population with mean μ and variance σ^2. So that:

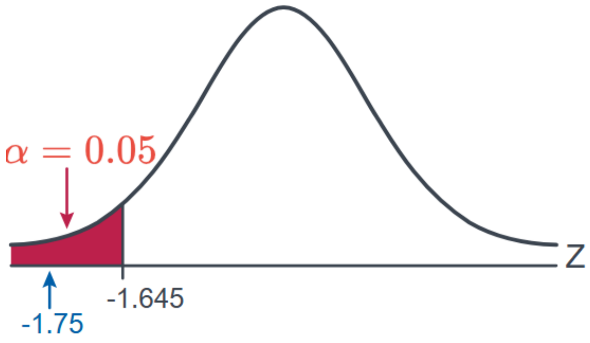
$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right) \text{ and } Z = \frac{\bar{X}-\mu}{\sigma/\sqrt{n}} \sim N(0,1)$$

2. The population variance $\sigma^2$is known.

Then, a $(1-\alpha)100\%$ **confidence interval for the mean** μ is:

$$\bar{x} \pm z_{\alpha/2}\left(\frac{\sigma}{\sqrt{n}}\right)$$

The interval, because it depends on Z, is often referred to as the Z**-interval for a mean**.



Hypothesis Testing

Solutions:

- First, remember H0: μ=85, alternative H1: μ<85. A **Type II error** means:"fail to reject "H0 " even though " H0 " is false“. So before calculating β, we must know for which values of X we fail to reject H0. For alpha=0.05, z=-1.65 and one left tailed test.

$$\frac{\bar{X}-85}{11.6/\sqrt{25}} < -1.645 \qquad Z = \frac{\bar{X}-\mu_0}{\sigma/\sqrt{n}}$$

Xbar=81.18 therefore, if X>81.18 fails to reject H0.

- Second, suppose the true mean is actually μ=90, therefore $H_0$is false, because the true mean is not 85.

$$\beta = P(\text{fail to reject } H_0 \mid \mu = 90)$$

- Third, $\beta = P(\bar{X} \geq 81.18 \mid \mu = 90)$

$$\beta = P(Z \geq -3.80)$$

Enregistrer

Z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
-3.9	0.00005	0.00005	0.00004	0.00004	0.00004	0.00004	0.00004	0.00004	0.00003	0.00003
-3.8	0.00007	0.00007	0.00007	0.00006	0.00006	0.00006	0.00006	0.00005	0.00005	0.00005
-3.7	0.00011	0.00010	0.00010	0.00010	0.00009	0.00009	0.00008	0.00008	0.00008	0.00008
-3.6	0.00016	0.00015	0.00015	0.00014	0.00014	0.00013	0.00013	0.00012	0.00012	0.00011
-3.5	0.00023	0.00022	0.00022	0.00021	0.00020	0.00019	0.00019	0.00018	0.00017	0.00017
-3.4	0.00034	0.00032	0.00031	0.00030	0.00029	0.00028	0.00027	0.00026	0.00025	0.00024
-3.3	0.00048	0.00047	0.00045	0.00043	0.00042	0.00040	0.00039	0.00038	0.00036	0.00035
-3.2	0.00069	0.00066	0.00064	0.00062	0.00060	0.00058	0.00056	0.00054	0.00052	0.00050
-3.1	0.00097	0.00094	0.00090	0.00087	0.00084	0.00082	0.00079	0.00076	0.00074	0.00071
-3.0	0.00135	0.00131	0.00126	0.00122	0.00118	0.00114	0.00111	0.00107	0.00104	0.00100

Solutions:

- Look at the value of Z in the table

$$P(Z < -3.80) = 0.00007$$

$$\beta = P(Z \geq -3.80)$$

$$\beta = 1 - 0.00007$$

$$\beta \approx 0.99993$$

Hypothesis Testing

Exercices: A manufacturer of hard safety hats for construction workers is concerned about the mean and the variation of the forces its helmets transmits to wearers when subjected to an external force. The manufacturer has designed the helmets so that the mean force transmitted by the helmets to the workers is 800 pounds (or less) with a standard deviation to be less than 40 pounds. Tests were run on a random sample of $n = 40$ helmets, and the sample mean and sample standard deviation were found to be 825 pounds and 48.5 pounds, respectively.
Do the data provide sufficient evidence, at the alpha=0.05 level, to conclude that the population standard deviation **exceeds** 40 pounds?

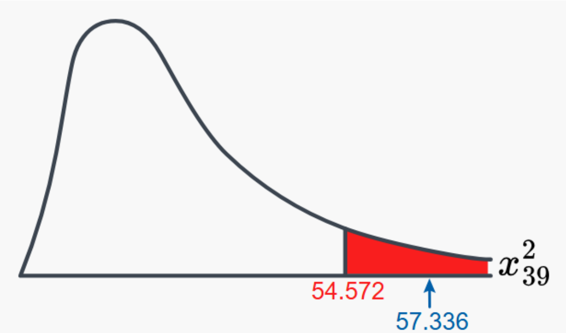
Hypothesis Testing

Solution. The manufacturer has designed the helmets so that the mean force transmitted by the helmets to the workers is 800 pounds (or less) with a standard deviation to be less than 40 pounds. Tests were run on a random sample of *n* = 40 helmets, and the sample mean and sample standard deviation were found to be 825 pounds and 48.5 pounds, respectively.
Do the data provide sufficient evidence, at the alpha=0.05 level, to conclude that the population standard deviation **exceeds** 40 pounds?

H0: $\sigma^2=40^2=1600$ H1: $\sigma^2>1600$

You have random sample of size *n* from a **normal population** with (unknown) mean μ and variance σ^2. You have to use the chi2-test when you test on the standard deviation.

$$\chi^2 = \frac{(40-1)48.5^2}{40^2} = 57.336$$



$$\chi^2 = \frac{(n-1)S^2}{\sigma_0^2}$$

Hypothesis Testing

Exercices: Is there sufficient evidence at the alpha=0.05 level to conclude that the variance of the fastest speed driven by male college students differs from the variance of the fastest speed driven by female college students?

Males *(X)*	Females *(Y)*
$n = 34$	$m = 29$
$\bar{x} = 105.5$	$\bar{y} = 90.9$
$s_x = 20.1$	$s_y = 12.2$

Hypothesis Testing

Solution: Is there sufficient evidence at the alpha=0.05 level to conclude that the variance of the fastest speed driven by male college students differs from the variance of the fastest speed driven by female college students?

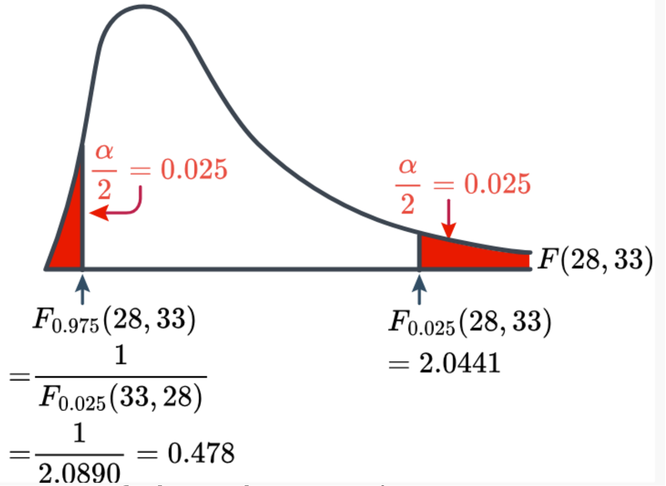
We compare **two population variances, so F-test**

$H_0\colon \sigma_X^2 = \sigma_Y^2$ and H1: $\sigma_X^2 \neq \sigma_Y^2$

Males (X)	Females (Y)
$n = 34$	$m = 29$
$\bar{x} = 105.5$	$\bar{y} = 90.9$
$s_x = 20.1$	$s_y = 12.2$

we can use the test statistic:

$$F = \frac{S_X^2}{S_Y^2} \qquad F = \frac{12.2^2}{20.1^2} = 0.368$$

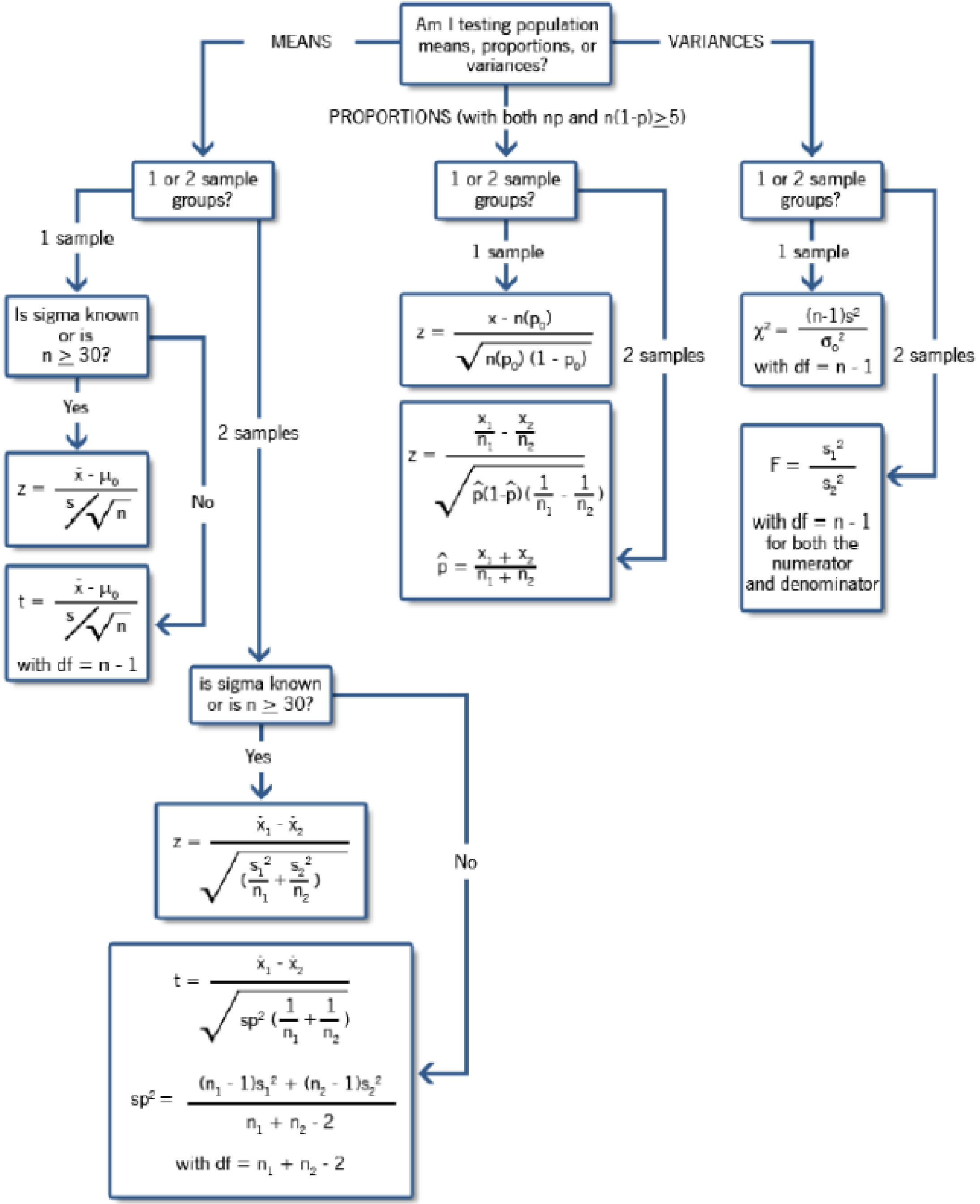


Now, we check the value in the F table using

$$F_{1-(\alpha/2)}(n-1, m-1) = \frac{1}{F_{\alpha/2}(m-1, n-1)}$$

Because the test statistic falls in the rejection region, we reject the null hypothesis in favor of the alternative hypothesis. There is sufficient evidence at the alpha=0.05 level to conclude that the population variances are not equal

Hypc … mary



Am I testing population means, proportions, or variances?
MEANS
PROPORTIONS (with both np and n(1-p)≥5)
VARIANCES
1 or 2 sample groups?
1 sample
2 samples
Is sigma known or is n ≥ 30?
Yes
No
$z = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$
$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$
with df = n - 1
is sigma known or is n ≥ 30?
$z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2})}}$
$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{sp^2 (\frac{1}{n_1} + \frac{1}{n_2})}}$
$sp^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$
with df = $n_1 + n_2 - 2$
$z = \frac{x - n(p_0)}{\sqrt{n(p_0)(1 - p_0)}}$
$z = \frac{\frac{x_1}{n_1} - \frac{x_2}{n_2}}{\sqrt{\hat{p}(1-\hat{p})(\frac{1}{n_1} - \frac{1}{n_2})}}$
$\hat{p} = \frac{x_1 + x_2}{n_1 + n_2}$
$\chi^2 = \frac{(n-1)s^2}{\sigma_0^2}$
with df = n - 1
$F = \frac{s_1^2}{s_2^2}$
with df = n - 1 for both the numerator and denominator

Hypothesis Testing: ANOVA

ANOVA stands for Analysis of Variance, a statistical test used to compare the means of three or more groups. It analyzes the variance within the group and between groups. ANOVA is a little misleading as it is concerned with investigating differences in means

Mathematically, ANOVA breaks down the total variability in the data into two components:

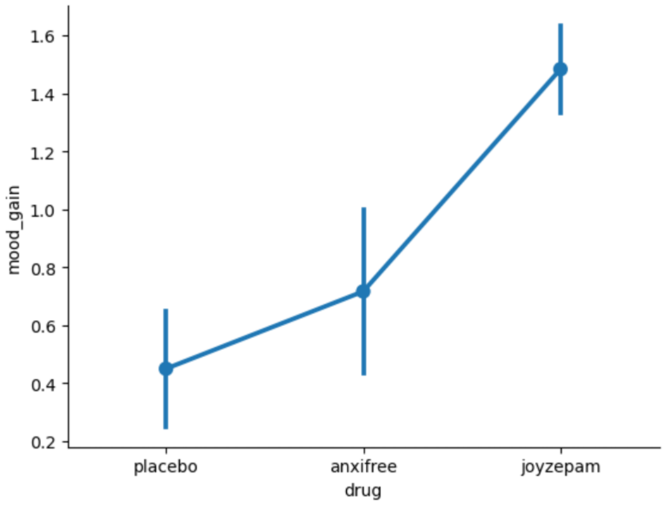
- **Within-Group Variability**: Variability caused by differences within individual groups, reflecting random fluctuations.

- **Between-Group Variability**: Variability caused by differences between the means of the different groups.

ANOVA

There are two types of ANOVA: one-way and two-way. Depending on the number of independent variables and how they interact with each other, both are used in different scenarios.

- **One-way**: A one-way ANOVA test is used when there is one independent variable with two or more groups. The objective is to determine whether a significant difference exists between the means of different groups. E.g. compare the effectiveness of the three different drugs (placebo, current drug, new drug)
    - **Null Hypothesis (H_0)**: All methods are equal (no difference in means).
    - **Alternative Hypothesis (H_1)**: At least one group's mean significantly differs



ANOVA

There are two types of ANOVA: one-way and two-way. Depending on the number of independent variables and how they interact with each other, both are used in different scenarios.

- **Two-way**: Used when there are two independent variables, each with two or more groups. The objective is to analyze how both independent variables influence the dependent variable.) E.g., relationship between teaching methods and study techniques and how they jointly affect student performance.

  - **The main effect of factor 1 (teaching method)**: Does the teaching method influence student exam scores?
  - **The main effect of factor 2 (study technique)**: Does the study technique affect exam scores?
  - **Interaction effect**: Does the effectiveness of the teaching method depend on the study technique used?

ANOVA: assumptions

All statistical tests have assumptions that must be met to ensure valid results.

- **Independence of observations:** The observations (data points) must be independent of each other. In the example, students' exam scores in one teaching method should not influence the scores of students in another method.

- **Homogeneity of variances:** The variances within each group should be approximately equal. ANOVA assumes that the variability of exam scores within each teaching method group is roughly the same. This can be tested using Levene's test, which checks for equal variances.

- **Normal distribution:** The data within each group should follow a normal distribution. In our teaching method example, the exam scores for students in each teaching group should ideally be normally distributed.

ANOVA: ~ F-test

ANOVA breaks down the total variability in the data into two components:

- **Within-Group Variability**: Variability caused by differences within individual groups, reflecting random fluctuations SS(E).
- **Between-Group Variability** : Variability caused by differences between the means of the different groups. SS(T)

$$SS(TO) = \sum_{i=1}^{m} \sum_{j=1}^{n_i} (X_{ij} - \overline{X}_{..})^2,$$ the total sum of squares;

$$SS(E) = \sum_{i=1}^{m} \sum_{j=1}^{n_i} (X_{ij} - \overline{X}_{i.})^2,$$ the sum of squares within treatments, groups, or classes, often called the error sum of squares;

$$SS(T) = \sum_{i=1}^{m} n_i (\overline{X}_{i.} - \overline{X}_{..})^2,$$ the sum of squares among the different treatments, groups, or classes, often called the between-treatment sum of squares.

Table 9.3-2 Analysis-of-variance table

Source	Sum of Squares (SS)	Degrees of Freedom	Mean Square (MS)	F Ratio
Treatment	SS(T)	$m-1$	$MS(T) = \frac{SS(T)}{m-1}$	$\frac{MS(T)}{MS(E)}$
Error	SS(E)	$n-m$	$MS(E) = \frac{SS(E)}{n-m}$	
Total	SS(TO)	$n-1$		

m= number of groups

ANOVA: examples

Teaching Method		
Lecture	Workshop	Online learning
80	55	70
85	34	65
78	43	74
83	54	77

The first step in the process is defining the hypothesis. State the null and alternative hypotheses:

- **Null Hypothesis (H_0)**: The means of exam scores for students across the three teaching methods are equal.
- **Alternative Hypothesis (H_1)**: At least one teaching method has a different mean exam score.

ANOVA: examples

Lecture : $(80+85+78+83)/4 = 326/4 = 81.5$

Workshop : $(55+34+43+54)/4 = 186/4 = 46.5$

Online Learning : $(70+65+74+77)/4 = 286/4 = 71.5$

$$SS(E) = \sum_{i=1}^{m}\sum_{j=1}^{n_i}(X_{ij} - \bar{X}_i)^2$$

Lecture :

$(80-81.5)^2 + (85-81.5)^2 + (78-81.5)^2 + (83-81.5)^2$

$= 2.25 + 12.25 + 12.25 + 2.25 = 29$

Workshop : 296

Online : 81

Table 9.3-2 Analysis-of-variance table

Source	Sum of Squares (SS)	Degrees of Freedom	Mean Square (MS)	F Ratio
Treatment	SS(T)	$m-1$	$\mathrm{MS(T)} = \dfrac{\mathrm{SS(T)}}{m-1}$	$\dfrac{\mathrm{MS(T)}}{\mathrm{MS(E)}}$
Error	SS(E)	$n-m$	$\mathrm{MS(E)} = \dfrac{\mathrm{SS(E)}}{n-m}$	
Total	SS(TO)	$n-1$		

$\mathrm{SS(TO)} = \sum_{i=1}^{m}\sum_{j=1}^{n_i}(X_{ij} - \overline{X}_{..})^2$, the total sum of squares;

$\mathrm{SS(E)} = \sum_{i=1}^{m}\sum_{j=1}^{n_i}(X_{ij} - \overline{X}_{i\cdot})^2$, the sum of squares within treatments, groups, or classes, often called the error sum of squares;

$\mathrm{SS(T)} = \sum_{i=1}^{m} n_i(\overline{X}_{i\cdot} - \overline{X}_{..})^2$, the sum of squares among the different treatments, groups, or classes, often called the between-treatment sum of squares.

ANOVA: examples

Mean all values

$$SS(T) = \sum_{i=1}^{m} n_i(\bar{X}_i - \bar{X}_{..})^2$$

Lecture : $4 \times (81.5 - 66.5)^2 = 4 \times 225 = 900$

Workshop : $4 \times (46.5 - 66.5)^2 = 4 \times 400 = 1600$

Online : $4 \times (71.5 - 66.5)^2 = 4 \times 25 = 100$

$$df_{between} = m - 1 = 3 - 1 = 2$$

$$df_{within} = N - m = 12 - 3 = 9$$

$$F = \frac{MS(T)}{MS(E)} = \frac{1300}{45.11} \approx 28.83$$

Table 9.3-2 Analysis-of-variance table

Source	Sum of Squares (SS)	Degrees of Freedom	Mean Square (MS)	*F* Ratio
Treatment	SS(T)	$m-1$	$MS(T) = \frac{SS(T)}{m-1}$	$\frac{MS(T)}{MS(E)}$
Error	SS(E)	$n-m$	$MS(E) = \frac{SS(E)}{n-m}$	
Total	SS(TO)	$n-1$		

	α = 0.05									
	d.f.N.									
d.f.D.	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
1	161.4	199.5	215.7	224.6	230.2	234.0	236.8	238.9	240.5	241.9
2	18.51	19.00	19.16	19.25	19.30	19.33	19.35	19.37	19.38	19.40
3	10.13	9.55	9.28	9.12	9.01	8.94	8.89	8.85	8.81	8.79
4	7.71	6.94	6.59	6.39	6.26	6.16	6.09	6.04	6.00	5.96
5	6.61	5.79	5.41	5.19	5.05	4.95	4.88	4.82	4.77	4.74
6	5.99	5.14	4.76	4.53	4.39	4.28	4.21	4.15	4.10	4.06
7	5.59	4.74	4.35	4.12	3.97	3.87	3.79	3.73	3.68	3.64
8	5.32	4.46	4.07	3.84	3.69	3.58	3.50	3.44	3.39	3.35
9	5.12	4.26	3.86	3.63	3.48	3.37	3.29	3.23	3.18	3.14

If we select α= 0.05, Fcrit=4.26.

We have F>>Fcrit, we reject H0, so **at least one method is significantly different** from the others

ANOVA: examples

ANOVA tells us whether there is a statistically significant difference between group means, but it doesn't specify which groups are significantly different from one another. This is the role **of post-hoc tests , use to perform pairwise comparisons between the groups to identify precisely where the differences exist**

ANOVA: exercice

Example: Let X1, X2, X3, X4 equal the cholesterol level of a woman under the age of 50, a man under 50, a woman 50, or older, and a man 50 or older, respectively. Assume that the distribution of Xi is N(μi, σ2), i = 1, 2, 3, 4. We shall test the null hypothesis H0: μ1 = μ2 = μ3 = μ4, using seven observations of each Xi

(a) Give a critical region for an α = 0.05 significance level.

(b) Construct an ANOVA table and state your conclusion, using the following data:

x1: 221 213 202 183 185 197 162
x2: 271 192 189 209 227 236 142
x3: 262 193 224 201 161 178 265
x4: 192 253 248 278 232 267 289

Table 9.3-2 Analysis-of-variance table

Source	Sum of Squares (SS)	Degrees of Freedom	Mean Square (MS)	F Ratio
Treatment	SS(T)	$m-1$	$\text{MS(T)} = \dfrac{\text{SS(T)}}{m-1}$	$\dfrac{\text{MS(T)}}{\text{MS(E)}}$
Error	SS(E)	$n-m$	$\text{MS(E)} = \dfrac{\text{SS(E)}}{n-m}$	
Total	SS(TO)	$n-1$		

(c) Give bounds on the p-value for this test

$$SS(T) = \sum_{i=1}^{m} n_i(\bar{X}_i - \bar{X}_{..})^2 \qquad SS(E) = \sum_{i=1}^{m}\sum_{j=1}^{n_i}(X_{ij} - \bar{X}_i)^2$$

ANOVA: exercice

Solution: Let X1, X2, X3, X4 equal the cholesterol level of a woman under the age of 50, a man under 50, a woman 50, or older, and a man 50 or older, respectively. Assume that the distribution of Xi is N(μi, σ2), i = 1, 2, 3, 4. We shall test the null hypothesis H0: μ1 = μ2 = μ3 = μ4, using seven observations of each Xi

(a) Give a critical region for an α = 0.05 significance level.

- **Number of groups** $k = 4$
- **Observations per group** $n_i = 7$
- **Total observations** $N = 28$
- Significance level: $\alpha = 0.05$

Between-groups: $df_{\text{between}} = k - 1 = 4 - 1 = 3$

Within-groups: $df_{\text{within}} = N - k = 28 - 4 = 24$

We are testing:

$$H_0 : \mu_1 = \mu_2 = \mu_3 = \mu_4$$

Need to search Fcrit to find the critical region. $F_{3,24}$=3.01.

Therefore, we reject H0 is F>3.01

ANOVA: exercice

Solution: Construct an ANOVA table and state your conclusion, using the following data:

$$\bar{X}_1 = \frac{221+213+202+183+185+197+162}{7} = \frac{1363}{7} \approx 194.71$$

$$\bar{X}_2 = \frac{271+192+189+209+227+236+142}{7} = \frac{1466}{7} \approx 209.43$$

$$\bar{X}_3 = \frac{262+193+224+201+161+178+265}{7} = \frac{1484}{7} \approx 211.99$$

$$\bar{X}_4 = \frac{192+253+248+278+232+267+289}{7} = \frac{1759}{7} \approx 251.29$$

$$\bar{X}_{\text{grand}} = \frac{\text{sum of all observations}}{N} = \frac{1363+1466+1484+1759}{28} = \frac{6072}{28} \approx 216.86$$

Between-Groups Sum of Squares (SSB)

$$SSB = n_i \sum_{i=1}^{k} (\bar{X}_i - \bar{X}_{\text{grand}})^2$$

$$SSB = 7\left[(194.71-216.86)^2 + (209.43-216.86)^2 + (211.99-216.86)^2 + (251.29-216.86)^2\right]$$

Within-Groups Sum of Squares (SSW)

$$SSW = \sum_{i=1}^{k} \sum_{j=1}^{n_i} (X_{ij} - \bar{X}_i)^2$$

$$SSW = 2386.4 + 10073.5 + 9689.0 + 6272.5 \approx 28421.4$$

$$MSB = \frac{SSB}{df_{\text{between}}} = \frac{12285.7}{3} \approx 4095.2$$

$$MSW = \frac{SSW}{df_{\text{within}}} = \frac{28421.4}{24} \approx 1184.2$$

$$F = \frac{MSB}{MSW} = \frac{4095.2}{1184.2} \approx 3.46$$

$$F = 3.46 > 3.01 \Rightarrow \text{Reject } H_0$$

ANOVA: exercice

Solution: Give bounds on the p-value for this test :

- F = 3.46, df1 = 3, df2 = 24
- From F-tables, $F_{0.05} = 3.01$ and $F_{0.025} \approx 4.06$

So the **p-value is between 0.05 and 0.025**:

$$0.025 < p\text{-value} < 0.05$$

Simple Linear Regression and Correlation

Session 5

Simple Linear Regression and Correlation

Session 5

Objectives:

- Introduction to Linear Models
- Understand how the method of least squares
- Analyze residuals to determine if the regression model is an adequate fit
- Test statistical hypotheses and construct confidence intervals on regression model parameters
- Use the regression model to make a prediction

Source:

- Applied-Statistics-and-Probability-for-Engineers-Douglas-C.-Montgomery-George-C.-Runger-Edisi-5-2011
- Lesson 5: Multiple Linear Regression | STAT 501

Covariance : Formal definition

Consider two random variables X and Y. Here, we define the covariance between X and Y, written Cov(X,Y). The covariance gives some information about how X and Y are statistically related, it measures how two variables vary together.

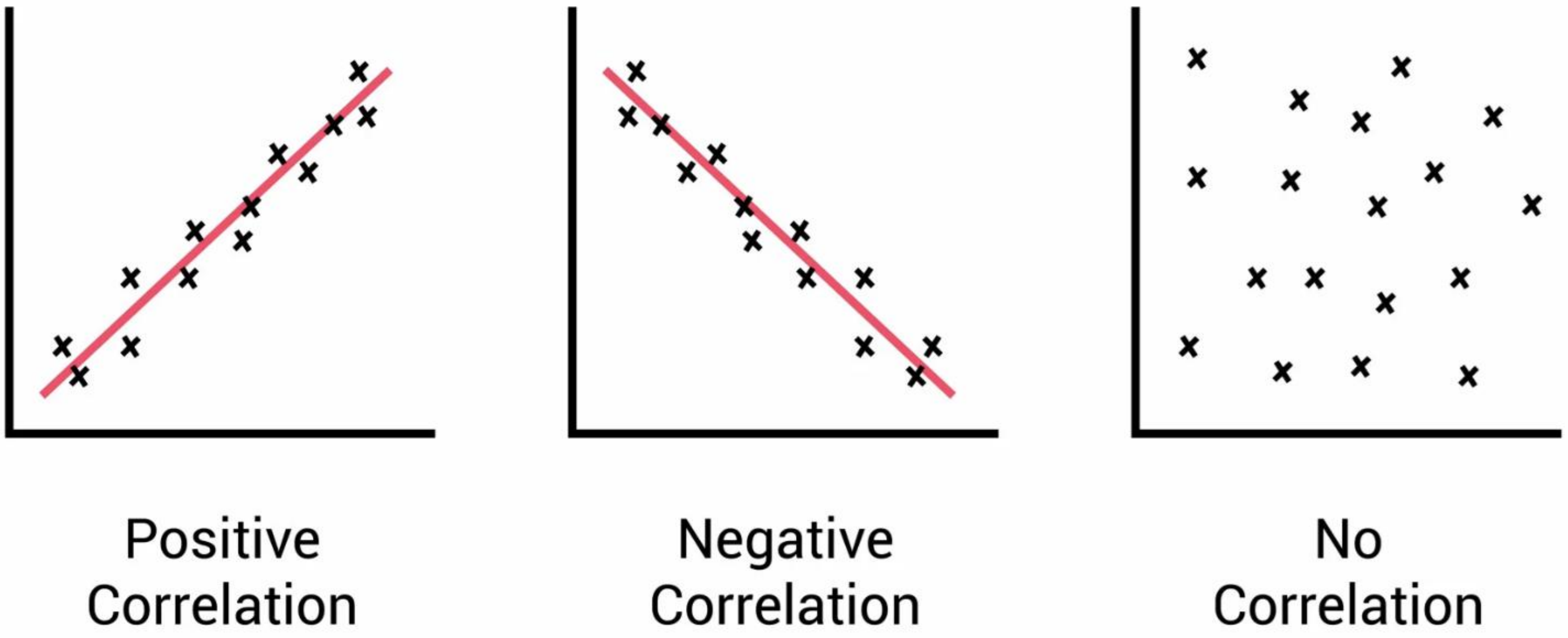
Can take any values from –inf to +inf

The **covariance** between the random variables X and Y, denoted as cov(X, Y) or σ_{XY}, is

$$\sigma_{XY} = E[(X - \mu_X)(Y - \mu_Y)] = E(XY) - \mu_X\mu_Y \qquad (5\text{-}28)$$

Correlation : Intuition

Correlation = normalized covariance ⇒ correlation takes values between **-1 and 1**



Correlation : Formal definition

The **correlation** between random variables X and Y, denoted as ρ_{XY}, is

$$\rho_{XY} = \frac{\operatorname{cov}(X, Y)}{\sqrt{V(X)V(Y)}} = \frac{\sigma_{XY}}{\sigma_X \sigma_Y} \tag{5-29}$$

Key difference : covariance depends on scale while correlation does not ⇒ correlation is comparable across datasets

Correlation : Pearson factor

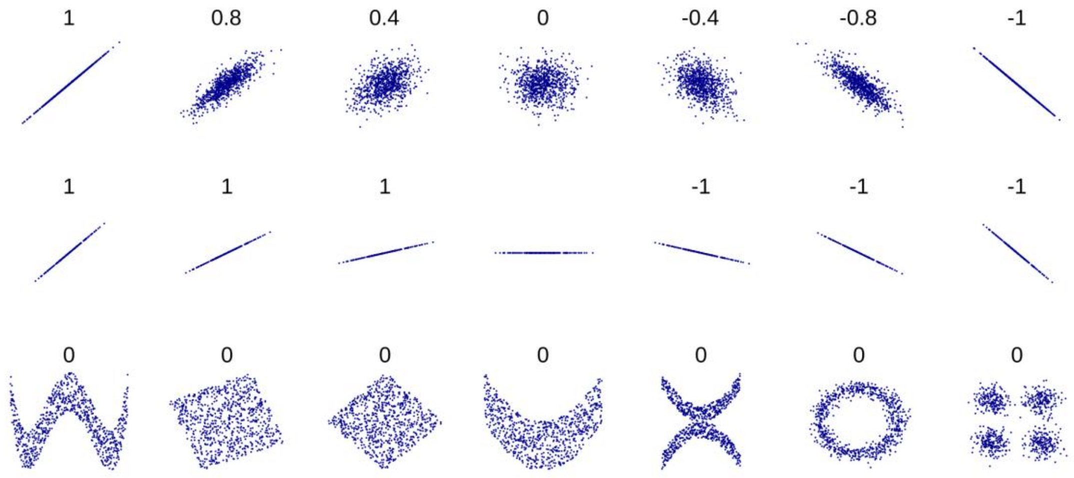
The **Pearson correlation coefficient** (**PCC**) is a correlation coefficient that measures linear correlation between two sets of data.

Assumptions:

- Linear relationship between variables
- Variables are continuous (interval or ratio scale)
- No strong outliers
- Approximately normally distributed (especially for inference

$$r = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2}\sqrt{\sum_{i=1}^{n}(y_i - \bar{y})^2}}$$

In python:
from scipy.stats import pearsonr
r, p_value = pearsonr(x, y)
r measures strength and direction
p-value tests whether the correlation is statistically significant



DenisBoigelot

Guess the Correlation

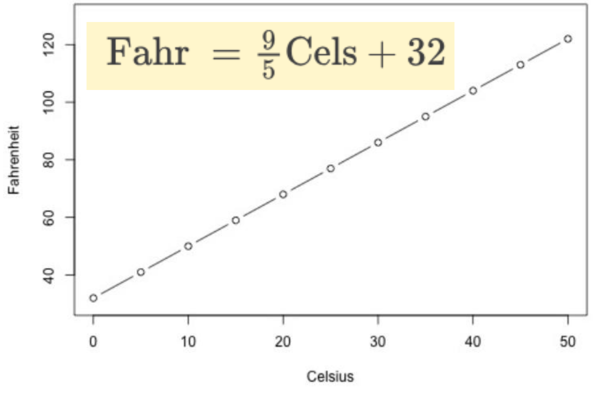
Linear regression

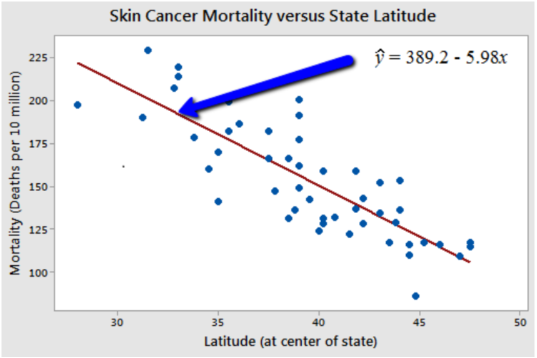
Often, we are interested in understanding the relationship between one variable, called the **dependent variable** or **response**, on a number of other variables, called the **independent variables** or **predictors**. The dependent variable is typically denoted by **y**, and the independent variables are denoted as $\mathbf{x_1, x_2, \ldots, x_p}$

Type of relationship:

- deterministic (or functional) relationships: the equation *exactly* describes the relationship between the two variables (e.g., if you know the temperature in degrees Celsius, you can use this equation to determine the temperature in degrees Fahrenheit *exactly*.

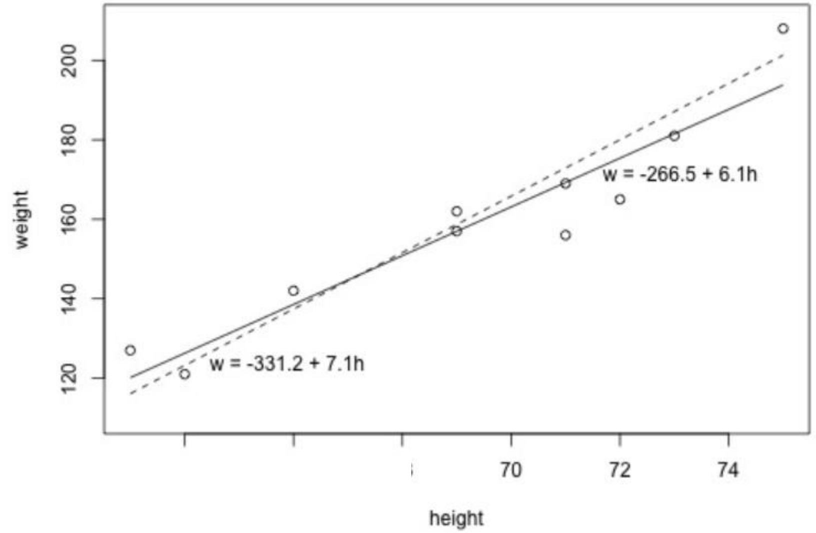
- statistical relationships: the relationship between the variables is not perfect.





Linear regression: What is the "Best Fitting Line"?

Which line , the solid line or the dashed line , do you think best summarizes the trend between height and weight?



- y_i denotes the observed response for experimental unit *i*
- x_i denotes the predictor value for experimental unit *i*
- $\hat{y}_i$ is the predicted response (or fitted value) for experimental unit *i*

Then, the equation for the best-fitting line is:

$$\hat{y}_i = b_0 + b_1 x_i$$

Response   Regressor or Predictor

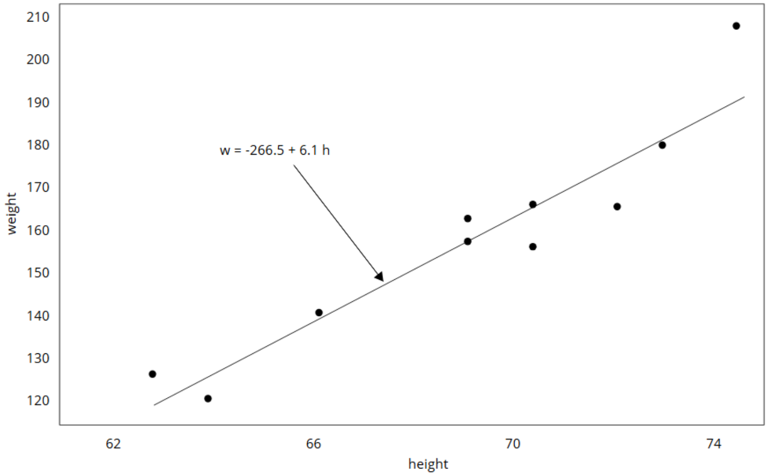
$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i \qquad i = 1, 2, \cdots, n$$

$\varepsilon_i \sim \mathrm{N}(0, \sigma^2)$

Intercept   Slope   Random error

Linear regression: What is the "Best Fitting Line"?

We have the data xi so we can derive for each line $\hat{y}_i$ using the line equation



i	x_i	y_i
1	63	127
2	64	121
3	66	142
4	69	157
5	69	162
6	71	156
7	71	169
8	72	165
9	73	181
10	75	208

$\hat{y}_i$

120.1 126.3 138.5 157.0 157.0 169.2 169.2 175.4 181.5 193.8

In general, when we use $\hat{y}_i = b_0 + b_1 x_i$ to predict the actual response y_i, we make a prediction error (or residual error) of size:

$$e_i = y_i - \hat{y}_i$$

Linear regression: What is the "Best Fitting Line"?

So we do it for both line.

Now, based on the least squares criterion, which equation best summarizes the data?

Of the two lines, the solid line is the best. However, is this equation guaranteed to be the best fitting line of all of the possible lines we didn't even consider?

$\hat{w} = -331.2 + 7.1h$ (the dashed line)

i	x_i	y_i	$\hat{y}_i$	$(y_i - \hat{y}_i)$	$(y_i - \hat{y}_i)^2$
1	63	127	116.1	10.9	118.81
2	64	121	123.2	-2.2	4.84
3	66	142	137.4	4.6	21.16
4	69	157	158.7	-1.7	2.89
5	69	162	158.7	3.3	10.89
6	71	156	172.9	-16.9	285.61
7	71	169	172.9	-3.9	15.21
8	72	165	180.0	-15.0	225.00
9	73	181	187.1	-6.1	37.21
10	75	208	201.3	6.7	44.89
					______ 766.5

$\hat{w} = -266.53 + 6.1376h$ (the solid line)

i	x_i	y_i	$\hat{y}_i$	$(y_i - \hat{y}_i)$	$(y_i - \hat{y}_i)^2$
1	63	127	120.139	6.8612	47.076
2	64	121	126.276	-5.2764	27.840
3	66	142	138.552	3.4484	11.891
4	69	157	156.964	0.0356	0.001
5	69	162	156.964	5.0356	25.357
6	71	156	169.240	-13.2396	175.287
7	71	169	169.240	-0.2396	0.057
8	72	165	175.377	-10.3772	107.686
9	73	181	181.515	-0.5148	0.265
10	75	208	193.790	14.2100	201.924
					______ 597.4

Linear regression: Least squares method

To determine the best fit, i.e., to determine b_0 and b_1, we can use the **LEAST SQUARES METHOD.**

We minimize the equation for the sum of the squared prediction errors:

$$Q = \sum_{i=1}^{n}(y_i - (b_0 + b_1 x_i))^2$$

Let take the derivative with respect to b_0 and b_1 and set to 0.

$$\frac{\partial Q}{\partial b_0} = \sum_{i=1}^{n} 2(y_i - b_0 - b_1 x_i)(-1) = 0$$

$$\sum_{i=1}^{n}(y_i - b_0 - b_1 x_i) = 0$$

$$n b_0 + b_1 \sum_{i=1}^{n} x_i = \sum_{i=1}^{n} y_i$$

$$b_0 = \frac{\sum y_i - b_1 \sum x_i}{n} = \bar{y} - b_1 \bar{x}$$

$$\frac{\partial Q}{\partial b_1} = \sum_{i=1}^{n} 2\left(y_i - (b_0 + b_1 x_i)\right) \frac{\partial}{\partial b_1}\left(y_i - (b_0 + b_1 x_i)\right)$$

$$\frac{\partial Q}{\partial b_1} = \sum_{i=1}^{n} 2\left(y_i - (b_0 + b_1 x_i)\right)(-x_i)$$

$$\frac{\partial Q}{\partial b_1} = -2 \sum_{i=1}^{n} x_i \left(y_i - b_0 - b_1 x_i\right)$$

$$\frac{\partial Q}{\partial b_1} = -2 \sum_{i=1}^{n} x_i y_i + 2 b_0 \sum_{i=1}^{n} x_i + 2 b_1 \sum_{i=1}^{n} x_i^2$$

$$\sum x_i y_i - b_0 \sum x_i - b_1 \sum x_i^2 = 0$$

$$b_1 = \frac{\sum x_i y_i - b_0 \sum x_i}{\sum x_i^2}$$

$$b_1 = \frac{\sum x_i y_i - (\bar{y} - b_1 \bar{x}) \sum x_i}{\sum x_i^2}$$

$$b_1 = \frac{\sum x_i y_i - \bar{y} \sum x_i + b_1 \bar{x} \sum x_i}{\sum x_i^2}$$

$$b_1 = \frac{\sum x_i y_i - n \bar{x} \bar{y} + b_1\, \bar{x}(n \bar{x})}{\sum x_i^2}$$

$$b_1 \left(\sum x_i^2 - n \bar{x}^2\right) = \sum x_i y_i - n \bar{x} \bar{y}$$

Linear regression: Least squares method

The **least squares estimates** of the intercept and slope in the simple linear regression model are

$$b_0 = \bar{y} - b_1\bar{x} \tag{11-7}$$

$$b_1 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n}(x_i - \bar{x})^2} \tag{11-8}$$

where $\bar{y} = (1/n)\sum_{i=1}^{n} y_i$ and $\bar{x} = (1/n)\sum_{i=1}^{n} x_i$.

Linear regression: Least squares method

Exercice : Using the least squares method determine the best fitting line.

$$b_0 = \bar{y} - b_1\bar{x}$$

$$b_1 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n}(x_i - \bar{x})^2}$$

Observation Number	Hydrocarbon Level x (%)	Purity y (%)
1	0.99	90.01
2	1.02	89.05
3	1.15	91.43
4	1.29	93.74
5	1.46	96.73
6	1.36	94.45
7	0.87	87.59
8	1.23	91.77
9	1.55	99.42
10	1.40	93.65
11	1.19	93.54
12	1.15	92.52
13	0.98	90.56
14	1.01	89.54
15	1.11	89.85
16	1.20	90.39
17	1.26	93.25
18	1.32	93.41
19	1.43	94.98
20	0.95	87.33

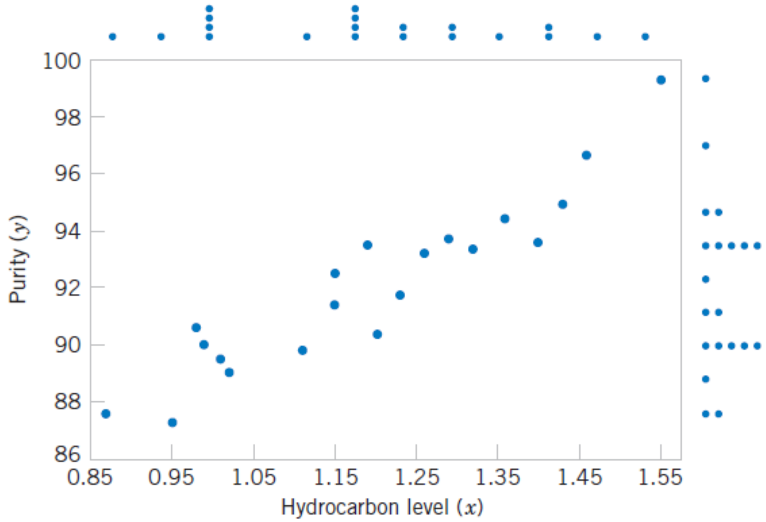


Figure 11-1 Scatter diagram of oxygen purity versus hydrocarbon level from Table 11-1.

Linear regression: Least squares method

Solution : Using the least squares method determine the best fitting line.

$$b_0 = \bar{y} - b_1\bar{x}$$

$$b_1 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n}(x_i - \bar{x})^2}$$

$$\bar{x} = \frac{23.92}{20} = 1.196 \qquad \bar{y} = \frac{1843.21}{20} = 92.16$$

$$\sum(x_i - \bar{x})(y_i - \bar{y}) \approx 10.17$$

$$\sum(x_i - \bar{x})^2 \approx 0.6805$$

$$b_1 = \frac{10.17}{0.6805} \approx 14.94$$

$$b_0 = 92.1605 - 14.94 \cdot 1.196 \approx 74.30$$

x	y	x - x̄	y - ȳ	(x-x̄)(y-ȳ)	(x-x̄)²
0.99	90.01	-0.206	-2.15	0.4439	0.0424
1.02	89.05	-0.176	-3.11	0.5474	0.0310
1.15	91.43	-0.046	-0.73	0.0336	0.0021
1.29	93.74	0.094	1.58	0.1485	0.0088
1.46	96.73	0.264	4.57	1.2050	0.0697
1.36	94.45	0.164	2.29	0.3756	0.0269
0.87	87.59	-0.326	-4.57	1.4908	0.1063
1.23	91.77	0.034	-0.39	-0.0133	0.0012
1.55	99.42	0.354	7.26	2.5680	0.1252
1.40	93.65	0.204	1.49	0.3039	0.0416
1.19	93.54	-0.006	1.38	-0.0083	0.000036
1.15	92.52	-0.046	0.36	-0.0166	0.0021
0.98	90.56	-0.216	-1.60	0.3456	0.0467
1.01	89.54	-0.186	-2.62	0.4873	0.0346
1.11	89.85	-0.086	-2.31	0.1987	0.0074
1.20	90.39	0.004	-1.77	-0.0071	0.000016
1.26	93.25	0.064	1.09	0.0698	0.0041
1.32	93.41	0.124	1.25	0.1550	0.0154
1.43	94.98	0.234	2.82	0.6599	0.0548
0.95	87.33	-0.246	-4.83	1.1872	0.0605

Linear regression: Least squares method

What is the Error Variance?

There is actually another unknown parameter in our regression model, σ^2 (the variance of the error term ϵ). The residuals $e_i = y_i - \hat{y}_i$ are used to obtain an estimate of σ^2. The sum of squares of the residuals, often called the **error sum of squares,** is

$$Y = \beta_0 + \beta_1 x + \epsilon$$

$$SS_E = \sum_{i=1}^{n} e_i^2 = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 \qquad (11\text{-}12)$$

We can show that the expected value of the error sum of squares is $E(SS_E) = (n - 2)\sigma^2$. Therefore an **unbiased estimator** of σ^2 is

$$\hat{\sigma}^2 = \frac{SS_E}{n - 2}$$

Linear regression: Least squares method

Exercice

$\bar{x} \approx 145.156/24 \approx 6.048 \qquad \bar{y} \approx 836.5/24 \approx 34.854$

$\hat{y} = -55.62 + 14.95x$

$$SS_E = \sum_{i=1}^{n} e_i^2 = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

$$\hat{\sigma}^2 = \frac{SS_E}{n-2}$$

Sale Price/1000	Taxes (Local, School), County)/1000	Sale Price/1000	Taxes (Local, School), County)/1000
25.9	4.9176	30.0	5.0500
29.5	5.0208	36.9	8.2464
27.9	4.5429	41.9	6.6969
25.9	4.5573	40.5	7.7841
29.9	5.0597	43.9	9.0384
29.9	3.8910	37.5	5.9894
30.9	5.8980	37.9	7.5422
28.9	5.6039	44.5	8.7951
35.9	5.8282	37.9	6.0831
31.5	5.3003	38.9	8.3607
31.0	6.2712	36.9	8.1400
30.9	5.9592	45.8	9.1416

(a) Assuming that a simple linear regression model is appropriate, obtain the least squares fit relating selling price to taxes paid. What is the estimate of σ^2?

(b) Find the mean selling price given that the taxes paid are $x = 7.50$.

(c) Calculate the fitted value of y corresponding to $x = 5.8980$. Find the corresponding residual.

Linear regression: Least squares method

Solution

$$b_0 = \bar{y} - b_1\bar{x}$$

$$b_1 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n}(x_i - \bar{x})^2}$$

$$\hat{\sigma}^2 = \frac{SS_E}{n-2}$$

a) $\bar{x} = \frac{\sum x_i}{24}, \quad \bar{y} = \frac{\sum y_i}{24}$

$$\bar{x} \approx 145.156/24 \approx 6.048 \quad \bar{y} \approx 836.5/24 \approx 34.854$$

$$\hat{y} = -55.62 + 14.95x$$

$$\hat{\sigma}^2 = \frac{\sum e_i^2}{22} \approx 5.21$$

b) If x=7.50 -> y=56.50
c) If x=5.8980 -> y=32.55. True value is 30.9, residual is -1.6551

Linear regression: R^2 factor

R^2 measures how well your regression model explains the variation in the dependent variable y, i.e., it tells you **the proportion of the total variation in y** that is explained by the model (linear regression).

It always ranges from 0 to 1:

- $R^2 = 0$ → the model explains none of the variation
- $R^2 = 1$ → the model explains all the variation

R^2 ×100 percent of the variation in *y* is 'explained by the variation in predictor *x*.

- The sum of squares of residuals, also called the residual sum of squares:

$$SS_{\text{res}} = \sum_i (y_i - f_i)^2 = \sum_i e_i^2 = \sum (y_i - \hat{y}_i)^2$$

- The total sum of squares (proportional to the variance of the data):

$$SS_{\text{tot}} = \sum_i (y_i - \bar{y})^2$$

The most general definition of the coefficient of determination is

$$R^2 = 1 - \frac{SS_{\text{res}}}{SS_{\text{tot}}}$$

Linear regression: R^2 factor

Caution #1

The coefficient of determination R^2 and the correlation coefficient *r* quantify the strength of a *linear* relationship. It is possible that R^2 = 0% and *r* = 0, suggesting there is no linear relation between *x* and *y*, and yet a perfect curved (or "curvilinear" relationship) exists.

Caution #2

A large R^2 value should not be interpreted as meaning that the estimated regression line fits the data well. Another function might better describe the trend in the data.

Caution #3

The coefficient of determination R^2 and the correlation coefficient *r* can both be greatly affected by just one data point (or a few data points).

Hypo Test for the Population Correlation Coefficient

Step 1: Hypotheses

- **Null hypothesis:**

$$H_0 : \rho = 0$$

- **Alternative hypothesis:**

$$H_a : \rho \neq 0 \quad (\text{or } \rho > 0,\ \rho < 0)$$

Step 2: Test Statistic

$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}}$$

- r: sample correlation coefficient
- n: sample size

Step 3: P-Value

- Use a **t-distribution with $n-2$ degrees of freedom**
- P-value = probability of observing a correlation as extreme as r if H_0 is true

Step 4: Decision

- If **P-value** < α → **Reject H_0**
  - Sufficient evidence of a **linear relationship** between x and y
- If **P-value** ≥ α → **Fail to reject H_0**
  - Insufficient evidence of a linear relationship

Hypo Test for the Population Correlation Coefficient

Exercice :

Let's perform the hypothesis test on the husband's age and wife's age data in which the sample correlation based on *n* = 170 couples is *r* = 0.939. To test $H_0: \rho = 0$ against the alternative $H_A: \rho \neq 0$, we obtain the following test statistic:

Step 2: Test Statistic

$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}}$$

- r: sample correlation coefficient
- n: sample size

Hypo Test for the Population Correlation Coefficient

Solution :

Step 2:

$$\begin{aligned} t^* &= \frac{r\sqrt{n-2}}{\sqrt{1-R^2}} \\ &= \frac{0.939\sqrt{170-2}}{\sqrt{1-0.939^2}} \\ &= 35.39 \end{aligned}$$

Step 3:

$$df = n - 2 = 168$$

x	P(X<= x)
35.3900	1.0000

The output tells us that the probability of getting a test-statistic smaller than 35.39 is greater than 0.999. Therefore, the probability of getting a test-statistic greater than 35.39 is less than 0.001.

Step 4:

P_value<<<<< alpha=0.05. We can reject the null hypothesis. There is sufficient statistical evidence to conclude that there is a significant linear relationship between a husband's age and his wife's age.

Hypo Test for the slope

Suppose we wish to test the hypothesis that the slope equals a constant, say, $\beta_{1,0}$. The appropriate hypotheses are

$$H_0\text{: } \beta_1 = \beta_{1,0}$$
$$H_1\text{: } \beta_1 \neq \beta_{1,0}$$

$$T_0 = \frac{\hat{\beta}_1 - \beta_{1,0}}{\sqrt{\hat{\sigma}^2 / S_{xx}}} \qquad S_{xx} = \sum (x_i - \bar{x})^2$$

Linear regression: uncertainty parameters

In simple linear regression the **estimated standard error of the slope** and the **estimated standard error of the intercept** are

$$se(\hat{\beta}_1) = \sqrt{\frac{\hat{\sigma}^2}{S_{xx}}} \quad \text{and} \quad se(\hat{\beta}_0) = \sqrt{\hat{\sigma}^2\left[\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}}\right]}$$

$$\hat{\sigma}^2 = \frac{SS_E}{n-2}$$

$$S_{xx} = \sum_{i=1}^{n} (x_i - \bar{x})^2$$

$$SS_E = \sum_{i=1}^{n} e_i^2 = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

$$b_0 = \bar{y} - b_1\bar{x}$$

$$b_1 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n}(x_i - \bar{x})^2}$$

$$\mathrm{Var}(A+B) = \mathrm{Var}(A) + \mathrm{Var}(B) + 2\,\mathrm{Cov}(A,B)$$

$$\mathrm{Var}(cX) = c^2\,\mathrm{Var}(X), \quad \mathrm{Cov}(A,cB) = c\,\mathrm{Cov}(A,B)$$

$$\mathrm{Var}(\hat{\beta}_0) = \mathrm{Var}(\bar{y}) + \mathrm{Var}(-\bar{x}\hat{\beta}_1) + 2\,\mathrm{Cov}(\bar{y}, -\bar{x}\hat{\beta}_1)$$

$$\sigma^2/n \quad \sigma^2/S_{xx} \quad \frac{\sigma^2\bar{x}}{S_{xx}}$$

Linear regression: uncertainty parameters

Exercice:

In simple linear regression the **estimated standard error of the slope** and the **estimated standard error of the intercept** are

$$se(\hat{\beta}_1) = \sqrt{\frac{\hat{\sigma}^2}{S_{xx}}} \quad \text{and} \quad se(\hat{\beta}_0) = \sqrt{\hat{\sigma}^2\left[\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}}\right]}$$

$$\hat{\sigma}^2 = \frac{SS_E}{n-2}$$

$$b_0 = \bar{y} - b_1\bar{x}$$

$$S_{xx} = \sum_{i=1}^{n} (x_i - \bar{x})^2$$

$$SS_E = \sum_{i=1}^{n} e_i^2 = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

$$b_1 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n}(x_i - \bar{x})^2}$$

$$T_0 = \frac{\hat{\beta}_1 - \beta_{1,0}}{\sqrt{\hat{\sigma}^2/S_{xx}}}$$

11-35. An article in *The Journal of Clinical Endocrinology and Metabolism* ["Simultaneous and Continuous 24-Hour Plasma and Cerebrospinal Fluid Leptin Measurements: Dissociation of Concentrations in Central and Peripheral Compartments" (2004, Vol. 89, pp. 258–265)] studied the demographics of simultaneous and continuous 24-hour plasma and cerebrospinal fluid leptin measurements. The data follow:

y = BMI (kg/m^2):	19.92	20.59	29.02	20.78	25.97	
	20.39	23.29	17.27	35.24		
x = Age (yr):	45.5	34.6	40.6	32.9	28.2	30.1
	52.1	33.3	47.0			

(a) Test for significance of regression using $\alpha = 0.05$. Find the P-value for this test. Can you conclude that the model specifies a useful linear relationship between these two variables?
(b) Estimate σ^2 and the standard deviation of $\hat{\beta}_1$.
(c) What is the standard error of the intercept in this model?

Linear regression: uncertainty parameters

Solution:

a)

$$\bar{x} = \frac{\sum x_i}{n} = \frac{344.3}{9} = 38.26$$

$$\bar{y} = \frac{\sum y_i}{n} = \frac{212.47}{9} = 23.61$$

$$S_{xx} = \sum (x_i - \bar{x})^2 = 560.41$$

$$S_{xy} = \sum (x_i - \bar{x})(y_i - \bar{y}) = 143.32$$

$$b_1 = \frac{S_{xy}}{S_{xx}} = \frac{143.32}{560.41} = 0.256$$

$$b_0 = \bar{y} - b_1\bar{x} = 23.61 - 0.256(38.26) = 13.82$$

$$\hat{y} = 13.828 + 0.256x$$

$$H_0 : b_1 = 0 \quad \text{vs} \quad H_a : b_1 \neq 0 \qquad T_0 = \frac{\hat{\beta}_1 - \beta_{1,0}}{\sqrt{\hat{\sigma}^2 / S_{xx}}}$$

$$s^2 = \frac{RSS}{n-2}$$

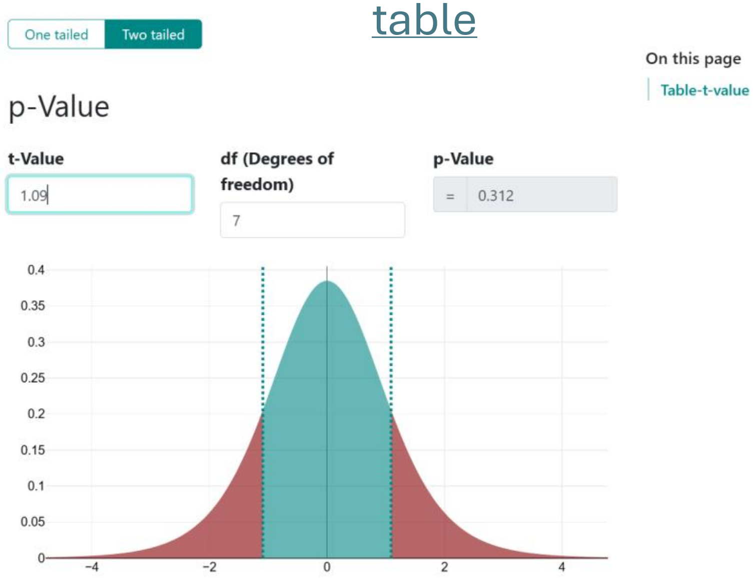
Residual sum of squares

$$RSS = \sum (y_i - \hat{y}_i)^2 = 214.9$$

$$\frac{s}{\sqrt{S_{xx}}} = \frac{5.54}{\sqrt{560.41}} = 0.234$$

t-distribution and t-table

$$t = \frac{b_1}{SE(b_1)} = \frac{0.256}{0.234} = 1.09$$



P_value=0.31 (two tailed because it can be > or < 0).

P_value>0.05 fail to reject H0. slope is **not statistically significant**, cannot conclude a useful linear relationship.

Linear regression: uncertainty parameters

Solution:
b)

$$SSE = \sum e_i^2 = 214.65$$

$$\sigma^2 = 214.65/7 \approx 30.664$$

$$S_{xx} = \sum (x_i - \bar{x})^2 = 560.41$$

$$S_{xy} = \sum (x_i - \bar{x})(y_i - \bar{y}) = 143.32$$

$$s_{b_1} = \sqrt{\frac{30.664}{560.27}} \approx \sqrt{0.0547} \approx 0.234$$

Solution:
c)

$$s_{b_0} = \sqrt{30.664\left(\frac{1}{9} + \frac{(38.26)^2}{560.27}\right)}$$

$$\hat{\sigma}^2 = \frac{SS_E}{n-2}$$

$$se(\hat{\beta}_1) = \sqrt{\frac{\hat{\sigma}^2}{S_{xx}}}$$

$$SS_E = \sum_{i=1}^{n} e_i^2 = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

$$se(\hat{\beta}_0) = \sqrt{\hat{\sigma}^2\left[\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}}\right]}$$

Confidence intervals

Exercice

An article in the *Journal of the American Ceramic Society* ["Rapid Hot-Pressing of Ultrafine PSZ Powders" (1991, Vol. 74, pp. 1547–1553)] considered the microstructure of the ultrafine powder of partially stabilized zirconia as a function of temperature. The data are shown below:

x Temperature (C): 1100 1200 1300 1100 1500 1200 1300

y Porosity (%): 30.8 19.2 6.0 13.5 11.4 7.7 3.6

(a) Fit the simple linear regression model using the method of least squares. Find an estimate of σ^2.

(b) Estimate the mean porosity for a temperature of 1400°C.

(c) Find the fitted value corresponding to $y = 11.4$ and the associated residual.

$$b_0 = \bar{y} - b_1\bar{x}$$

$$b_1 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n}(x_i - \bar{x})^2}$$

$$SS_E = \sum_{i=1}^{n} e_i^2 = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

$$\hat{\sigma}^2 = \frac{SS_E}{n-2}$$

Confidence intervals

Solution

a) mean_x=1242.86, mean_y=13.17

$$S_{xx} = \sum(x_i - \bar{x})^2 = 102040.82$$

$$S_{xy} = \sum(x_i - \bar{x})(y_i - \bar{y}) = -3486.53$$

$$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}} = \frac{-3486.53}{102040.82} = -0.03416$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1\bar{x}$$

$$\hat{y} = 55.626 - 0.03416x$$

Calculate the values and residuals

$$SSE = \sum e_i^2 = 386.66$$

$$\hat{\sigma}^2 = \frac{SSE}{n-2} = \frac{386.66}{5} = 77.33$$

Solution

b) $55.626 - 0.03416(1400)$

$= 7.80\%$

c) $x = 1500$

$$\hat{y}(1500) = 55.626 - 0.03416(1500)$$

$$e = y - \hat{y} = 11.4 - 4.39 = 7.01$$

Confidence Intervals on the Slope and Intercept

In addition to point estimates of the slope and intercept, it is possible to obtain **confidence interval** estimates of these parameters. The width of these confidence intervals is a measure of the overall quality of the regression line.

Under the assumption that the observations are normally and independently distributed, a $100(1 - \alpha)\%$ **confidence interval on the slope** β_1 in simple linear regression is

$$\hat{\beta}_1 - t_{\alpha/2,n-2}\sqrt{\frac{\hat{\sigma}^2}{S_{xx}}} \leq \beta_1 \leq \hat{\beta}_1 + t_{\alpha/2,n-2}\sqrt{\frac{\hat{\sigma}^2}{S_{xx}}} \qquad (11\text{-}29)$$

Similarly, a $100(1 - \alpha)\%$ **confidence interval on the intercept** β_0 is

$$\hat{\beta}_0 - t_{\alpha/2,n-2}\sqrt{\hat{\sigma}^2\left[\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}}\right]} \leq \beta_0 \leq \hat{\beta}_0 + t_{\alpha/2,n-2}\sqrt{\hat{\sigma}^2\left[\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}}\right]} \qquad (11\text{-}30)$$

Confidence Intervals on the mean and predicted value

- A confidence interval may be constructed on the mean response at a specified value of x. (e.g. What is the average BMI of people who are 40 years old?

It is for a group)
If you repeated the experiment many times at x0, W=what would the average outcome be?

- An important application of a regression model is predicting new or future observations *Y* corresponding to a specified level of the regressor variable x (e.g. I meet a 40-year-old patient. What BMI should I expect? The value of **one single future observation**

A $100(1-\alpha)\%$ **confidence interval about the mean response** at the value of $x = x_0$, say $\mu_{Y|x_0}$, is given by

$$\hat{\mu}_{Y|x_0} - t_{\alpha/2,n-2}\sqrt{\hat{\sigma}^2\left[\frac{1}{n} + \frac{(x_0-\bar{x})^2}{S_{xx}}\right]} \le \mu_{Y|x_0} \le \hat{\mu}_{Y|x_0} + t_{\alpha/2,n-2}\sqrt{\hat{\sigma}^2\left[\frac{1}{n} + \frac{(x_0-\bar{x})^2}{S_{xx}}\right]} \quad (11\text{-}31)$$

where $\hat{\mu}_{Y|x_0} = \hat{\beta}_0 + \hat{\beta}_1 x_0$ is computed from the fitted regression model.

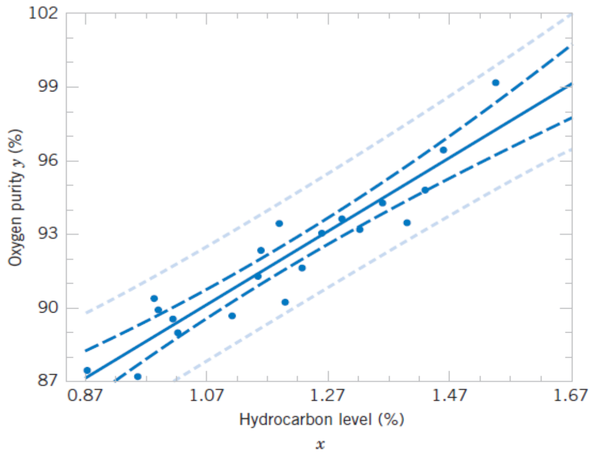
A $100(1-\alpha)$ % **prediction interval on a future observation** Y_0 at the value x_0 is given by

$$\hat{y}_0 - t_{\alpha/2,n-2}\sqrt{\hat{\sigma}^2\left[1 + \frac{1}{n} + \frac{(x_0-\bar{x})^2}{S_{xx}}\right]} \le Y_0 \le \hat{y}_0 + t_{\alpha/2,n-2}\sqrt{\hat{\sigma}^2\left[\boxed{1} + \frac{1}{n} + \frac{(x_0-\bar{x})^2}{S_{xx}}\right]} \quad (11\text{-}33)$$

The value $\hat{y}_0$ is computed from the regression model $\hat{y}_0 = \hat{\beta}_0 + \hat{\beta}_1 x_0$.

Differences

- A confidence interval may be constructed on the mean response at a specified value of x
  Mean response CI: "Where is the **true average** BMI at this age?" → narrow band along regression line.
- An important application of a regression model is predicting new or future observations *Y* corresponding to a specified level of the regressor variable
  Prediction interval: "Where will **this person's** BMI fall?" → wider band because individuals vary around the mean.



The solid middle line: regression line -> mean response

The inner dashed lines (closer to the line) -> confidence interval for the mean response
The outer dashed lines (wider band) -> prediction interval

Why the difference?

Confidence interval for the mean: only accounts for uncertainty in estimating the regression line.
Imagine the line is slightly uncertain because of limited data. If you repeated the experiment many times, the average x0 would lie in this interval 95% of the time.

Prediction interval: accounts for both the uncertainty in the regression line AND the natural variability of individual observations around the line.

So we must add extra variability, hence the "1 +" → the interval is wider

Confidence intervals

Exercice: Regression methods were used to analyze data from a study investigating the relationship between **roadway surface temperature** $(x)$and **pavement deflection** (y)

11-2. Regression methods were used to analyze the data from a study investigating the relationship between roadway surface temperature (x) and pavement deflection (y). Summary quantities were $n = 20$, $\sum y_i = 12.75$, $\sum y_i^2 = 8.86$, $\sum x_i = 1478$, $\sum x_i^2 = 143{,}215.8$, and $\sum x_i y_i = 1083.67$.

$$t_{0.005,18} = 2.878$$

$$b_1 = \frac{S_{xy}}{S_{xx}}$$

$$S_{xx} = \sum x^2 - \frac{(\sum x)^2}{n}$$

$$S_{xy} = \sum xy - \frac{(\sum x)(\sum y)}{n} =$$

Find a 99% confidence interval on each of the following:

(a) Slope

(b) Intercept

(c) Mean deflection when temperature x=85 °F

(d) Find a 99% prediction interval on pavement deflection when the temperature is 85 °F.

Under the assumption that the observations are normally and independently distributed, a $100(1 - \alpha)\%$ **confidence interval on the slope** β_1 in simple linear regression is

$$\hat{\beta}_1 - t_{\alpha/2,n-2}\sqrt{\frac{\hat{\sigma}^2}{S_{xx}}} \le \beta_1 \le \hat{\beta}_1 + t_{\alpha/2,n-2}\sqrt{\frac{\hat{\sigma}^2}{S_{xx}}} \tag{11-29}$$

Similarly, a $100(1 - \alpha)\%$ **confidence interval on the intercept** β_0 is

$$\hat{\beta}_0 - t_{\alpha/2,n-2}\sqrt{\hat{\sigma}^2\left[\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}}\right]} \le \beta_0 \le \hat{\beta}_0 + t_{\alpha/2,n-2}\sqrt{\hat{\sigma}^2\left[\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}}\right]} \tag{11-30}$$

A $100(1 - \alpha)\%$ **confidence interval about the mean response** at the value of $x = x_0$, say $\mu_{Y|x_0}$, is given by

$$\hat{\mu}_{Y|x_0} - t_{\alpha/2,n-2}\sqrt{\hat{\sigma}^2\left[\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}\right]} \le \mu_{Y|x_0} \le \hat{\mu}_{Y|x_0} + t_{\alpha/2,n-2}\sqrt{\hat{\sigma}^2\left[\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}\right]} \tag{11-31}$$

where $\hat{\mu}_{Y|x_0} = \hat{\beta}_0 + \hat{\beta}_1 x_0$ is computed from the fitted regression model.

A $100(1 - \alpha)$ % **prediction interval on a future observation** Y_0 at the value x_0 is given by

$$\hat{y}_0 - t_{\alpha/2,n-2}\sqrt{\hat{\sigma}^2\left[1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}\right]} \le Y_0 \le \hat{y}_0 + t_{\alpha/2,n-2}\sqrt{\hat{\sigma}^2\left[1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}\right]} \tag{11-33}$$

The value $\hat{y}_0$ is computed from the regression model $\hat{y}_0 = \hat{\beta}_0 + \hat{\beta}_1 x_0$.

Confidence intervals

Solution

$$\bar{x} = \frac{143}{20} = 7.15 \qquad S_{xx} = \sum x^2 - \frac{(\sum x)^2}{n} = 1478 - \frac{143^2}{20} = 455.55$$

$$\bar{y} = \frac{12.75}{20} = 0.6375 \qquad S_{xy} = \sum xy - \frac{(\sum x)(\sum y)}{n} = 1083.67 - \frac{143(12.75)}{20} = 992.52$$

$$b_1 = \frac{S_{xy}}{S_{xx}} = \frac{992.52}{455.55} = 2.178 \qquad b_0 = \bar{y} - b_1\bar{x} = 0.6375 - (2.178)(7.15) = -14.94$$

From t table: $t_{0.005,18} = 2.878$

$$\hat{\beta}_1 - t_{\alpha/2,n-2}\sqrt{\frac{\hat{\sigma}^2}{S_{xx}}} \qquad \sqrt{\frac{MSE}{S_{xx}}} = \sqrt{\frac{0.0181}{455.55}} = 0.0063$$

a) $2.160, \ 2.196$

$$\hat{\beta}_0 - t_{\alpha/2,n-2}\sqrt{\hat{\sigma}^2\left[\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}}\right]} \qquad \sqrt{0.0181\left(\frac{1}{20} + \frac{7.15^2}{455.55}\right)} = 0.054$$

b) $-15.10, \ -14.78$

Confidence intervals

Solution

c) $\hat{y}_0 = -14.94 + 2.178(85) = 170.19$

$$s_{\hat{y}_0} = 0.135\sqrt{\frac{1}{20} + \frac{(85 - 7.15)^2}{455.55}} = 0.491$$

$168.78,\ 171.60$

d) $$s_{pred} = 0.135\sqrt{1 + \frac{1}{20} + \frac{(85 - 7.15)^2}{455.55}} = 0.511$$

$$170.19 \pm (2.878)(0.511)$$

$168.72,\ 171.66$

A $100(1 - \alpha)\%$ **confidence interval about the mean response** at the value of $x = x_0$, say $\mu_{Y|x_0}$, is given by

$$\hat{\mu}_{Y|x_0} - t_{\alpha/2,n-2}\sqrt{\hat{\sigma}^2\left[\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}\right]}$$
$$\le \mu_{Y|x_0} \le \hat{\mu}_{Y|x_0} + t_{\alpha/2,n-2}\sqrt{\hat{\sigma}^2\left[\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}\right]} \qquad (11\text{-}31)$$

where $\hat{\mu}_{Y|x_0} = \hat{\beta}_0 + \hat{\beta}_1 x_0$ is computed from the fitted regression model.

A $100(1 - \alpha)$ % **prediction interval on a future observation** Y_0 at the value x_0 is given by

$$\hat{y}_0 - t_{\alpha/2,n-2}\sqrt{\hat{\sigma}^2\left[1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}\right]}$$
$$\le Y_0 \le \hat{y}_0 + t_{\alpha/2,n-2}\sqrt{\hat{\sigma}^2\left[1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}\right]} \qquad (11\text{-}33)$$

The value $\hat{y}_0$ is computed from the regression model $\hat{y}_0 = \hat{\beta}_0 + \hat{\beta}_1 x_0$.

Logistic regression

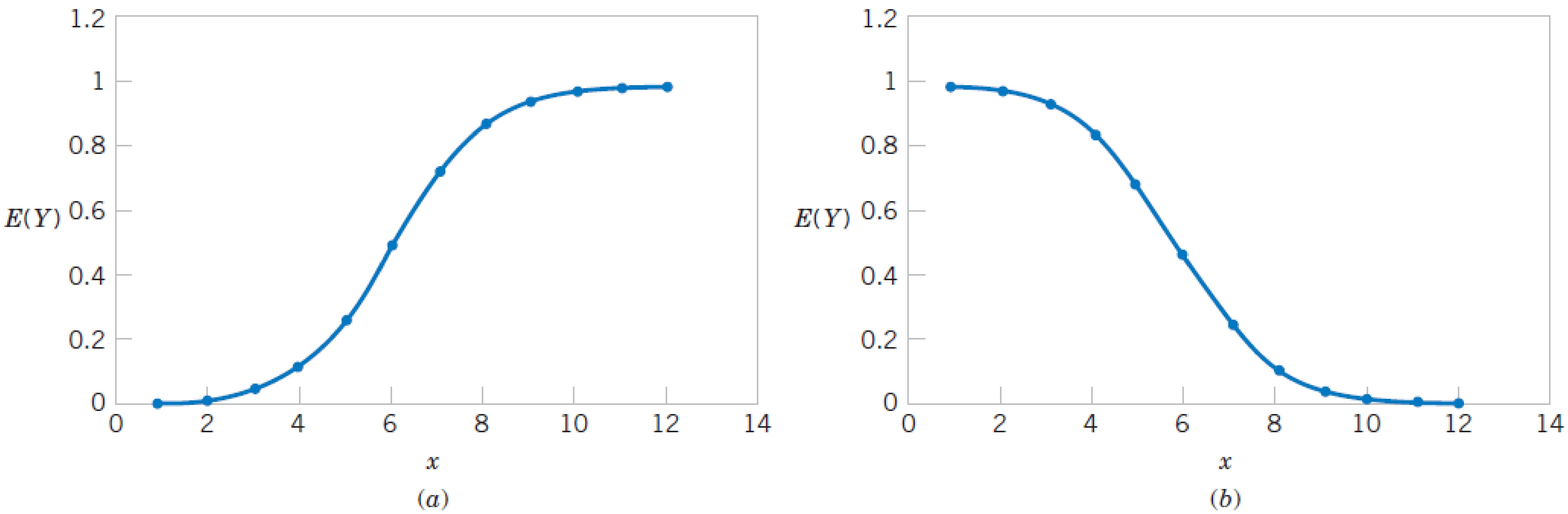


Figure 11-19 Examples of the logistic response function. (a) $E(Y) = 1/(1 + e^{-6.0-1.0x})$, (b) $E(Y) = 1/(1 + e^{-6.0+1.0x})$.

$$E(Y) = \frac{\exp(\beta_0 + \beta_1 x)}{1 + \exp(\beta_0 + \beta_1 x)} \qquad E(Y) = \frac{1}{1 + \exp[-(\beta_0 + \beta_1 x)]}$$

Multiple linear regression

Session 6

Multiple linear regression

Session 6

Objectives:

- Introduction to Multiple Linear Models
- Analyze residuals to determine if the regression model is an adequate fit
- Test statistical hypotheses and construct confidence intervals on regression model parameters
- Use the regression model to make a prediction

Sources:

- Applied-Statistics-and-Probability-for-Engineers-Douglas-C.-Montgomery-George-C.-Runger-Edisi-5-2011
- Lesson 5: Multiple Linear Regression | STAT 501

Multiple linear regression : intro

Many practical applications of regression analysis involve situations with more than one regressor, or predictor, variable. A regression model that includes two or more regressor variables is known as a **multiple regression model**.

Are a person's brain size and body size predictive of his or her intelligence?

Response (y): Performance IQ scores
Brain size: Potential predictor (x1)
Height in inches: Potential predictor (x2)
Weight in pounds: Potential predictor (x3)

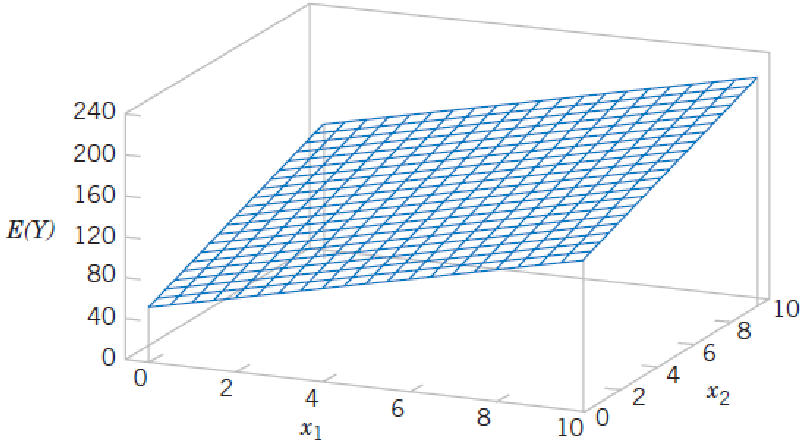
$$y_i = (\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \beta_3 x_{i3}) + \epsilon_i$$

Multiple linear regression : intro

In general, the **dependent variable** or **response** *Y* may be related to *k* **independent** or **regressor variables.** The model is called a multiple linear regression model with *k* regressor variables

$$Y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_k x_k + \epsilon$$

This model describes a hyperplane in the *k*-dimensional space of the regressor variables $\{x_j\}$. The parameter j represents the expected change in response *Y* per unit change in x_j when all the remaining regressors x_i $(i \;\; j)$ are held constant.



(a)

Multiple linear regression : intro

Note that models that are complex may often still be analyzed by multiple linear regression techniques. For example, consider the **cubic polynomial** model in one regressor variable.

$$Y = \beta_0 + \beta_1 x + \beta_2 x^2 + \beta_3 x^3 + \epsilon$$

If you put: $x_1 = x, x_2 = x^2, x_3 = x^3$

You obtain: $Y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \epsilon$

Multiple linear regression : intro

Interpreting the Coefficients

- β_0 ,the **constant (intercept) term**, represents the **baseline level of** Y, the expected value of Y when all $X_j = 0$.

Holding all other X-variables constant,

- β_j ,the **coefficient of** X_j ,measures how much the expected value of Y **increases or decreases** when X_j rises by **one unit**.
  - A positive β_j means Y increases as X_j increases;
  - A negative β_j means Y decreases as X_j increases.
  - If X_j increases by w units, then the expected value of Y changes by $\beta_j \times w$ units.

Multiple linear regression : Least-squares

The **method of least squares** may be used to estimate the regression coefficients in the multiple regression model. Suppose that *n* *k* observations are available

$$\begin{aligned} y_i &= \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_k x_{ik} + \epsilon_i \\ &= \beta_0 + \sum_{j=1}^{k} \beta_j x_{ij} + \epsilon_i \qquad i = 1, 2, \ldots, n \end{aligned}$$

LSE:

$$L = \sum_{i=1}^{n} \epsilon_i^2 = \sum_{i=1}^{n} \left(y_i - \beta_0 - \sum_{j=1}^{k} \beta_j x_{ij} \right)^2$$

Minimisation=derivative

$$\left. \frac{\partial L}{\partial \beta_0} \right|_{\hat\beta_0, \hat\beta_1, \ldots, \hat\beta_k} = -2 \sum_{i=1}^{n} \left(y_i - \hat\beta_0 - \sum_{j=1}^{k} \hat\beta_j x_{ij} \right) = 0$$

Not very nice notation...

$$\begin{array}{ccccccccc}
n\hat\beta_0 & + & \hat\beta_1 \sum_{i=1}^{n} x_{i1} & + & \hat\beta_2 \sum_{i=1}^{n} x_{i2} & + \cdots + & \hat\beta_k \sum_{i=1}^{n} x_{ik} & = & \sum_{i=1}^{n} y_i \\
\hat\beta_0 \sum_{i=1}^{n} x_{i1} & + & \hat\beta_1 \sum_{i=1}^{n} x_{i1}^2 & + & \hat\beta_2 \sum_{i=1}^{n} x_{i1} x_{i2} & + \cdots + & \hat\beta_k \sum_{i=1}^{n} x_{i1} x_{ik} & = & \sum_{i=1}^{n} x_{i1} y_i \\
\vdots & & \vdots & & \vdots & \vdots & \vdots & & \vdots \\
\hat\beta_0 \sum_{i=1}^{n} x_{ik} & + & \hat\beta_1 \sum_{i=1}^{n} x_{ik} x_{i1} & + & \hat\beta_2 \sum_{i=1}^{n} x_{ik} x_{i2} & + \cdots + & \hat\beta_k \sum_{i=1}^{n} x_{ik}^2 & = & \sum_{i=1}^{n} x_{ik} y_i
\end{array}$$

Multiple linear regression : Least-squares

Example: $Y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \epsilon$

Observation Number	Pull Strength y	Wire Length x_1	Die Height x_2	Observation Number	Pull Strength y	Wire Length x_1	Die Height x_2
1	9.95	2	50	14	11.66	2	360
2	24.45	8	110	15	21.65	4	205
3	31.75	11	120	16	17.89	4	400
4	35.00	10	550	17	69.00	20	600
5	25.02	8	295	18	10.30	1	585
6	16.86	4	200	19	34.93	10	540
7	14.38	2	375	20	46.59	15	250
8	9.60	2	52	21	44.88	15	290
9	24.35	9	100	22	54.12	16	510
10	27.50	8	300	23	56.63	17	590
11	17.08	4	412	24	22.13	6	100
12	37.00	11	400	25	21.15	5	400
13	41.95	12	500				

$$n = 25,\ \sum_{i=1}^{25} y_i = 725.82$$

$$\sum_{i=1}^{25} x_{i1} = 206,\ \sum_{i=1}^{25} x_{i2} = 8{,}294$$

$$\sum_{i=1}^{25} x_{i1}^2 = 2{,}396,\ \sum_{i=1}^{25} x_{i2}^2 = 3{,}531{,}848$$

$$\sum_{i=1}^{25} x_{i1}x_{i2} = 77{,}177,\ \sum_{i=1}^{25} x_{i1}y_i = 8{,}008.47,$$

$$\sum_{i=1}^{25} x_{i2}y_i = 274{,}816.71$$

$$n\hat{\beta}_0 + \hat{\beta}_1 \sum_{i=1}^{n} x_{i1} + \hat{\beta}_2 \sum_{i=1}^{n} x_{i2} = \sum_{i=1}^{n} y_i$$

$$\hat{\beta}_0 \sum_{i=1}^{n} x_{i1} + \hat{\beta}_1 \sum_{i=1}^{n} x_{i1}^2 + \hat{\beta}_2 \sum_{i=1}^{n} x_{i1}x_{i2} = \sum_{i=1}^{n} x_{i1}y_i$$

$$\hat{\beta}_0 \sum_{i=1}^{n} x_{i2} + \hat{\beta}_1 \sum_{i=1}^{n} x_{i1}x_{i2} + \hat{\beta}_2 \sum_{i=1}^{n} x_{i2}^2 = \sum_{i=1}^{n} x_{i2}y_i$$

$$25\hat{\beta}_0 + 206\hat{\beta}_1 + 8294\hat{\beta}_2 = 725.82$$

$$206\hat{\beta}_0 + 2396\hat{\beta}_1 + 77{,}177\hat{\beta}_2 = 8{,}008.47$$

$$8294\hat{\beta}_0 + 77{,}177\hat{\beta}_1 + 3{,}531{,}848\hat{\beta}_2 = 274{,}816.71$$

A Matrix Formulation of the Multiple Regression Model

In the multiple regression setting, because of the potentially large number of predictors, it is more efficient to use matrices to define the regression model and the subsequent analyses. Here, we review basic matrix algebra, as well as learn some of the more important multiple regression formulas in matrix form.

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i \longrightarrow \begin{aligned} y_1 &= \beta_0 + \beta_1 x_1 + \epsilon_1 \\ y_2 &= \beta_0 + \beta_1 x_2 + \epsilon_2 \\ &\vdots \\ y_n &= \beta_0 + \beta_1 x_n + \epsilon_n \end{aligned}$$

$$\underbrace{\begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}}_{Y} = \underbrace{\begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{bmatrix}}_{=X} \underbrace{\begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix}}_{\beta} + \underbrace{\begin{bmatrix} \epsilon_1 \\ \epsilon_2 \\ \vdots \\ \epsilon_n \end{bmatrix}}_{+\epsilon}$$

Remember: T**wo matrices can be multiplied together only if** the number of columns of the first matrix equals the number of rows of the second matrix.

A Matrix Formulation of the Multiple Regression Model

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_k x_{ik} + \epsilon_i \qquad i = 1, 2, \ldots, n$$

This model is a system of n equations that can be expressed in matrix notation as

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}$$

where

$$\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} \qquad \mathbf{X} = \begin{bmatrix} 1 & x_{11} & x_{12} & \cdots & x_{1k} \\ 1 & x_{21} & x_{22} & \cdots & x_{2k} \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & x_{n1} & x_{n2} & \cdots & x_{nk} \end{bmatrix} \qquad \boldsymbol{\beta} = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_k \end{bmatrix} \quad \text{and} \quad \boldsymbol{\epsilon} = \begin{bmatrix} \epsilon_1 \\ \epsilon_2 \\ \vdots \\ \epsilon_n \end{bmatrix}$$

A Matrix Formulation of the Multiple Regression Model

We wish to find the vector of least squares estimators $\hat{\beta}$, that minimizes

$$L = \sum_{i=1}^{n} \epsilon_i^2 = \boldsymbol{\epsilon}'\boldsymbol{\epsilon} = (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})'(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})$$

The least squares estimator $\hat{\boldsymbol{\beta}}$ is the solution for $\boldsymbol{\beta}$ in the equations

$$\frac{\partial L}{\partial \boldsymbol{\beta}} = \mathbf{0}$$

$$(y - X\beta)' = y' - \beta' X'$$

$$L = y'y - 2\beta' X' y + \beta' X' X \beta$$

$$y' X \beta = \beta' X' y$$

$$\frac{\partial}{\partial \beta}(y'y) = 0$$
$$\frac{\partial}{\partial \beta}(-2\beta' X' y) = -2X' y$$
$$\frac{\partial}{\partial \beta}(\beta' X' X \beta) = 2X' X \beta$$

$$\frac{\partial L}{\partial \boldsymbol{\beta}} = -2X' y + 2X' X \beta = 0$$

$$X' X \beta = X' y$$

$$\hat{\beta} = (X' X)^{-1} X' y$$

A Matrix Formulation of the Multiple Regression Model

We wish to find the vector of least squares estimators $\hat{\beta}$, that minimizes

$$L = \sum_{i=1}^{n} \epsilon_i^2 = \boldsymbol{\epsilon}'\boldsymbol{\epsilon} = (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})'(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})$$

The least squares estimator $\hat{\boldsymbol{\beta}}$ is the solution for $\boldsymbol{\beta}$ in the equations

$$\frac{\partial L}{\partial \boldsymbol{\beta}} = \mathbf{0}$$

Equations to be solved

Normal Equations

$$\mathbf{X}'\mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{X}'\mathbf{y}$$

Least Square Estimate of $\boldsymbol{\beta}$

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$$

Remember $SS_E = \sum_{i=1}^{n} e_i^2 = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$

the apostrophe $'$ means **transpose.**

$$X = \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix} \quad (3 \times 2) \qquad X' = \begin{bmatrix} 1 & 3 & 5 \\ 2 & 4 & 6 \end{bmatrix} \quad (2 \times 3)$$

Is an $\hat{\beta}$ unbiased estimator?

$$\hat{\beta} = (X'X)^{-1}X'y$$

$$y = X\beta + \epsilon$$

$$\hat{\beta} = (X'X)^{-1}X'y$$

$$\hat{\beta} = (X'X)^{-1}X'(X\beta + \varepsilon)$$

$$\hat{\beta} = (X'X)^{-1}(X'X\beta + X'\varepsilon)$$

$$(X'X)^{-1}X'X = I,$$

$$\hat{\beta} = (X'X)^{-1}X'X\beta + (X'X)^{-1}X'\varepsilon$$

$$\hat{\beta} = \beta + (X'X)^{-1}X'\varepsilon$$

β is constant:

$$E[\beta] = \beta$$

$$E[\hat{\beta}] = E[\beta + (X'X)^{-1}X'\varepsilon]$$

$$E[\hat{\beta}] = \beta + (X'X)^{-1}X'E[\varepsilon]$$

ε is random → $E[\varepsilon] = 0$

$$E[\hat{\beta}] = \beta + (X'X)^{-1}X' \cdot 0$$

Multiple linear regression : Least-squares

Example:

$$\mathbf{X}'\mathbf{X} = \begin{bmatrix} 1 & 1 & \cdots & 1 \\ 2 & 8 & \cdots & 5 \\ 50 & 110 & \cdots & 400 \end{bmatrix} \begin{bmatrix} 1 & 2 & 50 \\ 1 & 8 & 110 \\ \vdots & \vdots & \vdots \\ 1 & 5 & 400 \end{bmatrix}$$

$$= \begin{bmatrix} 25 & 206 & 8{,}294 \\ 206 & 2{,}396 & 77{,}177 \\ 8{,}294 & 77{,}177 & 3{,}531{,}848 \end{bmatrix}$$

$$\mathbf{X}'\mathbf{y} = \begin{bmatrix} 1 & 1 & \cdots & 1 \\ 2 & 8 & \cdots & 5 \\ 50 & 110 & \cdots & 400 \end{bmatrix} \begin{bmatrix} 9.95 \\ 24.45 \\ \vdots \\ 21.15 \end{bmatrix} = \begin{bmatrix} 725.82 \\ 8{,}008.47 \\ 274{,}816.71 \end{bmatrix}$$

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$$

$$\begin{bmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \\ \hat{\beta}_2 \end{bmatrix} = \begin{bmatrix} 25 & 206 & 8{,}294 \\ 206 & 2{,}396 & 77{,}177 \\ 8{,}294 & 77{,}177 & 3{,}531{,}848 \end{bmatrix}^{-1} \begin{bmatrix} 725.82 \\ 8{,}008.37 \\ 274{,}811.31 \end{bmatrix}$$

$$= \begin{bmatrix} 0.214653 & -0.007491 & -0.000340 \\ -0.007491 & 0.001671 & -0.000019 \\ -0.000340 & -0.000019 & +0.0000015 \end{bmatrix} \begin{bmatrix} 725.82 \\ 8{,}008.47 \\ 274{,}811.31 \end{bmatrix}$$

$$= \begin{bmatrix} 2.26379143 \\ 2.74426964 \\ 0.01252781 \end{bmatrix}$$

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\,\mathbf{X}'\mathbf{y}$$

Observation Number	Pull Strength y	Wire Length x_1	Die Height x_2	Observation Number	Pull Strength y	Wire Length x_1	Die Height x_2
1	9.95	2	50	14	11.66	2	360
2	24.45	8	110	15	21.65	4	205
3	31.75	11	120	16	17.89	4	400
4	35.00	10	550	17	69.00	20	600
5	25.02	8	295	18	10.30	1	585
6	16.86	4	200	19	34.93	10	540
7	14.38	2	375	20	46.59	15	250
8	9.60	2	52	21	44.88	15	290
9	24.35	9	100	22	54.12	16	510
10	27.50	8	300	23	56.63	17	590
11	17.08	4	412	24	22.13	6	100
12	37.00	11	400	25	21.15	5	400
13	41.95	12	500				

$$\mathbf{X} = \begin{bmatrix} 1 & 2 & 50 \\ 1 & 8 & 110 \\ 1 & 11 & 120 \\ 1 & 10 & 550 \\ 1 & 8 & 295 \\ 1 & 4 & 200 \\ 1 & 2 & 375 \\ 1 & 2 & 52 \\ 1 & 9 & 100 \\ 1 & 8 & 300 \\ 1 & 4 & 412 \\ 1 & 11 & 400 \\ 1 & 12 & 500 \\ 1 & 2 & 360 \\ 1 & 4 & 205 \\ 1 & 4 & 400 \\ 1 & 20 & 600 \\ 1 & 1 & 585 \\ 1 & 10 & 540 \\ 1 & 15 & 250 \\ 1 & 15 & 290 \\ 1 & 16 & 510 \\ 1 & 17 & 590 \\ 1 & 6 & 100 \\ 1 & 5 & 400 \end{bmatrix} \quad \mathbf{y} = \begin{bmatrix} 9.95 \\ 24.45 \\ 31.75 \\ 35.00 \\ 25.02 \\ 16.86 \\ 14.38 \\ 9.60 \\ 24.35 \\ 27.50 \\ 17.08 \\ 37.00 \\ 41.95 \\ 11.66 \\ 21.65 \\ 17.89 \\ 69.00 \\ 10.30 \\ 34.93 \\ 46.59 \\ 44.88 \\ 54.12 \\ 56.63 \\ 22.13 \\ 21.15 \end{bmatrix}$$

Multiple linear regression : σ^2 estimator

Just as in simple linear regression, it is important to estimate σ^2 , the variance of the error term in a multiple regression model.

p: number of parameters, n number of data

$$\hat{\sigma}^2 = \frac{\sum_{i=1}^{n} e_i^2}{n-p} = \frac{SS_E}{n-p}$$

Remember for simple linear regression, you have 2 parameters (b0,b1), therefore n-2 !!

Covariance matrix

$$\mathbf{C} = (\mathbf{X}'\mathbf{X})^{-1} = \begin{bmatrix} C_{00} & C_{01} & C_{02} \\ C_{10} & C_{11} & C_{12} \\ C_{20} & C_{21} & C_{22} \end{bmatrix}$$

$$\text{cov}(\hat{\boldsymbol{\beta}}) = \sigma^2(\mathbf{X}'\mathbf{X})^{-1} = \sigma^2\mathbf{C}$$

The estimates of the variances of these regression coefficients are obtained by replacing σ^2 with an estimate. When σ^2 is replaced by its estimate $\hat{\sigma}^2$, the square root of the estimated variance of the jth regression coefficient is called the **estimated standard error** of $\hat{\beta}_j$ or $se(\hat{\beta}_j) = \sqrt{\hat{\sigma}^2 C_{jj}}$. These standard errors are a useful measure of the **precision of estimation** for the regression coefficients; small standard errors imply good precision.

$$SE(\hat{\beta}_j) = \sqrt{\hat{\sigma}^2 \cdot [(X^T X)^{-1}]_{jj}}$$

Multiple linear regression : σ^2 estimator

Exercice: A regression model is to be developed for predicting the ability of soil to absorb chemical contaminants. Ten observations have been taken on a soil absorption index (*y*) and two regressors: *x*1 amount of extractable iron ore and *x*2

$$(\mathbf{X}'\mathbf{X})^{-1} = \begin{bmatrix} 1.17991 & -7.30982\text{ E-3} & 7.3006\text{ E-4} \\ -7.30982\text{ E-3} & 7.9799\text{ E-5} & -1.23713\text{ E-4} \\ 7.3006\text{ E-4} & -1.23713\text{ E-4} & 4.6576\text{ E-4} \end{bmatrix},$$

$$\mathbf{X}'\mathbf{y} = \begin{bmatrix} 220 \\ 36{,}768 \\ 9{,}965 \end{bmatrix}$$

(a) Estimate the regression coefficients in the model specified above.

(b) What is the predicted value of the absorption index y when $x_1 = 200$ and $x_2 = 50$?

Multiple linear regression : σ^2 estimator

Solution: $\hat{\beta} = (X'X)^{-1}X'y$

$$(\mathbf{X'X})^{-1} = \begin{bmatrix} 1.17991 & -7.30982\text{ E-3} & 7.3006\text{ E-4} \\ -7.30982\text{ E-3} & 7.9799\text{ E-5} & -1.23713\text{ E-4} \\ 7.3006\text{ E-4} & -1.23713\text{ E-4} & 4.6576\text{ E-4} \end{bmatrix},$$

$$\mathbf{X'y} = \begin{bmatrix} 220 \\ 36{,}768 \\ 9{,}965 \end{bmatrix}$$

(a) Estimate the regression coefficients in the model specified above.

(b) What is the predicted value of the absorption index y when $x_1 = 200$ and $x_2 = 50$?

$$\hat{\beta} = \begin{bmatrix} -1.9122 \\ 0.09309 \\ 0.25323 \end{bmatrix}$$

$$\hat{y} = -1.9122 + 0.09309\,x_1 + 0.25323\,x_2$$

$$\hat{y} = -1.9122 + 0.09309(200) + 0.25323(50)$$

$$= 29.37$$

Multiple linear regression : σ² estimator

Exercice:

12-4. You have fit a multiple linear regression model and the $(\mathbf{X}'\mathbf{X})^{-1}$ matrix is:

$$(\mathbf{X}'\mathbf{X})^{-1} = \begin{bmatrix} 0.893758 & -0.0282448 & -0.0175641 \\ -0.028245 & 0.0013329 & 0.0001547 \\ -0.017564 & 0.0001547 & 0.0009108 \end{bmatrix}$$

(a) How many regressor variables are in this model?

(b) If the error sum of squares is 307 and there are 15 observations, what is the estimate of σ^2?

(c) What is the standard error of the regression coefficient $\hat{\beta}_1$?

$$\hat{\sigma}^2 = \frac{\sum_{i=1}^{n} e_i^2}{n-p} = \frac{SS_E}{n-p}$$

$$SS_E = \sum_{i=1}^{n} e_i^2 = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

$$SE(\hat{\beta}_j) = \sqrt{\hat{\sigma}^2 \cdot [(X^T X)^{-1}]_{jj}}$$

Multiple linear regression : σ^2 estimator

Solution:

a) First column is for intercept, so you have 3 columns, you have two regressor variables

b) $$\hat{\sigma}^2 = \frac{307}{15-3} = \frac{307}{12}$$

c) $$SE(\hat{\beta}_j) = \sqrt{\hat{\sigma}^2 \cdot [(X^T X)^{-1}]_{jj}}$$

J=1 -> (2,2) of the matrix 0.0013329

$$SE(\hat{\beta}_1) = \sqrt{25.5833 \times 0.0013329} = \sqrt{0.034087}$$

12-4. You have fit a multiple linear regression model and the $(\mathbf{X'X})^{-1}$ matrix is:

$$(\mathbf{X'X})^{-1} = \begin{bmatrix} 0.893758 & -0.0282448 & -0.0175641 \\ -0.028245 & 0.0013329 & 0.0001547 \\ -0.017564 & 0.0001547 & 0.0009108 \end{bmatrix}$$

(a) How many regressor variables are in this model?

(b) If the error sum of squares is 307 and there are 15 observations, what is the estimate of σ^2?

(c) What is the standard error of the regression coefficient $\hat{\beta}_1$?

$$\hat{\sigma}^2 = \frac{\sum_{i=1}^{n} e_i^2}{n-p} = \frac{SS_E}{n-p}$$

Multiple linear regression : Test for Significance of Regression

The test for significance of regression is a test to determine whether a linear relationship exists between the response variable *y* and a subset of the regressor variables $x_1, x_2, \ldots, x_k$

$$H_0\colon \beta_1 = \beta_2 = \cdots = \beta_k = 0$$
$$H_1\colon \beta_j \neq 0 \quad \text{for at least one } j$$

Rejection of $H_0\colon \beta_1 = \beta_2 = \cdots = \beta_k = 0$ implies that at least one of the regressor variables $x_1, x_2, \ldots, x_k$ contributes significantly to the model.

Statistical test:

$$F_0 = \frac{SS_R/k}{SS_E/(n-p)} = \frac{MS_R}{MS_E}$$

Source of Variation	Sum of Squares	Degrees of Freedom	Mean Square	F_0
Regression	SS_R	k	MS_R	MS_R/MS_E
Error or residual	SS_E	$n-p$	MS_E	
Total	SS_T	$n-1$		

p=# parameters=k+1

MS=MS/dof

Multiple linear regression : Test for Significance of Regression

Total Sum of Squares (TSS) – total variation in y around the mean:

$$TSS = \sum (y_i - \bar{y})^2$$

Regression Sum of Squares (RSS or SSR) – variation explained by the regression:

$$RSS = \sum (\hat{y}_i - \bar{y})^2$$

Error Sum of Squares (SSE) – unexplained variation (residuals):

$$SSE = \sum (y_i - \hat{y}_i)^2$$

$$SS_E = SS_T - SS_R$$

$$SS_E = \mathbf{y}'\mathbf{y} - \frac{\left(\sum_{i=1}^{n} y_i\right)^2}{n} - \left[\hat{\boldsymbol{\beta}}'\mathbf{X}'\mathbf{y} - \frac{\left(\sum_{i=1}^{n} y_i\right)^2}{n}\right]$$

$$SS_R = \hat{\boldsymbol{\beta}}'\mathbf{X}'\mathbf{y} - \frac{\left(\sum_{i=1}^{n} y_i\right)^2}{n}$$

$$SSR = (X\hat{\beta} - \bar{y}\mathbf{1})'(X\hat{\beta} - \bar{y}\mathbf{1})$$

$$SSR = (X\hat{\beta})'(X\hat{\beta}) - 2\bar{y}\mathbf{1}'X\hat{\beta} + \bar{y}^2\mathbf{1}'\mathbf{1}$$

$$= \hat{\beta}'X'X\hat{\beta} \quad - 2\bar{y}\sum y_i \quad + n\bar{y}^2$$

$$SSR = \hat{\beta}'X'X\hat{\beta} - 2n\bar{y}^2 + n\bar{y}^2$$

$$SSR = \hat{\beta}'X'X\hat{\beta} - n\bar{y}^2$$

$$SS_R = \hat{\boldsymbol{\beta}}'\mathbf{X}'\mathbf{y} - \frac{\left(\sum_{i=1}^{n} y_i\right)^2}{n}$$

$$\mathbf{1}'\hat{y} = \hat{y}_1 + \hat{y}_2 + \cdots + \hat{y}_n$$

$$\mathbf{1}'\mathbf{1} = n$$

$$\mathbf{1}'X\hat{\beta} = \sum \hat{y}_i$$

The regression line passes through the point (x_bar,y_bar)

$$\sum \hat{y}_i = \sum y_i$$

$$2\bar{y}\sum y_i = 2n\bar{y}^2$$

$$X'X\hat{\beta} = X'y$$

Multiple linear regression : Test for Significance of Regression

Example: We will test for significance of regression (with 0.05) using the wire bond pull strength data.

$$SS_T = \mathbf{y}'\mathbf{y} - \frac{\left(\sum_{i=1}^{n} y_i\right)^2}{n} = 27{,}178.5316 - \frac{(725.82)^2}{25}$$
$$= 6105.9447$$

$$SS_R = \hat{\boldsymbol{\beta}}'\mathbf{X}'\mathbf{y} - \frac{\left(\sum_{i=1}^{n} y_i\right)^2}{n} = 27{,}063.3581 - \frac{(725.82)^2}{25}$$
$$= 5990.7712$$

$$SS_E = SS_T - SS_R = \mathbf{y}'\mathbf{y} - \hat{\boldsymbol{\beta}}'\mathbf{X}'\mathbf{y} = 115.1716$$

$$f_0 = \frac{MS_R}{MS_E} = \frac{2995.3856}{5.2352} = 572.17$$

$$f_0 > f_{0.05,2,22} = 3.44$$

$$\mathbf{X}'\mathbf{y} = \begin{bmatrix} 1 & 1 & \cdots & 1 \\ 2 & 8 & \cdots & 5 \\ 50 & 110 & \cdots & 400 \end{bmatrix} \begin{bmatrix} 9.95 \\ 24.45 \\ \vdots \\ 21.15 \end{bmatrix} = \begin{bmatrix} 725.82 \\ 8{,}008.47 \\ 274{,}816.71 \end{bmatrix}$$

$$\hat{y} = 2.26379 + 2.74427x_1 + 0.01253x_2$$

$$\mathbf{X} = \begin{bmatrix} 1 & 2 & 50 \\ 1 & 8 & 110 \\ 1 & 11 & 120 \\ 1 & 10 & 550 \\ 1 & 8 & 295 \\ 1 & 4 & 200 \\ 1 & 2 & 375 \\ 1 & 2 & 52 \\ 1 & 9 & 100 \\ 1 & 8 & 300 \\ 1 & 4 & 412 \\ 1 & 11 & 400 \\ 1 & 12 & 500 \\ 1 & 2 & 360 \\ 1 & 4 & 205 \\ 1 & 4 & 400 \\ 1 & 20 & 600 \\ 1 & 1 & 585 \\ 1 & 10 & 540 \\ 1 & 15 & 250 \\ 1 & 15 & 290 \\ 1 & 16 & 510 \\ 1 & 17 & 590 \\ 1 & 6 & 100 \\ 1 & 5 & 400 \end{bmatrix} \quad \mathbf{y} = \begin{bmatrix} 9.95 \\ 24.45 \\ 31.75 \\ 35.00 \\ 25.02 \\ 16.86 \\ 14.38 \\ 9.60 \\ 24.35 \\ 27.50 \\ 17.08 \\ 37.00 \\ 41.95 \\ 11.66 \\ 21.65 \\ 17.89 \\ 69.00 \\ 10.30 \\ 34.93 \\ 46.59 \\ 44.88 \\ 54.12 \\ 56.63 \\ 22.13 \\ 21.15 \end{bmatrix}$$

Observation Number	Pull Strength y	Wire Length x_1	Die Height x_2	Observation Number	Pull Strength y	Wire Length x_1	Die Height x_2
1	9.95	2	50	14	11.66	2	360
2	24.45	8	110	15	21.65	4	205
3	31.75	11	120	16	17.89	4	400
4	35.00	10	550	17	69.00	20	600
5	25.02	8	295	18	10.30	1	585
6	16.86	4	200	19	34.93	10	540
7	14.38	2	375	20	46.59	15	250
8	9.60	2	52	21	44.88	15	290
9	24.35	9	100	22	54.12	16	510
10	27.50	8	300	23	56.63	17	590
11	17.08	4	412	24	22.13	6	100
12	37.00	11	400	25	21.15	5	400
13	41.95	12	500				

We reject the null hypothesis and conclude that pull strength is linearly related to either wire length or die height, or both.

Multiple linear regression : Test on individual regression coefficient

Testing hypotheses on the individual regression coefficients. Such tests would be useful in determining the potential value of each of the regressor variables in the regression model. For example, the model might be more effective with the inclusion of additional variables or perhaps with the deletion of one or more of the regressors presently in the model.

The hypothesis to test if an individual regression coefficient, say β_j equals a value β_{j0} is

$$H_0: \beta_j = \beta_{j0}$$
$$H_1: \beta_j \neq \beta_{j0}$$

The test statistic for this hypothesis is

$$T_0 = \frac{\hat{\beta}_j - \beta_{j0}}{\sqrt{\sigma^2 C_{jj}}} = \frac{\hat{\beta}_j - \beta_{j0}}{se(\hat{\beta}_j)}$$

$$SE(\hat{\beta}_j) = \sqrt{\hat{\sigma}^2 \cdot [(X^T X)^{-1}]_{jj}}$$

The null hypothesis H0: $\beta_j=\beta_{j0}$ is rejected if $|t_0|>t_{\alpha/2,n-p}$
Note that if H0: $\beta_j=0$ can not be rejected, this means that we can delete the regressor from our model.

Multiple linear regression : Test on individual regression coefficient

Example. Consider the wire bond pull strength data, and suppose that we want to test the hypothesis that the regression coefficient for *x*2 (die height) is zero

$$T_0 = \frac{\hat{\beta}_j - \beta_{j0}}{\sqrt{\sigma^2 C_{jj}}} = \frac{\hat{\beta}_j - \beta_{j0}}{se(\hat{\beta}_j)}$$

Hypothesis

$$H_0\colon \beta_2 = 0$$
$$H_1\colon \beta_2 \neq 0$$

$$SE(\hat{\beta}_j) = \sqrt{\hat{\sigma}^2 \cdot [(X^T X)^{-1}]_{jj}}$$

$$t_0 = \frac{\hat{\beta}_2}{\sqrt{\hat{\sigma}^2 C_{22}}}$$

$$SS_E = SS_T - SS_R = \mathbf{y}'\mathbf{y} - \hat{\boldsymbol{\beta}}'\mathbf{X}'\mathbf{y} = 115.1716$$

$$\hat{\sigma}^2 = \frac{\sum_{i=1}^{n} e_i^2}{n-p} = \frac{SS_E}{n-p}$$

$$(X'X)^{-1} \approx \begin{bmatrix} 0.2162 & -0.00755 & -0.000344 \\ -0.00755 & 0.001684 & -0.0000191 \\ -0.000344 & -0.0000191 & 0.00000151 \end{bmatrix}$$

n-p=25-k+1=25-2+1=22

$$A^{-1} = \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix}^{-1}$$

$$t_0 = \frac{\hat{\beta}_2}{\sqrt{\hat{\sigma}^2 C_{22}}}$$

$$= \frac{0.01253}{\sqrt{(5.2352)(0.0000015)}} = 4.477$$

$$\frac{1}{\det A} \begin{pmatrix} ei - fh & ch - bi & bf - ce \\ fg - di & ai - cg & cd - af \\ dh - eg & bg - ah & ae - bd \end{pmatrix}$$

Since $t_{0.025,22}$ =2.074, we reject *H*0 and conclude that the variable *x*2 (die height) contributes significantly to the model

Multiple linear regression : Test on individual regression coefficient

Exercice. You have fit a regression model with two regressors to a data set that has 20 observations. The total sum of squares is 1000 and the model sum of squares is 750.

(a) What is the value of R2 for this model?

(b) What is the value of the *F*-statistic for testing the significance of regression? What conclusions would you draw about this model if α= 0.05? What if α=0.01?

$$F_0 = \frac{SS_R/k}{SS_E/(n-p)} = \frac{MS_R}{MS_E}$$

For $\alpha = 0.05$, the critical F ≈ 3.59

For $\alpha = 0.01$, the critical F ≈ 6.93

- The sum of squares of residuals, also called the residual sum of squares:

$$SS_{\text{res}} = \sum_i (y_i - f_i)^2 = \sum_i e_i^2 = \sum (y_i - \hat{y}_i)^2$$

- The total sum of squares (proportional to the variance of the data):

$$SS_{\text{tot}} = \sum_i (y_i - \bar{y})^2$$

The most general definition of the coefficient of determination is

$$R^2 = 1 - \frac{SS_{\text{res}}}{SS_{\text{tot}}}$$

Multiple linear regression : Test on individual regression coefficient

Solution. You have fit a regression model with two regressors to a data set that has 20 observations. The total sum of squares is 1000 and the model sum of squares is 750.

(a) n=20, p=2, SST=1000, SSR=750 so therefore SSE=250

$$R^2 = \frac{SSR}{SST} = \frac{750}{1000} = 0.75$$

(b) F score

$$F = \frac{SSR/p}{SSE/(n-p-1)} = \frac{750/2}{250/17} = \frac{375}{14.70588}$$

For $\alpha = 0.05$, the critical F ≈ 3.59

For $\alpha = 0.01$, the critical F ≈ 6.93

Since F=25.5>6.93, the model is significant at both 0.05 and 0.01 levels.

Multiple linear regression : confidence intervals

Confidence Interval on a Regression Coefficient

A $100(1-\alpha)$ % **confidence interval on the regression coefficient** β_j, $j = 0, 1, \ldots, k$ in the multiple linear regression model is given by

$$\hat{\beta}_j - t_{\alpha/2,n-p}\sqrt{\hat{\sigma}^2 C_{jj}} \le \beta_j \le \hat{\beta}_j + t_{\alpha/2,n-p}\sqrt{\hat{\sigma}^2 C_{jj}} \qquad (12\text{-}35)$$

Confidence Interval on the Mean Response

For the multiple linear regression model, a $100(1-\alpha)$% **confidence interval on the mean response** at the point $x_{01}, x_{02}, \ldots, x_{0k}$ is

$$\hat{\mu}_{Y|\mathbf{x}_0} - t_{\alpha/2,n-p}\sqrt{\hat{\sigma}^2 \mathbf{x}_0'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_0} \le \mu_{Y|\mathbf{x}_0} \le \hat{\mu}_{Y|\mathbf{x}_0} + t_{\alpha/2,n-p}\sqrt{\hat{\sigma}^2 \mathbf{x}_0'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_0} \qquad (12\text{-}39)$$

Prediction Interval

A $100(1-\alpha)$% **prediction interval for this future observation** is

$$\hat{y}_0 - t_{\alpha/2,n-p}\sqrt{\hat{\sigma}^2(1 + \mathbf{x}_0'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_0)} \le Y_0 \le \hat{y}_0 + t_{\alpha/2,n-p}\sqrt{\hat{\sigma}^2(1 + \mathbf{x}_0'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_0)} \qquad (12\text{-}41)$$

Multiple linear regression : confidence intervals

Exemple:

We will construct a 95% confidence interval on the parameter β_1 in the wire bond pull strength problem. The point estimate of β_1 is $\hat{\beta}_1 = 2.74427$ and the diagonal element of $(\mathbf{X}'\mathbf{X})^{-1}$ corresponding to β_1 is $C_{11} = 0.001671$. The estimate of σ^2 is $\hat{\sigma}^2 = 5.2352$, and $t_{0.025,22} = 2.074$. Therefore, the 95% CI on β_1

Confidence Interval on a Regression Coefficient

A $100(1-\alpha)$% **confidence interval on the regression coefficient** β_j, $j = 0, 1, \ldots, k$ in the multiple linear regression model is given by

$$\hat{\beta}_j - t_{\alpha/2,n-p}\sqrt{\hat{\sigma}^2 C_{jj}} \le \beta_j \le \hat{\beta}_j + t_{\alpha/2,n-p}\sqrt{\hat{\sigma}^2 C_{jj}} \tag{12-35}$$

$$2.74427 - (2.074)\sqrt{(5.2352)(.001671)} \le \beta_1 \le 2.74427 + (2.074)\sqrt{(5.2352)(.001671)}$$

$$2.55029 \le \beta_1 \le 2.93825$$

Multiple linear regression : confidence intervals

Exemple: construct a 95% CI on the mean pull strength for a wire bond with wire length $x1$ =8 and die height $x2$ =275

Confidence Interval on the Mean Response

For the multiple linear regression model, a $100(1-\alpha)\%$ **confidence interval on the mean response** at the point $x_{01}, x_{02}, \ldots, x_{0k}$ is

$$\hat{\mu}_{Y|\mathbf{x}_0} - t_{\alpha/2,n-p}\sqrt{\hat{\sigma}^2\mathbf{x}_0'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_0}$$
$$\leq \mu_{Y|\mathbf{x}_0} \leq \hat{\mu}_{Y|\mathbf{x}_0} + t_{\alpha/2,n-p}\sqrt{\hat{\sigma}^2\mathbf{x}_0'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_0} \qquad (12\text{-}39)$$

$$\mathbf{x}_0 = \begin{bmatrix} 1 \\ 8 \\ 275 \end{bmatrix}$$

The estimated mean response at this point is found from Equation 12-36 as

$$\hat{\mu}_{Y|\mathbf{x}_0} = \mathbf{x}_0'\hat{\beta} = [1 \quad 8 \quad 275]\begin{bmatrix} 2.26379 \\ 2.74427 \\ 0.01253 \end{bmatrix} = 27.66$$

$$\begin{bmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \\ \hat{\beta}_2 \end{bmatrix} = \begin{bmatrix} 25 & 206 & 8{,}294 \\ 206 & 2{,}396 & 77{,}177 \\ 8{,}294 & 77{,}177 & 3{,}531{,}848 \end{bmatrix}^{-1}\begin{bmatrix} 725.82 \\ 8{,}008.37 \\ 274{,}811.31 \end{bmatrix}$$

$$= \begin{bmatrix} 0.214653 & -0.007491 & -0.000340 \\ -0.007491 & 0.001671 & -0.000019 \\ -0.000340 & -0.000019 & +0.0000015 \end{bmatrix}\begin{bmatrix} 725.82 \\ 8{,}008.47 \\ 274{,}811.31 \end{bmatrix}$$

$$= \begin{bmatrix} 2.26379143 \\ 2.74426964 \\ 0.01252781 \end{bmatrix}$$

The variance of $\hat{\mu}_{Y|\mathbf{x}_0}$ is estimated by

$$\hat{\sigma}^2\mathbf{x}_0'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_0 = 5.2352\,[1\ 8\ 275]$$
$$\times \begin{bmatrix} .214653 & -.007491 & -.000340 \\ -.007491 & .001671 & -.000019 \\ -.000340 & -.000019 & .0000015 \end{bmatrix}\begin{bmatrix} 1 \\ 8 \\ 275 \end{bmatrix}$$
$$= 5.2352\,(0.0444) = 0.23244$$

Therefore, a 95% CI on the mean pull strength at this point is found from Equation 12-39 as

$$27.66 - 2.074\sqrt{0.23244} \leq \mu_{Y|\mathbf{x}_0} \leq 27.66 + 2.074\sqrt{0.23244}$$

which reduces to

$$26.66 \leq \mu_{Y|\mathbf{x}_0} \leq 28.66$$

Multiple linear regression : confidence intervals

Exemple: Suppose that the engineer in Example 12-1 wishes to construct a 95% prediction interval on the wire bond pull strength when the wire length is $x_1 = 8$ and the die height is $x_2 = 275$. Note that $\mathbf{x}_0' = [1 \quad 8 \quad 275]$, and the point estimate of the pull strength is $\hat{y}_0 = \mathbf{x}_0'\hat{\boldsymbol{\beta}} = 27.66$.

Prediction Interval

A $100(1-\alpha)\%$ **prediction interval** for this future observation is

$$\hat{y}_0 - t_{\alpha/2,n-p}\sqrt{\hat{\sigma}^2(1 + \mathbf{x}_0'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_0)} \le Y_0 \le \hat{y}_0 + t_{\alpha/2,n-p}\sqrt{\hat{\sigma}^2(1 + \mathbf{x}_0'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_0)} \tag{12-41}$$

From the previous we calculated $\mathbf{x}_0'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_0 = 0.04444$.

$$27.66 - 2.074\sqrt{5.2352(1 + 0.0444)} \le Y_0 \le 27.66 + 2.074\sqrt{5.2352(1 + 0.0444)}$$

$$22.81 \le Y_0 \le 32.51$$

Multiple linear regression : Multicollinearity

What is it?

In multiple regression analysis, we typically look for relationships between the response variable Y and the predictor variables x_j. However, in many cases, the predictor variables themselves are also correlated with one another. When these correlations among predictors are particularly strong, the phenomenon is referred to as **multicollinearity**.

The issue:

Multicollinearity can have serious effects on the estimates of the regression coefficients and on the general applicability of the estimated model.

Test:

To test for multicollinearity we can use the Variance Inflation Factor (VIF). A VIF of 1 will mean that the variables are not correlated; a VIF between 1 and 5 shows that variables are moderately correlated, and a VIF between 5 and 10 will mean that variables are highly correlated

$$VIF(\beta_j) = \frac{1}{(1 - R_j^2)} \qquad j = 1, 2, \ldots, k$$

Multiple linear regression : Multicollinearity

$$VIF(\beta_j) = \frac{1}{(1 - R_j^2)} \qquad j = 1, 2, \ldots, k$$

- Quantifies **how much multicollinearity inflates the variance** of coefficient βj.
- Rj comes from regressing predictor Xj on all other predictors.
- Larger Rj² → predictor is highly correlated with others → larger VIF.

$$X_1 = \alpha_0 + \alpha_2 X_2 + \alpha_3 X_3 + \cdots + \alpha_p X_p + u$$

$$R_1^2 = \frac{SS_{\text{explained}}}{SS_{\text{total}}} = 1 - \frac{SS_{\text{residual}}}{SS_{\text{total}}}$$

Compares actual values to mean

$$SST = \sum_{i=1}^{n} (y_i - \bar{y})^2$$

Compares actual values to predicted values

$$SSE = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

Multiple linear regression : Multicollinearity

What happens if you have two variables?

$$= \sum (y_i - \hat{y}_i)^2$$

$$R_1^2 = \frac{\mathrm{SS_{explained}}}{\mathrm{SS_{total}}} = 1 - \frac{\mathrm{SS_{residual}}}{\mathrm{SS_{total}}}$$

$$X_1 = a_0 + a_1 X_2 \qquad a_1 = \frac{\sum (X_2 - \bar{X}_2)(X_1 - \bar{X}_1)}{\sum (X_2 - \bar{X}_2)^2}$$

$$\mathrm{Cov}(X_1, X_2) = \frac{1}{n} \sum (X_1 - \bar{X}_1)(X_2 - \bar{X}_2)$$

$$a_1 = \frac{\mathrm{Cov}(X_1, X_2)}{\mathrm{Var}(X_2)} \qquad \mathrm{Var}(X_2) = \frac{1}{n} \sum (X_2 - \bar{X}_2)^2$$

$$\hat{X}_1 = a_0 + a_1 X_2$$

$$\hat{X}_1 - \bar{X}_1 = a_1 (X_2 - \bar{X}_2)$$

$$SS_{\text{explained}} = \sum (\hat{X}_1 - \bar{X}_1)^2 = a_1^2 \sum (X_2 - \bar{X}_2)^2$$

$$SS_{\text{total}} = \sum (X_1 - \bar{X}_1)^2$$

$$R_1^2 = \frac{a_1^2 \sum (X_2 - \bar{X}_2)^2}{\sum (X_1 - \bar{X}_1)^2}$$

$$R_1^2 = \frac{\mathrm{Cov}(X_1, X_2)^2}{\mathrm{Var}(X_1)\mathrm{Var}(X_2)} \qquad \mathrm{Corr}(X_1, X_2) = \frac{\mathrm{Cov}(X_1, X_2)}{\sqrt{\mathrm{Var}(X_1)\mathrm{Var}(X_2)}}$$

$$R_1^2 = (\mathrm{Corr}(X_1, X_2))^2$$

Multiple linear regression : Multicollinearity

Exercice: A researcher wants to study how study habits affect students' final exam scores. The dataset for 8 students is as follows:

Student	Y: Exam Score	X_1: Hours Studied	X_2: Hours in Review Sessions
1	88	10	2
2	92	12	3
3	85	8	1
4	95	14	4
5	80	7	1
6	89	11	3
7	78	6	1
8	94	13	4

1. **Fit a multiple regression model:** $Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \epsilon$.
2. **Calculate the correlation** between X_1 and X_2.
3. **Compute the Variance Inflation Factor (VIF)** for X_1 and X_2:
   - $\mathrm{VIF}_j = \frac{1}{1-R_j^2}$, where R_j^2 comes from regressing predictor X_j on the other predictor.
4. **Interpret the results:**
   - Are X_1 and X_2 strongly correlated?
   - Do the VIF values indicate multicollinearity?

Multiple linear regression : Multicollinearity

Solution:

1) $\boldsymbol{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}$

$$\mathbf{Y} = \begin{bmatrix} 88 \\ 92 \\ 85 \\ 95 \\ 80 \\ 89 \\ 78 \\ 94 \end{bmatrix}, \quad \mathbf{X} = \begin{bmatrix} 1 & 10 & 2 \\ 1 & 12 & 3 \\ 1 & 8 & 1 \\ 1 & 14 & 4 \\ 1 & 7 & 1 \\ 1 & 11 & 3 \\ 1 & 6 & 1 \\ 1 & 13 & 4 \end{bmatrix}, \quad \boldsymbol{\beta} = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \end{bmatrix}$$

$$\mathbf{X}^\top \mathbf{X} = \begin{bmatrix} 8 & 81 & 19 \\ 81 & 881 & 236 \\ 19 & 236 & 59 \end{bmatrix}$$

$$\mathbf{X}^\top \mathbf{Y} = \begin{bmatrix} 701 \\ 7235 \\ 1940 \end{bmatrix}$$

$$(X^\top X)^{-1} \approx \begin{bmatrix} 60.91 & -3.13 & 2.10 \\ -3.13 & 0.76 & -0.47 \\ 2.10 & -0.47 & 0.48 \end{bmatrix}$$

$$Y = 60.91 + 3.13X_1 - 2.10X_2$$

Student	Y: Exam Score	X_1: Hours Studied	X_2: Hours in Review Sessions
1	88	10	2
2	92	12	3
3	85	8	1
4	95	14	4
5	80	7	1
6	89	11	3
7	78	6	1
8	94	13	4

Multiple linear regression : Multicollinearity

Solution:

2) $$r_{X_1,X_2} = \frac{\mathrm{Cov}(X_1, X_2)}{\sigma_{X_1}\sigma_{X_2}}$$

$$\sigma_{X_1} = \sqrt{\frac{\sum(X_1 - \bar{X}_1)^2}{n-1}} \approx 2.90$$

$$\sigma_{X_2} = \sqrt{11.874/7} = \sqrt{1.696} \approx 1.30$$

$$\mathrm{Cov}(X_1, X_2) = \frac{\sum_{i=1}^{n}(X_{1i} - \bar{X}_1)(X_{2i} - \bar{X}_2)}{n-1} = 25.625/7$$

Solution:

3) $$\mathrm{VIF}_{X_1} = \frac{1}{1 - R_1^2}, \quad \mathrm{VIF}_{X_2} = \frac{1}{1 - R_2^2}$$

$$R_1^2 = r^2_{X_1,X_2}$$

$$\mathrm{VIF}_{X_1} = \frac{1}{1 - 0.94} = \frac{1}{0.06}$$ =16.6

$$R_2^2 = r^2_{X_2,X_1} = r^2_{X_1,X_2}$$

$$R_1^2 = \frac{\mathrm{Cov}(X_1, X_2)^2}{\mathrm{Var}(X_1)\mathrm{Var}(X_2)}$$

Multiple linear regression : Multicollinearity

Exercice:

1. Write the estimated regression equation using the reported coefficient estimates.

2. Does the 95% confidence interval for X2 include zero?
   a) State the lower and upper bounds.
   b) Explain what this implies about the statistical significance of X2.
3. Based on the regression output, is there evidence of a relationship between the dependent variable and the explanatory variables? Explain using the appropriate test statistic.
4. Are all individual coefficients statistically significant? Explain your answer.

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.855921721
R Square	0.732601993
Adjusted R Square	0.662845991
Standard Error	7.067993765
Observations	30

ANOVA

	df	*SS*	*MS*	*F*	*Significance F*
Regression	6	3147.966342	524.6611	10.50235	1.24041E-05
Residual	23	1149.000325	49.95654		
Total	29	4296.966667			

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*Lower 99.0%*	*Upper 99.0%*
Intercept	10.78707639	11.58925724	0.930782	0.361634	-13.18712868	34.76128	-21.747859	43.32201173
X1	0.613187608	0.160983115	3.809018	0.000903	0.280168664	0.946207	0.161254	1.06512125
X2	-0.073050143	0.13572469	-0.538223	0.595594	-0.353818055	0.207718	-0.4540749	0.307974622
X3	0.320332116	0.168520319	1.900852	0.069925	-0.028278721	0.668943	-0.152761	0.793425219
X4	0.081732134	0.221477677	0.369031	0.71548	-0.376429347	0.539894	-0.5400301	0.703494319
X5	0.038381447	0.146995442	0.261106	0.796334	-0.265701791	0.342465	-0.3742841	0.451046997
X6	-0.217056682	0.178209471	-1.217986	0.235577	-0.585711058	0.151598	-0.7173505	0.283237125

Y = Overall rating of supervisor

X_1 = Handles employee complaints

X_2 = Does not allow special privileges

X_3 = Opportunity to learn new things

X_4 = Raises based on performance

X_5 = Too critical of poor performance

X_6 = Rate of advancing to better jobs

Multiple linear regression : Multicollinearity

Solution:

1. Write the estimated regression equation using the reported coefficient estimates.

$$\hat{Y} = 10.7871 + 0.6132X_1 - 0.0731X_2 + 0.3203X_3 + 0.0817X_4 + 0.0384X_5 - 0.2171X_6$$

1. Does the 95% confidence interval for X2 include zero?
   a) State the lower and upper bounds. CI: $[-0.3538,\ 0.2077]$
   b) Explain what this implies about the statistical significance of X2.
   Include 0 -> X_2 is not statistically significant at 5% (fail to reject $H_0: \beta_2 = 0$)

Based on the regression output, is there evidence of a relationship between the dependent variable and the explanatory variables? Explain using the appropriate test statistic.

Yes — **overall F-test is significant**: $F = 10.502$, Significance F $= 1.24 \times 10^{-5}$ →reject "no relationship" at 5%.

Are all individual coefficients statistically significant? Explain your answer.

No. **Only** X_1 is significant at 5% (p = 0.000903). X_3 is marginal (p ≈ 0.0699). Others are not significant.

Analysis of Categorical Data

Session 7

Analysis of Categorical Data

Session 7

Objectives:

- include categorical variables with quantitative predictors in a single regression model and interpret coefficients.

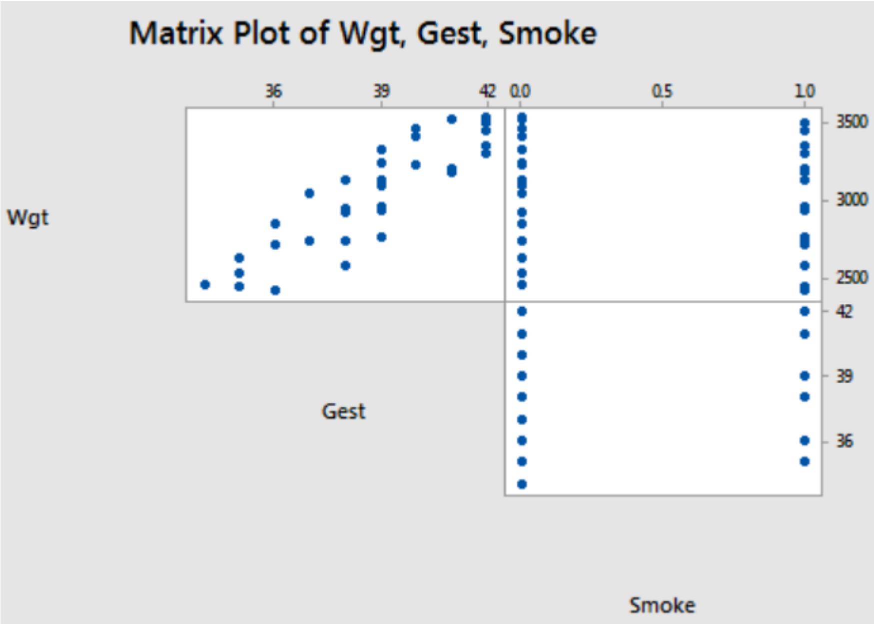
Sources:

- Applied-Statistics-and-Probability-for-Engineers-Douglas-C.-Montgomery-George-C.-Runger-Edisi-5-2011
- Lesson 8: Categorical Predictors | STAT 501

Analysis of Categorical Data

Example: Researchers interested in answering about the relationship:

- Response (*y*): birth weight (**Weight**) in grams of baby
- Potential predictor **Smoking** status of mother (yes or no)
- Potential predictor length of gestation (**Gest**) in weeks



- Positive linear relationship between the length of gestation and birth weight: as the length of gestation increases, the birth weight of babies tends to increase.
- However, It is hard to see if any kind of relationship exists between birth weight and smoking status, or between the length of gestation and smoking status

Analysis of Categorical Data

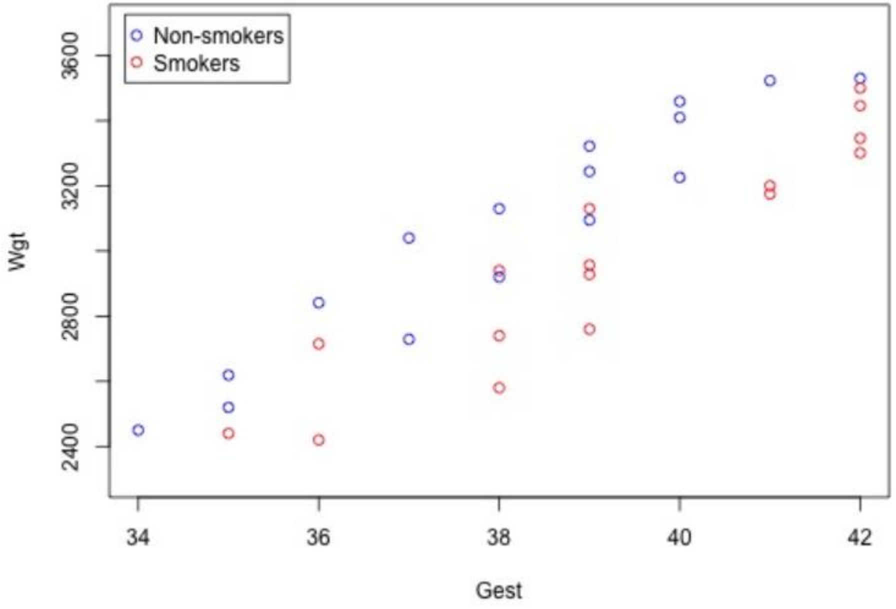
A "**binary predictor**" is a variable that takes on only two possible values. Here are a few common examples of binary predictor variables that you are likely to encounter in your own research:

- Gender (male, female)
- Smoking status (smoker, nonsmoker)
- Treatment (yes, no)
- Health status (diseased, healthy)
- Company status (private, public)

To include a qualitative variable in a regression model, we have to "**code**" the variable, that is, assign a unique number to each of the possible categories. A common coding scheme is to use what's called a "**zero-one-indicator variable**." Using such a variable here, we code the binary predictor **Smoking** as:

X_{i2}=1 if mother *i* smokes

X_{i2}=0 if mother *i* does not smoke



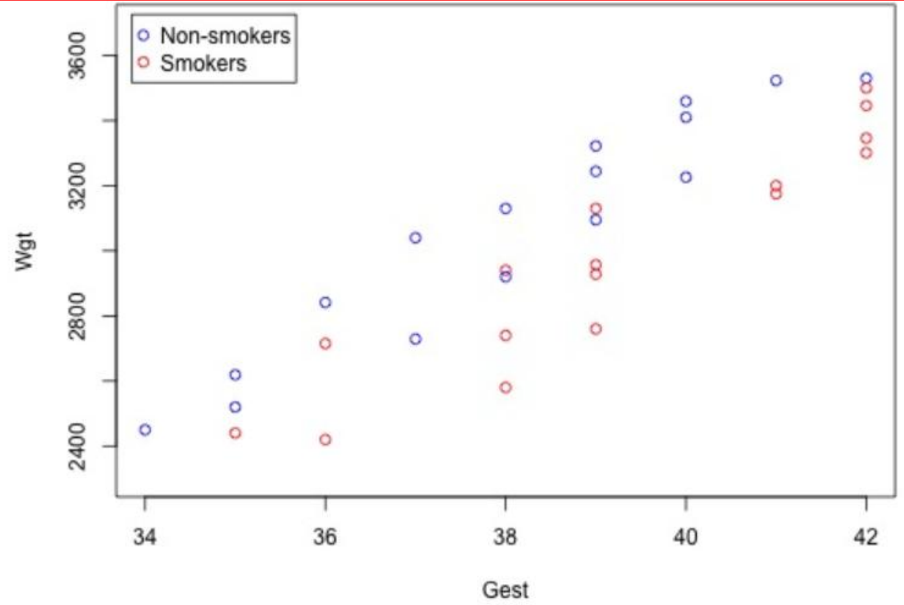
Analysis of Categorical Data

There might be two distinct linear trends in the data : one for smoking mothers and one for non-smoking mothers. Therefore, a **first-order model** with **one binary and one quantitative predictor** appears to be a natural model to formulate for these data. That is:

$$y_i = (\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2}) + \epsilon_i$$

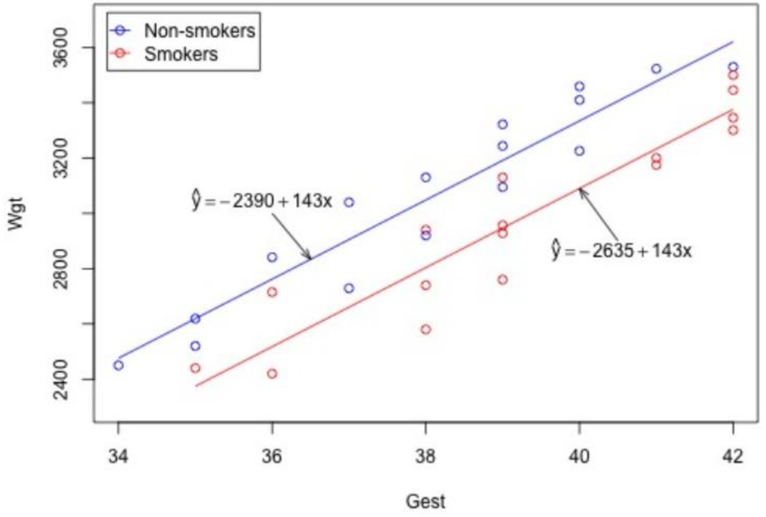
where:

- y_i is the birth weight of baby *i* in *grams*
- x_{i1} is the length of gestation of baby *i* in *weeks*
- $x_{i2} = 1$, if the mother smoked during pregnancy, and $x_{i2} = 0$, if she did not



Therefore, you see 2 differents models/fits because

$$\mu_Y = \beta_0 + \beta_1 x_{i1} \qquad \mu_Y = (\beta_0 + \beta_2) + \beta_1 x_{i1}$$



So you have the same slope but different intercept

Analysis of Categorical Data

Why Not Two Separate Regressions?

Because having a single model has advantages:

- Uses all data. This implies that confidence intervals be narrower
- Allows hypothesis testing $H_0 : \beta_2 = 0$ If rejected → groups differ
- Simpler interpretation

Impact of using a different coding

In the birth weight example, we coded the qualitative variable **Smoking** by creating a (0, 1) indicator variable that took on the value 1 for smoking mothers and 0 for non-smoking mothers. What if we had instead used (1, -1) coding?

- $x_{i2} = 1$, if the mother smokes
- $x_{i2} = -1$, if mother does not smoke

You can determine the meaning of any regression coefficient by seeing what effect changing the value of the predictor has on the mean response.

No smoking:

$$\mu_Y = (\beta_0 - \beta_2) + \beta_1 x_{i1}$$

Smoking:

$$\mu_Y = (\beta_0 + \beta_2) + \beta_1 x_{i1}$$

Conclusions:

- Intercept and group coefficient change numerically
- Slopes and predictions do not change
- Only interpretation changes

Impact of using a different coding: example

Imagine, you have 2 groups with 2 predictors (age and smoker). You want to predict lung capacity. You have therefore two slopes:

- Non-smoker: y=50 +2*Age
- Smoker: y=40+2*Age

Difference of 10 years.

(0,1) Coding (NS,S)

$y=\beta 0+\beta 1 * Age+\beta 2*D$

Non smoker: $y=50+2*Age$ -> $\beta 0=50$, $\beta 1=2$
Smoker: $y=40+2*Age= \beta 0+\beta 1 * Age+\beta 2=50+2*Age+ \beta 2$
-> $\beta 2=-10$

Model final $y=50+2Age-10D$

Prediction (50yrs, smoker):
$y=50+2*50-10=140$

(1,-1) Coding (NS,S)

$y=\alpha 0+\alpha 1 * Age+\alpha 2 * Z$

Non smoker: $y=50+2*Age=\alpha 0+\alpha 1 * Age+\alpha 2$
Smoker: $y=40+2*Age =\alpha 0+\alpha 1 * Age-\alpha 2$
-> $(\alpha 0 + \alpha 2)=50$ and $(\alpha 0 - \alpha 2)=40$
-> $2*\alpha 0=90$ -> $\alpha 0=45$ and $\alpha 2=5$

Model final: $y=45+2Age+5Z$

Prediction (50yrs, smoker):
$y=45+2*50+5*(-1)=45+100-5=140$

Smoker: 40+2*Age=140 : **SLOPE and PREDICTION ARE THE SAME**

What happens if you have multiple variables

What if my categorical variable contains three levels (or more)?

For example, restaurant rate: Good, medium, bad

Maybe I can do Good=2, medium=1, bad=0?

NO THIS WOULD RESULT IN COLINEARITY!!

Solution is to set up a series of dummy variable. In general, for k levels, you need k-1 dummy variables

$$D_1 = \begin{cases} 1 & \text{Medium} \\ 0 & \text{otherwise} \end{cases}$$

$$D_2 = \begin{cases} 1 & \text{Good} \\ 0 & \text{otherwise} \end{cases}$$

Model:

$$y = \beta_0 + \beta_1 x + \beta_2 D_1 + \beta_3 D_2$$

Bad:

$$y = \beta_0 + \beta_1 x$$

Medium:

$$y = \beta_0 + \beta_1 x + \beta_2$$

Good:

$$y = \beta_0 + \beta_1 x + \beta_3$$

$$x_1 = \begin{cases} 1 \text{ if sales region B} \\ 0 \text{ otherwise} \end{cases} \quad x_2 = \begin{cases} 1 \text{ if sales region C} \\ 0 \text{ otherwise} \end{cases} \quad x_3 = \begin{cases} 1 \text{ if sales region D} \\ 0 \text{ otherwise} \end{cases}$$

What happens if you have multiple variables

Exemple:

Consider a regression study involving a dependent variable y, a quantitative independent variable x1 , and a categorical independent variable with three possible levels (level 1, level 2, and level 3).

- How many dummy variables are required to represent the categorical variable?
- Write a multiple regression equation relating x1 and the categorical variable to y.
- Interpret the parameters in your regression equation

What happens if you have multiple variables

Solution:

. Consider a regression study involving a dependent variable y, a quantitative independent variable x1 , and a categorical independent variable with three possible levels (level 1, level 2, and level 3).

- How many dummy variables are required to represent the categorical variable?

You have three level, therefore you need 2 dummy variables.

- Write a multiple regression equation relating x1 and the categorical variable to y.

$$D_2 = \begin{cases} 1 & \text{if level 2} \\ 0 & \text{otherwise} \end{cases} \qquad D_3 = \begin{cases} 1 & \text{if level 3} \\ 0 & \text{otherwise} \end{cases} \qquad y = \beta_0 + \beta_1 x_1 + \beta_2 D_2 + \beta_3 D_3 + \varepsilon$$

- Interpret the parameters in your regression equation

•β0: expected value of y when x1=0 for level 1.
•β1: expected change in y for a 1-unit increase in x1, holding the categorical level fixed (same slope across levels in this additive model).
•β2: difference in expected y between level 2 and level 1, at the same x1. (Shift in intercept for level 2 vs level 1.)
•β3: difference in expected y between level 3 and level 1, at the same x1. (Shift in intercept for level 3 vs level 1.)

Level 1: $E(y) = \beta_0 + \beta_1 x_1$
Level 2: $E(y) = (\beta_0 + \beta_2) + \beta_1 x_1$
Level 3: $E(y) = (\beta_0 + \beta_3) + \beta_1 x_1$

What happens if you have multiple variables

Example:

Management proposed the following regression model to predict sales at a fast-food outlet.

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \epsilon$$

where

$$x_1 = \text{number of competitors within one mile}$$
$$x_2 = \text{population within one mile (1000s)}$$
$$x_3 = \begin{cases} 1 \text{ if drive-up window present} \\ 0 \text{ otherwise} \end{cases}$$
$$y = \text{sales (\$1000s)}$$

The following estimated regression equation was developed after 20 outlets were surveyed.

$$\hat{y} = 10.1 - 4.2x_1 + 6.8x_2 + 15.3x_3$$

a. What is the expected amount of sales attributable to the drive-up window?
b. Predict sales for a store with two competitors, a population of 8000 within one mile, and no drive-up window.
c. Predict sales for a store with one competitor, a population of 3000 within one mile, and a drive-up window.

What happens if you have multiple variables

Solution:

a) The coeff of x3 is 15.3. Therefore, it is the difference in expected sales between stores WITH and WITHOUT drive-up. Having a drive-up window increases expected sales by 15.3k USD.

b) Two competitors (x1=2), a population of 8000 (x2=8) within one mile, and no drive-up window (x3=0).

y=10.1-4.2*2+6.8*8+15.3*0=56.1k

c) One competitor (x1=1), a population of 3000 (x2=3) within one mile, and with drive-up window (x3=1).

y=10.1-4.2*1+6.8*3+15.3*1=41.6k

$$\hat{y} = 10.1 - 4.2x_1 + 6.8x_2 + 15.3x_3$$

x_1 = number of competitors within one mile

x_2 = population within one mile (1000s)

$$x_3 = \begin{cases} 1 \text{ if drive-up window present} \\ 0 \text{ otherwise} \end{cases}$$

y = sales ($1000s)

What happens if you have multiple variables

Example:

Repair Time in Hours	Months Since Last Service	Type of Repair	Repairperson
2.9	2	Electrical	Dave Newton
3.0	6	Mechanical	Dave Newton
4.8	8	Electrical	Bob Jones
1.8	3	Mechanical	Dave Newton
2.9	2	Electrical	Dave Newton
4.9	7	Electrical	Bob Jones
4.2	9	Mechanical	Bob Jones
4.8	8	Mechanical	Bob Jones
4.4	4	Electrical	Bob Jones
4.5	6	Electrical	Dave Newton

a. Ignore for now the months since the last maintenance service (x_1) and the repairperson who performed the service. Develop the estimated simple linear regression equation to predict the repair time (y) given the type of repair (x_2). Recall that $x_2 = 0$ if the type of repair is mechanical and 1 if the type of repair is electrical.

b. Does the equation that you developed in part (a) provide a good fit for the observed data? Explain.

c. Ignore for now the months since the last maintenance service and the type of repair associated with the machine. Develop the estimated simple linear regression equation to predict the repair time given the repairperson who performed the service. Let $x_3 = 0$ if Bob Jones performed the service and $x_3 = 1$ if Dave Newton performed the service.

d. Does the equation that you developed in part (c) provide a good fit for the observed data? Explain.

What happens if you have multiple variables

Repair Time in Hours	Months Since Last Service	Type of Repair	Repairperson
2.9	2	Electrical	Dave Newton
3.0	6	Mechanical	Dave Newton
4.8	8	Electrical	Bob Jones
1.8	3	Mechanical	Dave Newton
2.9	2	Electrical	Dave Newton
4.9	7	Electrical	Bob Jones
4.2	9	Mechanical	Bob Jones
4.8	8	Mechanical	Bob Jones
4.4	4	Electrical	Bob Jones
4.5	6	Electrical	Dave Newton

Solution:

a) $x_2 = \begin{cases} 0 & \text{Mechanical} \\ 1 & \text{Electrical} \end{cases}$ $\quad y = \beta_0 + \beta_2 x_2 + \varepsilon$

$$\hat{\beta}_0 = \bar{y}_{\text{Mech}} \qquad \bar{y}_{\text{Mech}} = \frac{3.0 + 1.8 + 4.2 + 4.8}{4} = \frac{13.8}{4} = 3.45$$

$$\hat{\beta}_2 = \bar{y}_{\text{Elec}} - \bar{y}_{\text{Mech}} = 4.0667 - 3.45 = 0.6167$$

$$\hat{y} = 3.45 + 0.6167\,x_2$$

b) $SST = \sum (y_i - \bar{y})^2 = 10.476$

$$SSE = \sum (y_i - \hat{y}_i)^2 = 9.5633$$

$$R^2 = 1 - \frac{SSE}{SST} = 1 - \frac{9.5633}{10.476} \approx 0.087$$

So only about **8.7%** of the variation in repair time is explained by repair type alone.

c) $x_3 = \begin{cases} 0 & \text{Bob Jones} \\ 1 & \text{Dave Newton} \end{cases}$ $\quad y = \beta_0 + \beta_3 x_3 + \varepsilon$

$$\bar{y}_{\text{Bob}} = \frac{4.8 + 4.9 + 4.2 + 4.8 + 4.4}{5} = \frac{23.1}{5} = 4.62$$

$$\bar{y}_{\text{Dave}} = \frac{2.9 + 3.0 + 1.8 + 2.9 + 4.5}{5} = \frac{15.1}{5} = 3.02$$

$$\hat{\beta}_0 = \bar{y}_{\text{Bob}} = 4.62$$

$$\hat{\beta}_3 = \bar{y}_{\text{Dave}} - \bar{y}_{\text{Bob}} = 3.02 - 4.62 = -1.60$$

$$\hat{y} = 4.62 - 1.60\,x_3$$

What happens if you have multiple variables

Solution:

d) $\hat{y} = 4.62 - 1.60\,x_3$

$$SST = \sum(y_i - \bar{y})^2 = 10.476$$

$$SSE = \sum(y_i - \hat{y}_i)^2 = 4.076$$

$$R^2 = 1 - \frac{4.076}{10.476} \approx 0.611$$

So repairperson alone explains about 61.1% of variability, it’s a fairly good fit for a one-variable model

Repair Time in Hours	Months Since Last Service	Type of Repair	Repairperson
2.9	2	Electrical	Dave Newton
3.0	6	Mechanical	Dave Newton
4.8	8	Electrical	Bob Jones
1.8	3	Mechanical	Dave Newton
2.9	2	Electrical	Dave Newton
4.9	7	Electrical	Bob Jones
4.2	9	Mechanical	Bob Jones
4.8	8	Mechanical	Bob Jones
4.4	4	Electrical	Bob Jones
4.5	6	Electrical	Dave Newton

What happens if you have multiple variables

Repair Time in Hours	Months Since Last Service	Type of Repair	Repairperson
2.9	2	Electrical	Dave Newton
3.0	6	Mechanical	Dave Newton
4.8	8	Electrical	Bob Jones
1.8	3	Mechanical	Dave Newton
2.9	2	Electrical	Dave Newton
4.9	7	Electrical	Bob Jones
4.2	9	Mechanical	Bob Jones
4.8	8	Mechanical	Bob Jones
4.4	4	Electrical	Bob Jones
4.5	6	Electrical	Dave Newton

Example:
This problem is an extension of the situation described in the previous exercice

a. Develop the estimated regression equation to predict the repair time given the number of months since the last maintenance service, the type of repair, and the repairperson who performed the service.

b. Interpret the the fitting parameters: $\hat{y} = 2.9626 + 0.2914\,(\text{Months}) - 1.1024\,(\text{Mechanical}) - 0.6091\,(\text{Dave})$

c. At the .05 level of significance, test whether the estimated regression equation developped in part (a) represents a significant relationship between the independent variables and the dependent variable.

d. Is the addition of the independent variable x3, the repairperson who performed the service, statistically significant? Use α = 0.05. What explanation can you give for the results observed? $\hat{y} = 2.6459 + 0.3053\,(\text{Months}) - 1.1347\,(\text{Mechanical})$

What happens if you have multiple variables

Solution:

a) We have 3 predictors:

- x1 = months since last service (quantitative)
- x2= type of repair
  - 0 = mechanical
  - 1 = electrical
- x3 = repairperson
  - 0 = Bob Jones
  - 1 = Dave Newton

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \varepsilon$$

Then, using computer tools you can derive the free parameters

$$\hat{y} = 1.8602 + 0.2914x_1 + 1.1024x_2 - 0.6091x_3$$

Repair Time in Hours	Months Since Last Service	Type of Repair	Repairperson
2.9	2	Electrical	Dave Newton
3.0	6	Mechanical	Dave Newton
4.8	8	Electrical	Bob Jones
1.8	3	Mechanical	Dave Newton
2.9	2	Electrical	Dave Newton
4.9	7	Electrical	Bob Jones
4.2	9	Mechanical	Bob Jones
4.8	8	Mechanical	Bob Jones
4.4	4	Electrical	Bob Jones
4.5	6	Electrical	Dave Newton

b)

- b1=0.2914: each additional month since service increases expected repair time by 0.291 hours, holding repair type and person fixed.
- b2=1.1024b_2=1.1024b2=1.1024: electrical repairs take 1.102 hours longer than mechanical, holding months and person fixed.
- b3=−0.6091b_3=-0.6091b3=−0.6091: Dave’s repairs are 0.609 hours shorter than Bob’s, holding months and type fixed (but we will test if this is significant

What happens if you have multiple variables

Solution:

c) $\hat{y} = 1.8602 + 0.2914x_1 + 1.1024x_2 - 0.6091x_3$

$$H_0 : \beta_1 = \beta_2 = \beta_3 = 0$$

$$H_a : \text{at least one } \beta_j \neq 0$$

Repair Time in Hours	Months Since Last Service	Type of Repair	Repairperson
2.9	2	Electrical	Dave Newton
3.0	6	Mechanical	Dave Newton
4.8	8	Electrical	Bob Jones
1.8	3	Mechanical	Dave Newton
2.9	2	Electrical	Dave Newton
4.9	7	Electrical	Bob Jones
4.2	9	Mechanical	Bob Jones
4.8	8	Mechanical	Bob Jones
4.4	4	Electrical	Bob Jones
4.5	6	Electrical	Dave Newton

$$\bar{y} = \frac{1}{10}\sum y_i = \frac{38.2}{10} = 3.82$$

$$SST = \sum (y_i - \bar{y})^2 = 10.476$$

$$\mathrm{SSE} = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 = 1.0455$$

$$SSR = SST - SSE = 10.476 - 1.0455 = 9.4305$$

$$MSR = \frac{SSR}{3} = \frac{9.4305}{3} = 3.1435$$

$$MSE = \frac{SSE}{6} = \frac{1.0455}{6} = 0.17425$$

$n = 10$

number of predictors $p = 3$

df(regression) $= p = 3$

df(error) $= n - p - 1 = 10 - 3 - 1 = 6$

df(total) $= n - 1 = 9$

$$F = \frac{MSR}{MSE} = \frac{3.1435}{0.17425} = 18.040$$

0.05

df₁/df₂	1	2	3	4	5	6
1	161.448	18.513	10.128	7.709	6.608	5.987
2	199.500	19.000	9.552	6.944	5.786	5.143
3	215.707	19.164	9.277	6.591	5.409	4.757

18.04>4.757⇒Reject H0

The regression is significant at the 0.05 level (there is a significant relationship between the predictors and repair time).

What happens if you have multiple variables

Solution:

d) $\hat{y} = 1.8602 + 0.2914x_1 + 1.1024x_2 - 0.6091x_3$

$$F = \frac{MSR}{MSE} \qquad \text{SSE} = \sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

$$H_0 : \beta_3 = 0 \qquad H_a : \beta_3 \neq 0$$

$$\hat{y} = 2.9626 + 0.2914\,(\text{Months}) - 1.1024\,(\text{Mechanical}) - 0.6091\,(\text{Dave})$$

$$\hat{y} = 2.6459 + 0.3053\,(\text{Months}) - 1.1347\,(\text{Mechanical})$$

The idea is to compare a the full model with (4 predictors) and the restricted model without x3 (3 predictors).
If x3 has no effect, the difference of SSE between the full (F) and the restricted (R) should be small!!

We have 10 observations, therefore, dof for the full model is 10-4=6 and fore the without 10-3=7. Between both model you have 1dof diffrence.

So we look at: $F = \frac{(SSE_R - SSE_F)/df_1}{MSE_F}$, we compare the signal gained by adding the variable to the noise level of the model

$$\frac{SSE_R - SSE_F}{1} = 0.4296$$

$$MSE_F = \frac{SSE_F}{df_{E,F}} = \frac{1.0455}{6}$$

$$F = 2.465$$

0.05

df₁/df₂	1	2	3	4	5	6
1	161.448	18.513	10.128	7.709	6.608	5.987

$$2.465 < 5.987$$

Fail to reject H0. At the 0.05 level of significance:

The addition of x3 (repairperson) is NOT statistically significant.

What happens if you have multiple variables

Example:
A 10-year study conducted by the American Heart Association provided data on how age, blood pressure, and smoking relate to the risk of strokes. Assume that the following data are from a portion of this study. Risk is interpreted as the probability (times 100) that the patient will have a stroke over the next 10-year period. For the smoking variable, define a dummy variable with 1 indicating a smoker and 0 indicating a nonsmoker.

- Develop an estimated regression equation that relates risk of a stroke to the person's age, blood pressure, and whether the person is a smoker.

$$\hat{\beta}_0 = -91.7595, \quad \hat{\beta}_1 = 1.07674, \quad \hat{\beta}_2 = 0.25181, \quad \hat{\beta}_3 = 8.73987$$

- Is smoking a significant factor in the risk of a stroke? Explain. Use α=0 .05.

$$\hat{y} = -110.9423 + 1.31500\,\text{Age} + 0.29640\,\text{Pressure}$$

- What is the probability of a stroke over the next 10 years for Art Speen, a 68-year-old smoker who has blood pressure of 175? What action might the physician recommend for this patient?

Risk	**Age**	**Pressure**	**Smoker**
12	57	152	No
24	67	163	No
13	58	155	No
56	86	177	Yes
28	59	196	No
51	76	189	Yes
18	56	155	Yes
31	78	120	No
37	80	135	Yes
15	78	98	No
22	71	152	No
36	70	173	Yes
15	67	135	Yes
48	77	209	Yes
15	60	199	No
36	82	119	Yes
8	66	166	No
34	80	125	Yes
3	62	117	No
37	59	207	Yes

What happens if you have multiple variables

Solution:

- Develop an estimated regression equation that relates risk of a stroke to the person's age, blood pressure, and whether the person is a smoker.

$$Y = \beta_0 + \beta_1(\text{Age}) + \beta_2(\text{Pressure}) + \beta_3(\text{Smoker}) + \varepsilon$$

$$X_i = [1,\ \text{Age}_i,\ \text{Pressure}_i,\ \text{Smoker}_i] \qquad \hat{\beta} = (X'X)^{-1}X'y$$

$$X'X = \begin{pmatrix} \sum 1 & \sum x_1 & \sum x_2 & \sum x_3 \\ \sum x_1 & \sum x_1^2 & \sum x_1x_2 & \sum x_1x_3 \\ \sum x_2 & \sum x_1x_2 & \sum x_2^2 & \sum x_2x_3 \\ \sum x_3 & \sum x_1x_3 & \sum x_2x_3 & \sum x_3^2 \end{pmatrix}$$

$$X'X = \begin{pmatrix} 20 & 1389 & 3142 & 10 \\ 1389 & 98223 & 216370 & 733 \\ 3142 & 216370 & 513838 & 1624 \\ 10 & 733 & 1624 & 10 \end{pmatrix}$$

$$(X'X)^{-1}$$

$$\begin{pmatrix} 6.992919628 & -0.0689598706 & -0.0156070627 & 0.5964258683 \\ -0.0689598706 & 0.0008311846 & 0.0000950867 & -0.0074080404 \\ -0.0156070627 & 0.0000950867 & 0.0000617218 & -0.0013864191 \\ 0.5964258683 & -0.0074080404 & -0.0013864191 & 0.2717379534 \end{pmatrix}$$

$$X'y = \begin{pmatrix} \sum y \\ \sum x_1y \\ \sum x_2y \\ \sum x_3y \end{pmatrix} = \begin{pmatrix} 539 \\ 39198 \\ 88251 \\ 368 \end{pmatrix}$$

$$\hat{\beta}_0 = -91.7595, \quad \hat{\beta}_1 = 1.07674, \quad \hat{\beta}_2 = 0.25181, \quad \hat{\beta}_3 = 8.73987$$

Risk	Age	Pressure	Smoker
12	57	152	0
24	67	163	0
13	58	155	0
56	86	177	1
28	59	196	0
51	76	189	1
18	56	155	1
31	78	120	0
37	80	135	1
15	78	98	0
22	71	152	0
36	70	173	1
15	67	135	1
48	77	209	1
15	60	199	0
36	82	119	1
8	66	166	0
34	80	125	1
3	62	117	0
37	59	207	1

What happens if you have multiple variables

Solution:

b) $\hat{y} = -91.7595 + 1.07674(\text{Age}) + 0.25181(\text{Pressure}) + 8.73987(\text{Smoker})$

$\hat{y} = -110.9423 + 1.31500\,\text{Age} + 0.29640\,\text{Pressure}$

$H_0 : \beta_3 = 0 \qquad H_a : \beta_3 \neq 0$

$$F = \frac{MSR}{MSE}$$

$$\text{SSE} = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

The idea is to compare a the full model with (4 predictors) and the restricted model without x3 (3 predictors).
If x3 has no effect, the difference of SSE between the full (F) and the restricted (R) should be small!!

$$F = \frac{(SSE_R - SSE_F)/q}{SSE_F/(n-k)}$$

SSE_R = SSE from reduced model

SSE_F = SSE from full model

q = number of restrictions (here 1)

k = number of parameters in full model (4)

$n = 20$

$$F = \frac{(811.3097 - 530.2104)/1}{530.2104/16}$$

$F \approx 8.48$

Degrees of freedom 1: 1

Degrees of freedom 2: 16

Probability level: 0.05

Calculate!

Critical F-value: 4.49399842

$8.25 > 4.49$

Reject H0. At the 0.05 level of significance:

The addition of x3 (smoker) is statistically significant.

What happens if you have multiple variables

Solution:

c) What is the probability of a stroke over the next 10 years for Art Speen, a 68-year-old smoker who has blood pressure of 175? What action might the physician recommend for this patient?

$$\hat{y} = -91.7595 + 1.07674(\text{Age}) + 0.25181(\text{Pressure}) + 8.73987(\text{Smoker})$$

$$\hat{y} = -91.7595 + 1.07674(68) + 0.25181(175) + 8.73987(1)$$

$$\hat{y} \approx 34.27$$

Bayesian inference

Session 8

Bayesian inference

Session 8

Objectives:

- Difference between frequentist and Bayesian
- Apply Bayesian inference for linear regression

Bayesian vs frequentist

There are two main approaches in statistics: frequentist (or classical) methods and Bayesian methods. Most of the methods we have discussed so far are frequentist. It is important to understand both approaches.

Bayesian vs frequentist

Frequentist statistics is based on the idea of long-run frequencies. It assumes that repeated sampling from a population will yield a stable estimate of parameters. Key concepts include:

- **Point Estimation:** Frequentist methods provide point estimates of parameters: single values that serve as the best estimate of the true population parameter.
- **Confidence Intervals:** These intervals quantify the uncertainty associated with point estimates by stating a range within which the true parameter is expected to lie.
- **Hypothesis Testing:** This involves making inferences about populations based on sample data, commonly using p-values to determine significance.

Frequentist methods are often seen as more objective, focusing on properties of estimators based on repeated sampling **(solely on observed data)**.

Bayes theorem

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

- P(A|B) is a conditional probability: the probability of event A occurring given that B is true.
- P(A|B) is is also a conditional probability: the probability of event B occurring given that A is true
- P(A) and P(B) are the probabilities of observing A and B respectively

Example:
A certain disease affects about 1out of 10,000 people. There is a test to check whether the person has the disease. The test is quite accurate. In particular, we know that the probability that the test result is positive, given that the person does not have the disease, is only 2 percent; the probability that the test result is negative, given that the person has the disease, is only 1 percent.

A random person gets tested for the disease and the result comes back positive. What is the probability that the person has the disease?

Bayes Theorem

Solution:

D=event that the person has the disease
T= event that the test result is true/positive.

P(D)=1/10000
P(T|Dc)=0.02
P(Tc|D)=0.01 -> P(T|D)=0.99

We want P(D|T) ?

Bayes' Theorem:

P(D|T)=P(T|D)*P(D)/P(T)

P(T)=P(T|D)*P(D)+P(T|Dc)*P(Dc)=0.99*1/10000+0.02*9999/10000= 0.020097

So: P(D|T)=0.99*1/10000/ 0.020097= 0.0049=**0.5%**

$$P(B) = \sum_{i=1}^{k} P(B \mid E_i)\, P(E_i).$$

$$P(B|A) = \frac{P(A|B)P(B)}{P(A)}.$$

Bayes theorem: interpretation

For Bayesian linear regression, we want to understand how model parameters θ relate to observed data $X.$ The goal is to solve the probability of the observed set of parameter values, given the set of input variables X

$$\underbrace{P(\theta|X)}_{\text{A posterior probability}} = \frac{\overbrace{P(X|\theta)}^{\text{Likelihood}}\overbrace{p(\theta)}^{\text{A prior probability}}}{\underbrace{P(X)}_{\text{Marginal probability}}}$$

- p(θ): Prior

-> Belief about parameters before seing data. It is not a single probability but a distribution!!

- p(X|θ): Likehood

-> How well parameters explain the data

- p(θ |X): Posterior

-> Update belief about parameters after observing data

- p(X): Evidence

-> Normalization (often ignored in practice

Bayesian vs frequentist

Frequentist statistics is based on the idea of long-run frequencies. It assumes that repeated sampling from a population will yield a stable estimate of parameters. Key concepts include:

- **Point Estimation:** Frequentist methods provide point estimates of parameters: single values that serve as the best estimate of the true population parameter.
- **Confidence Intervals:** These intervals quantify the uncertainty associated with point estimates by stating a range within which the true parameter is expected to lie.
- **Hypothesis Testing:** This involves making inferences about populations based on sample data, commonly using p-values to determine significance.

Frequentist methods are often seen as more objective, focusing on properties of estimators based on repeated sampling **(solely on observed data)**.

Bayesian statistics, on the other hand, is built on the principle of updating beliefs in the presence of new evidence. It incorporates prior information, combining it with the likelihood of the observed data to derive posterior distributions of the parameters. Key concepts include:

- **Prior Distribution:** This represents beliefs about parameters before observing any data.
- **Likelihood Function:** This describes how probable the observed data is given the parameters.
- **Posterior Distribution:** This is the updated belief about parameters after observing data and is calculated using Bayes' Theorem.

Bayesian methods, on the other hand, allow for the incorporation of prior knowledge and are often seen as more subjective, reflecting an individual's beliefs.

Are you Bayesian or frequentist

Experiment:

A coin has already been flipped. You do **not** see the result. What is the probability of heads?

Are you Bayesian or frequentist

Experiment:
A coin has already been flipped. You do **not** see the result. What is the probability of heads?

The coin already landed either heads or tails

Therefore p=0 or 1

- The outcome is fixed
- Probability is about repeated experiments
- No notion of belief

I believe the coin is probably fair. I have a belief/prior.

Therefore p=0.5

- You use a prior belief
- Probability represents uncertainty
- You reason before seeing data

Bayesian vs frequentist: example

You flip a coin **10 times** and get **8 heads, 2 tails**.
Goal: estimate probability of heads p.

Frequentist Result

Estimate using data only:
p=0.8
You obtain an estimate point not a distribution

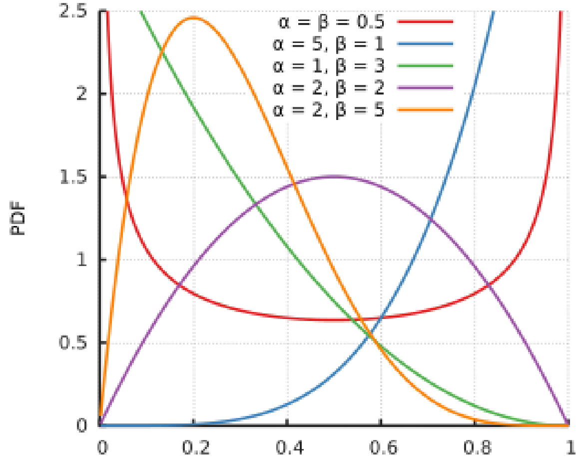
Then derive a confidence interval: (invented) e.g.
0.43-0.95

Conclusion
“You cannot say, 95% chance p is in this interval **BUT** if we repeated this experiment many times, 95% of such intervals would contain the true p.”

p is fixed, I just estimate it using repeated sampling. If you have new data, just update the estimate, there is no memory.

Bayesian vs frequentist: example

You flip a coin **10 times** and get **8 heads, 2 tails**.
Goal: estimate probability of heads p.



Bayesian Result
We have two output so let use the Beta distribution B(α,β) α=prior success

Prior:
Start with prior: your coin is fair therefore we need a centered Beta distribution.
p~Beta(2,2)

Likelihood: Binomial $P(D|p) = \binom{n}{k} p^k (1-p)^{n-k}$ $P(D \mid p) \propto p^k (1-p)^{n-k} \propto p^8 (1-p)^2$

$$\mathbb{E}[p] = \frac{\alpha}{\alpha + \beta}$$

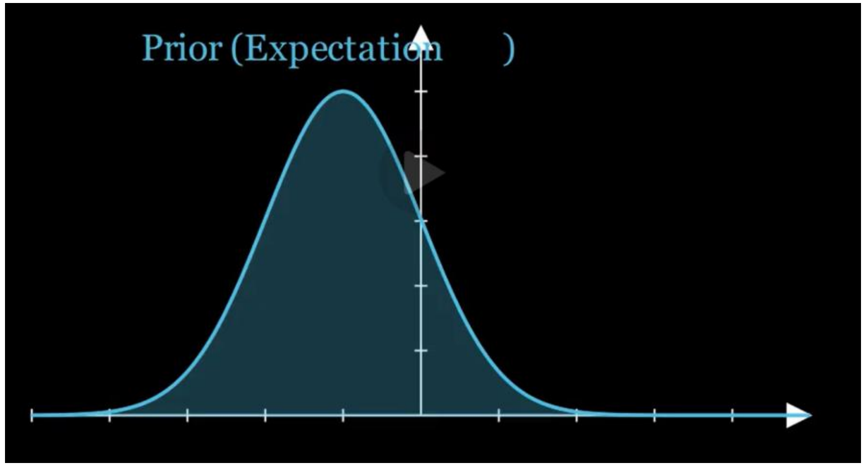
Posterior: $P(p \mid D) \propto P(D \mid p)\, P(p) \propto p^8 (1-p)^2 \cdot p^1 (1-p)^1 \propto p^9 (1-p)^3 \sim \mathrm{Beta}(10,4)$

The peak of the distribution is 0.71 (10/14)
Frequentist:
- Uses only the data → 8/10=0.8

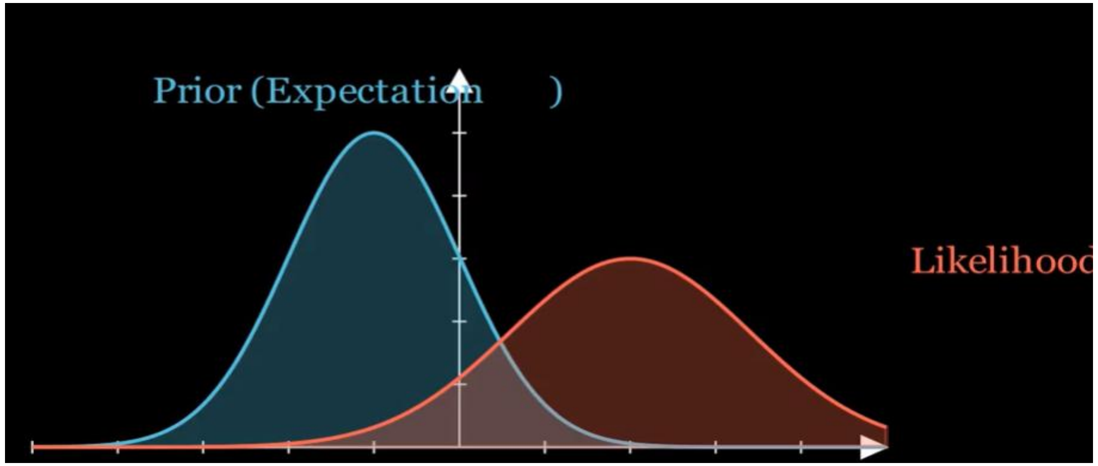
Bayesian:
- Combines prior + data
- The prior (Beta(2,2)) pulls the estimate toward 0.5

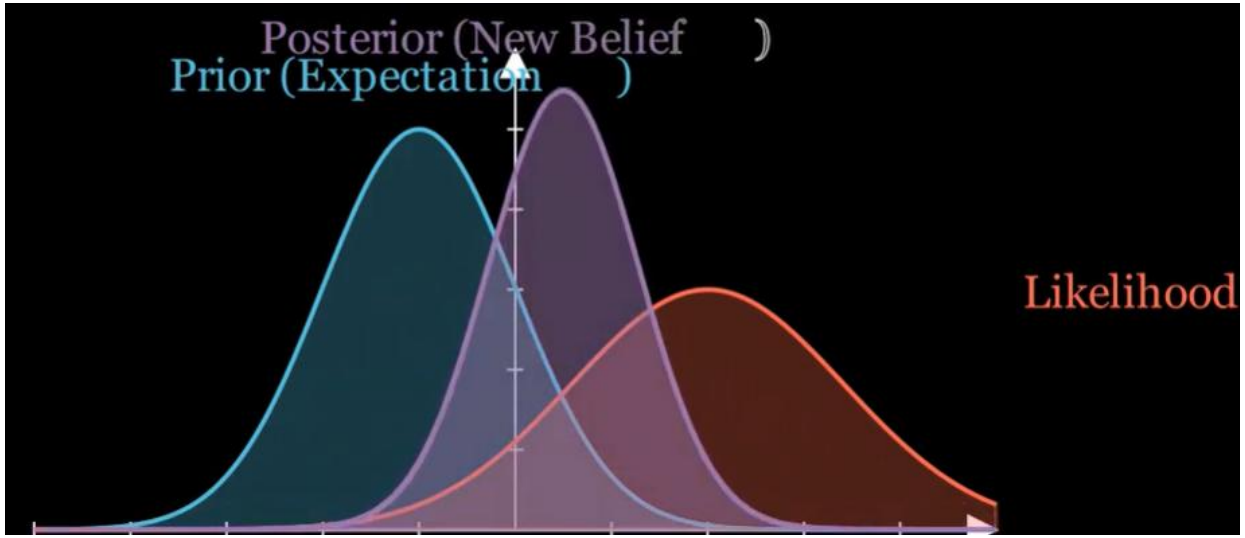
Bayesian vs frequentist: example

Prior, what you know from your belief



how well parameters explain the data





Posterior: updated belief about parameters after observing data

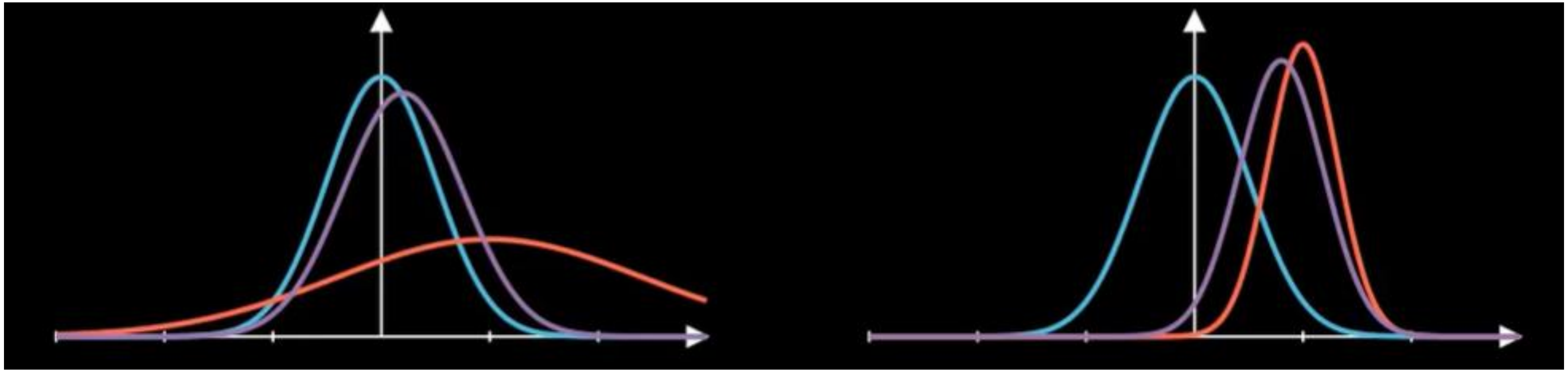
This a probability distribution, the peak is the best guest, the width the uncertainties

Bayesian vs frequentist: example

Bayesian methods are very good to handle uncertainties.

- If you data are very noisy, your likelihood is very spread out. Your posterior distribution will rely more on the priors (your experience)

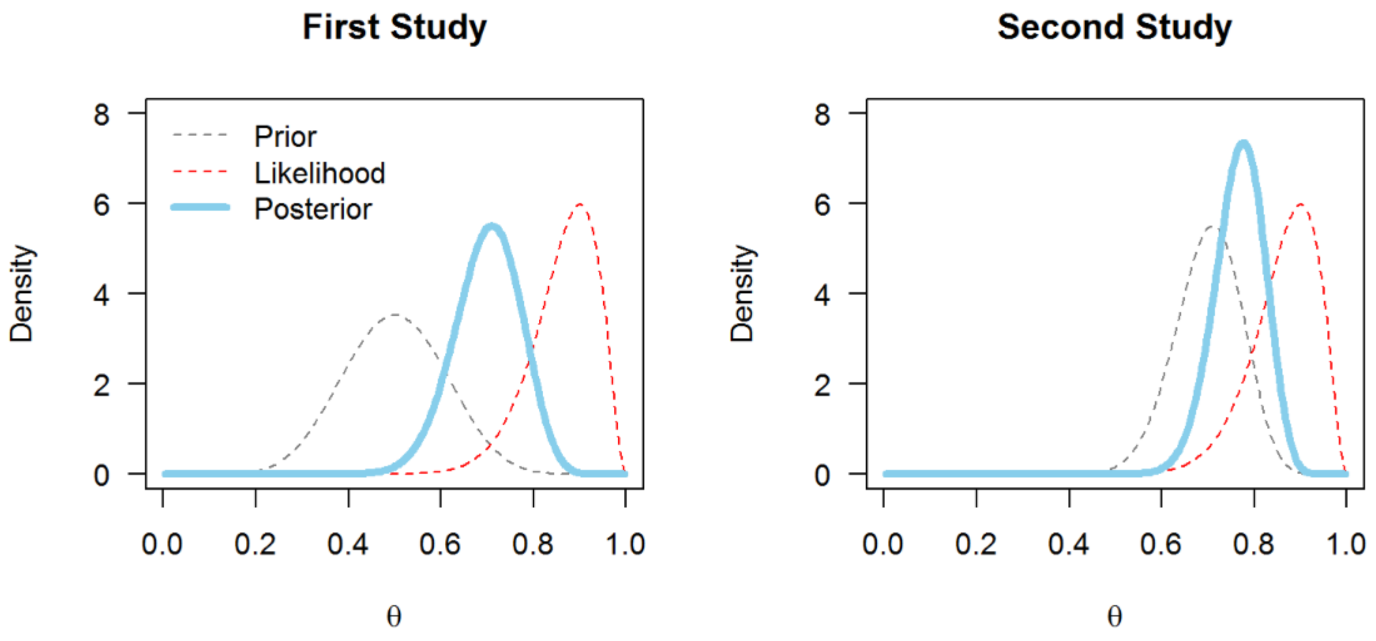
- If you data are very good data (small uncertainties), your likelihood is very narrow. Your posterior distribution will rely more on the data and shrinks the influence of the priors



Bayesian vs frequentist: example

Today's posterior is tomorrow's prior" (Lindley 1970)

After collecting some data and updating your prior, you now have a posterior belief of something. If you wish to collect more data, you do not use your original prior (because it no longer reflects your belief), but you instead use the posterior from your previous study as the prior for your current one. Then, you collect some data, update your priors into your posteriors…and so on.



Bayesian vs frequentist: example

	Frequentist Regression	Bayesian Regression
Assumptions	Parameters are fixed, unknown constants	Parameters are random variables, incorporating prior knowledge through prior distributions
Parameter Estimation	Uses methods like OLS or MLE	Combines prior distribution with likelihood function using Bayes' theorem to obtain a posterior distribution
Interpretation of Results	Focuses on point estimates and confidence intervals	Provides posterior distribution for a comprehensive understanding of uncertainty
Handling of Prior Information	Does not explicitly incorporate prior information	Incorporates prior knowledge or beliefs through prior probability distributions

Bayesian methods naturally quantify uncertainty through probability distributions, while frequentist methods rely on intervals or test statistics to represent uncertainty

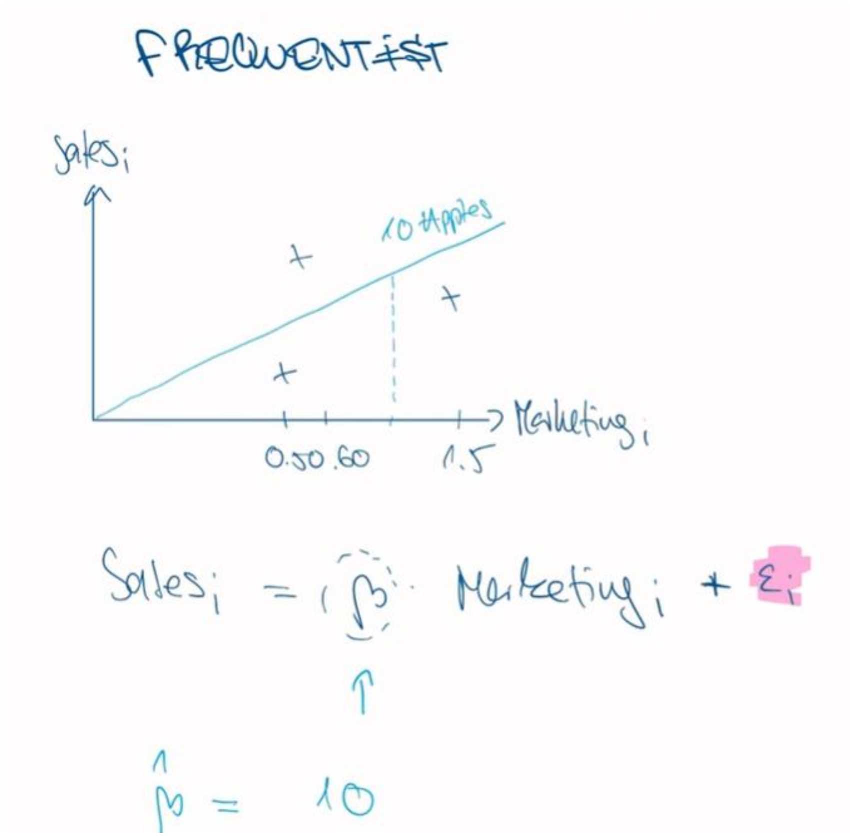
Pros
Simple and standard (what most people learn first)
No need to assume anything beforehand
Fast and works well with lots of data
Seen as more "objective"
Cons
Can't directly say probability of a parameter
Confidence intervals are kind of unintuitive
No clean way to incorporate prior knowledge
Updating = basically redo everything

Pros
Can use prior knowledge (useful if you already know something)
Gives a full picture (not just one number, but uncertainty too)
Easy to update step by step as new data comes in
Interpretation feels natural ("there's a 95% chance…")
Cons
You have to choose a prior (can be subjective)
Can get computationally heavy
With little data, results can depend a lot on the prior

Bayesian vs frequentist: linear regression

Let imagine, you want to spent USD per customer and see how much more apples do you sell?

Of course, you do not have data for all the customers, but imagine you have data only for 3 customers



The **least squares estimates** of the intercept and slope in the simple linear regression model are

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} \quad (11\text{-}7)$$

$$\hat{\beta}_1 = \frac{\sum_{i=1}^{n} y_i x_i - \frac{\left(\sum_{i=1}^{n} y_i\right)\left(\sum_{i=1}^{n} x_i\right)}{n}}{\sum_{i=1}^{n} x_i^2 - \frac{\left(\sum_{i=1}^{n} x_i\right)^2}{n}} \quad (11\text{-}8)$$

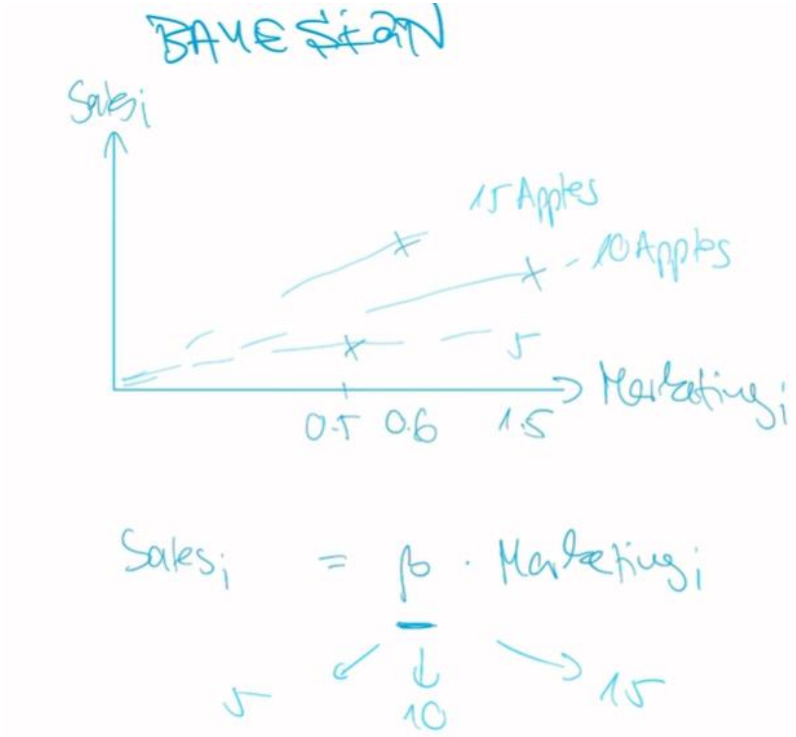
where $\bar{y} = (1/n)\sum_{i=1}^{n} y_i$ and $\bar{x} = (1/n)\sum_{i=1}^{n} x_i$.

[Bayesian vs. Frequentist Regression]

Bayesian vs frequentist: linear regression

Let imagine, you want to spent USD per customer and see how much more apples do you sell?

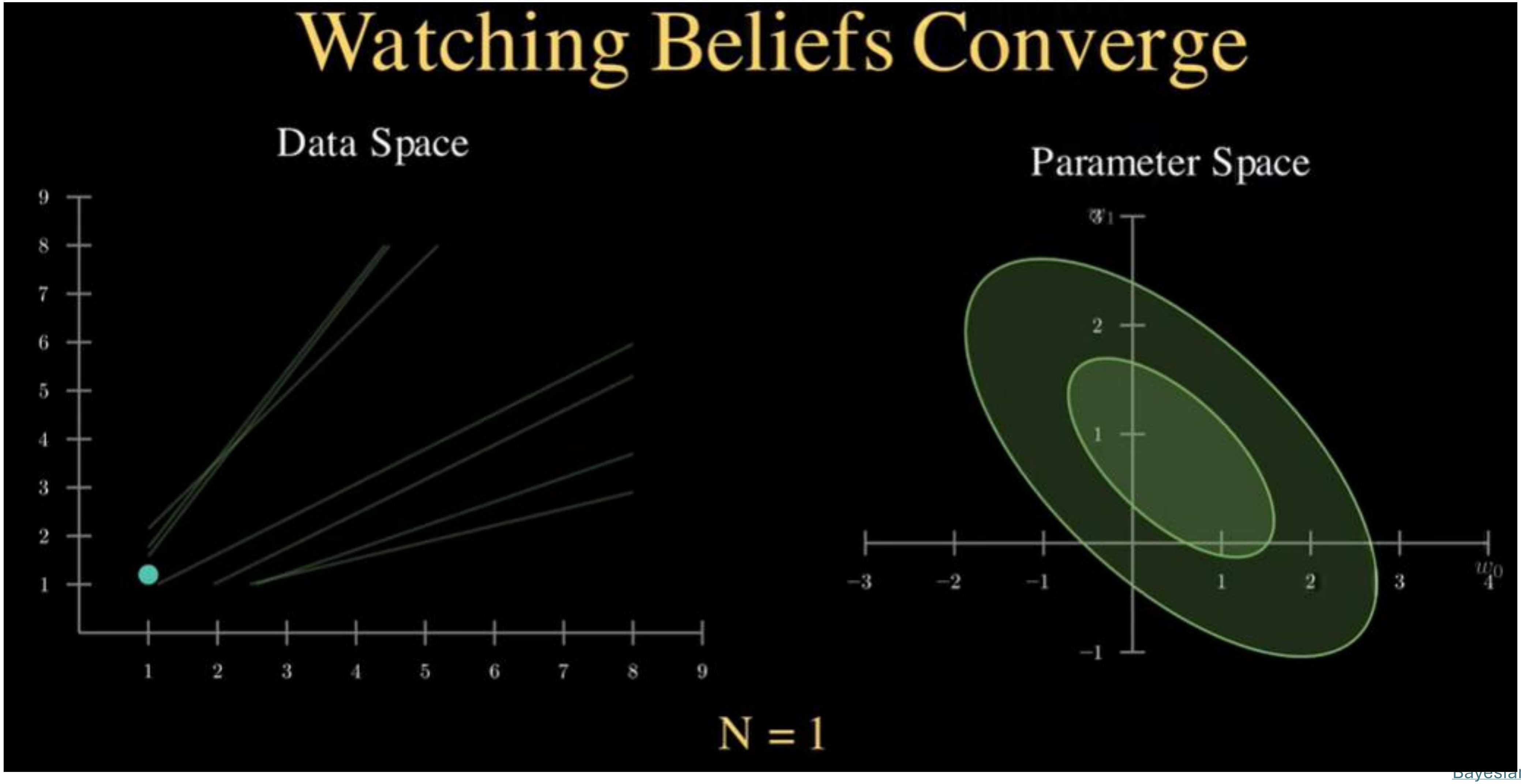
Of course, you do not have data for all the customers, but imagine you have data only for 3 customers



There is not a true value for β, this parameter varies with the data

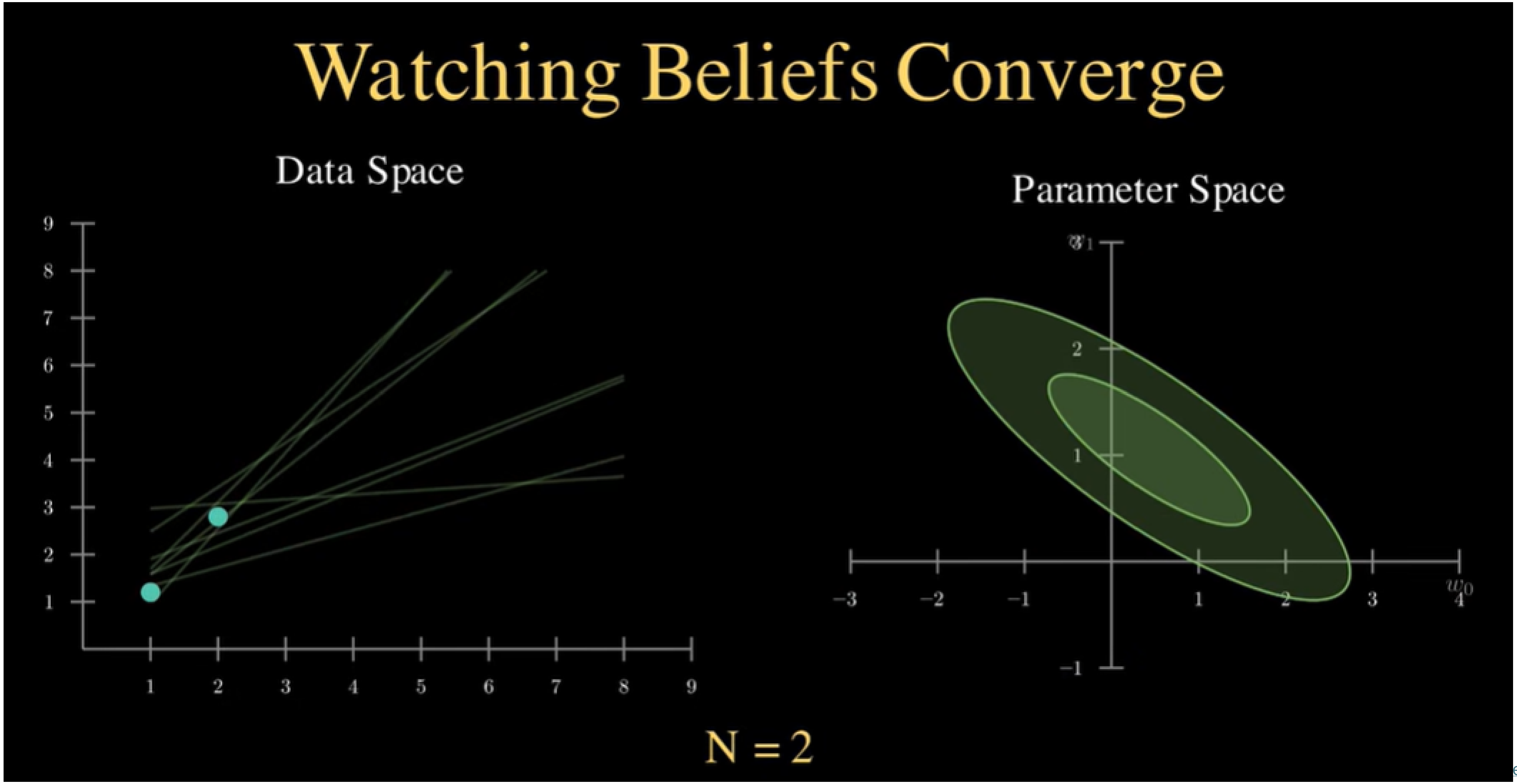
[Bayesian vs. Frequentist Regression]()

Bayesian vs frequentist: linear regression



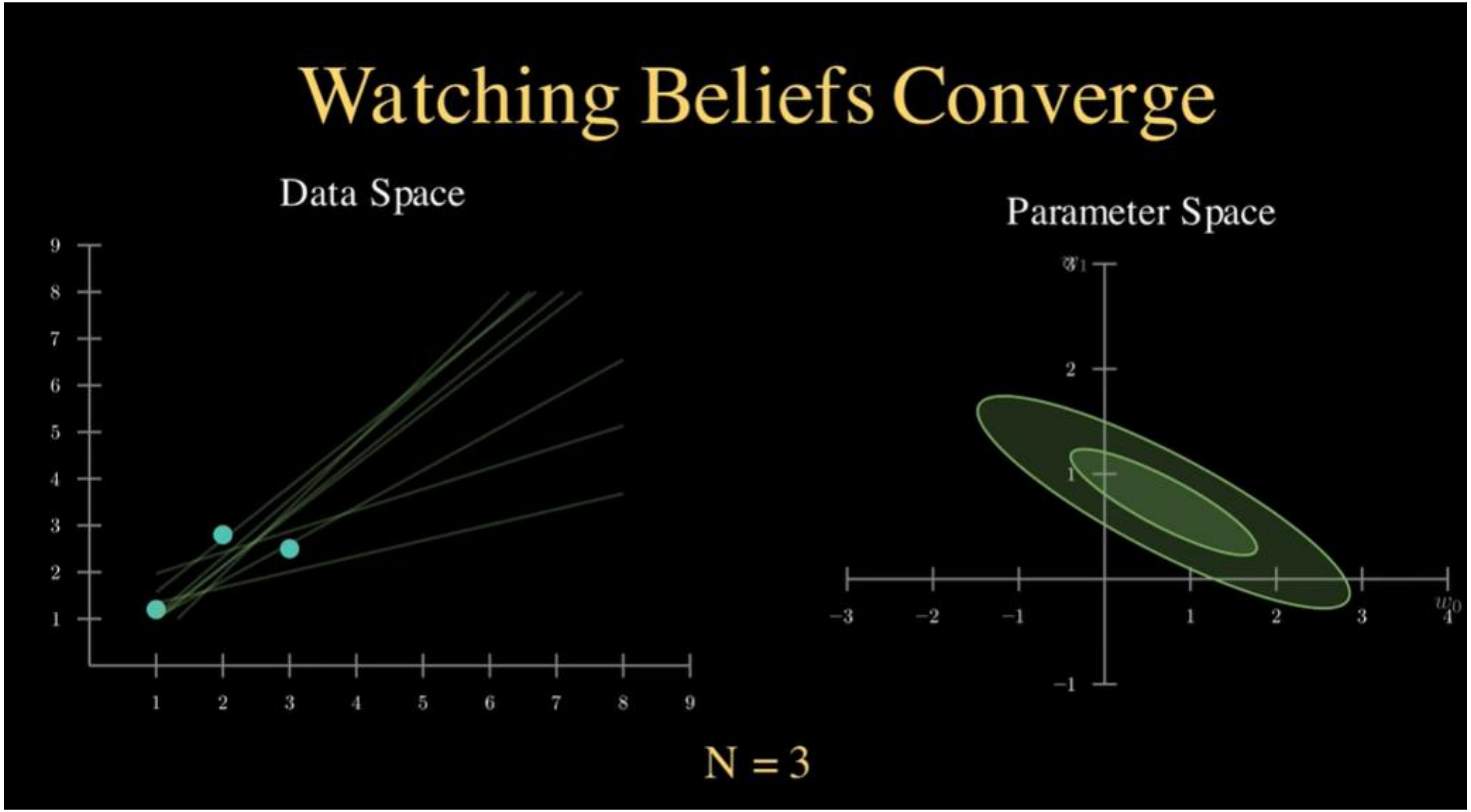
Bayesian Linear Regression

Bayesian vs frequentist: linear regression



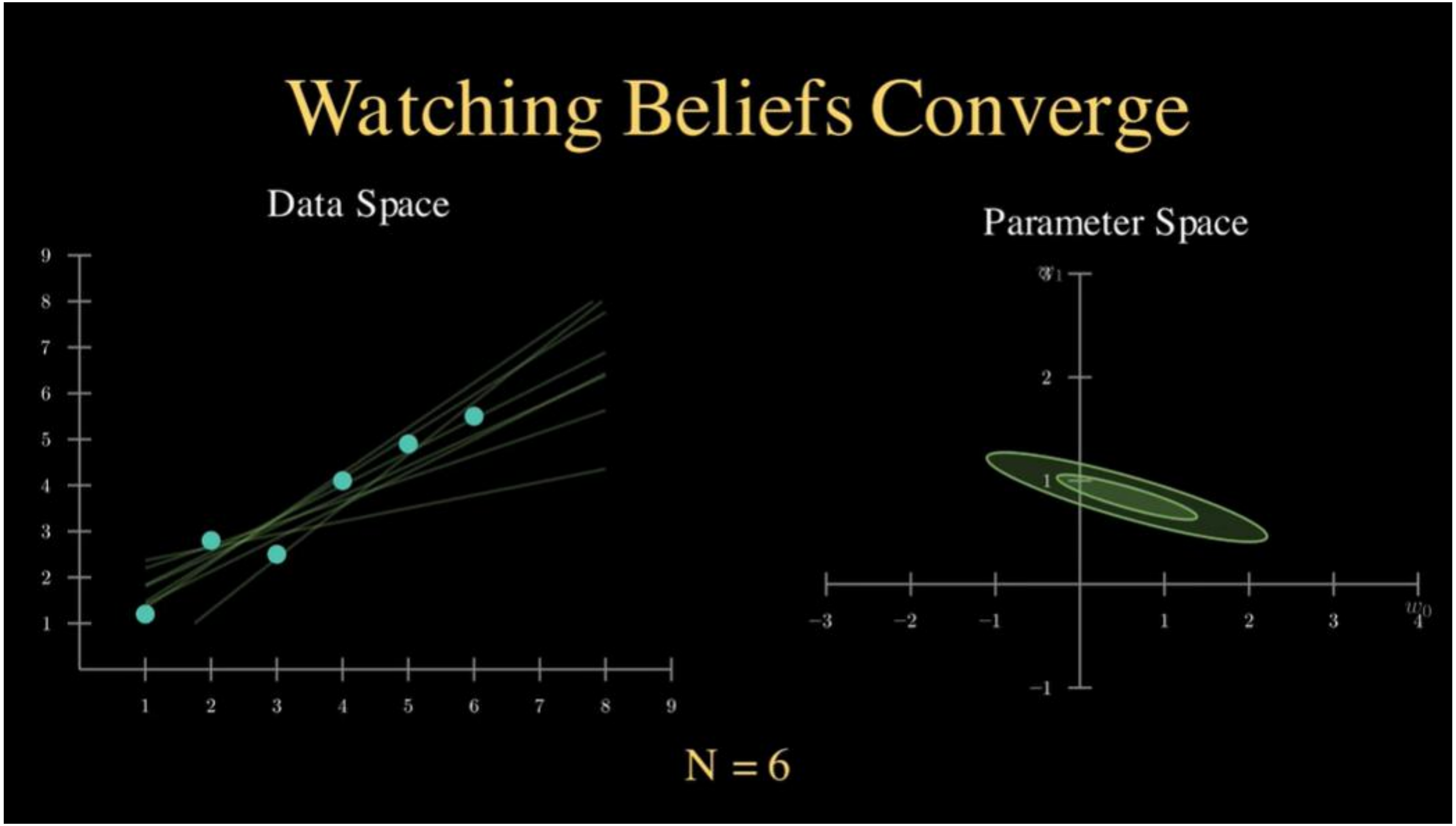
ear Regression

Bayesian vs frequentist: linear regression



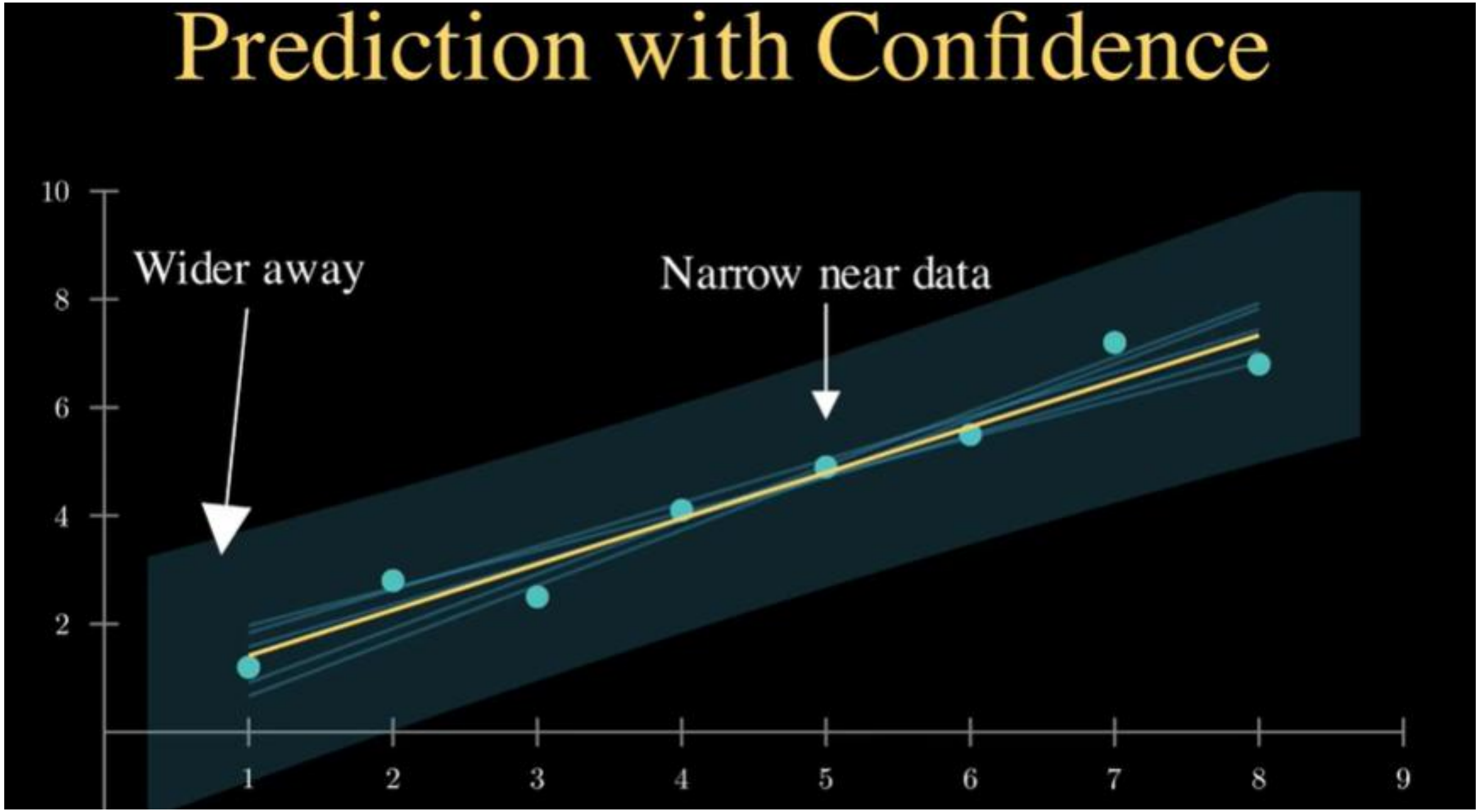
Bayesian Linear Regression

Bayesian vs frequentist: linear regression



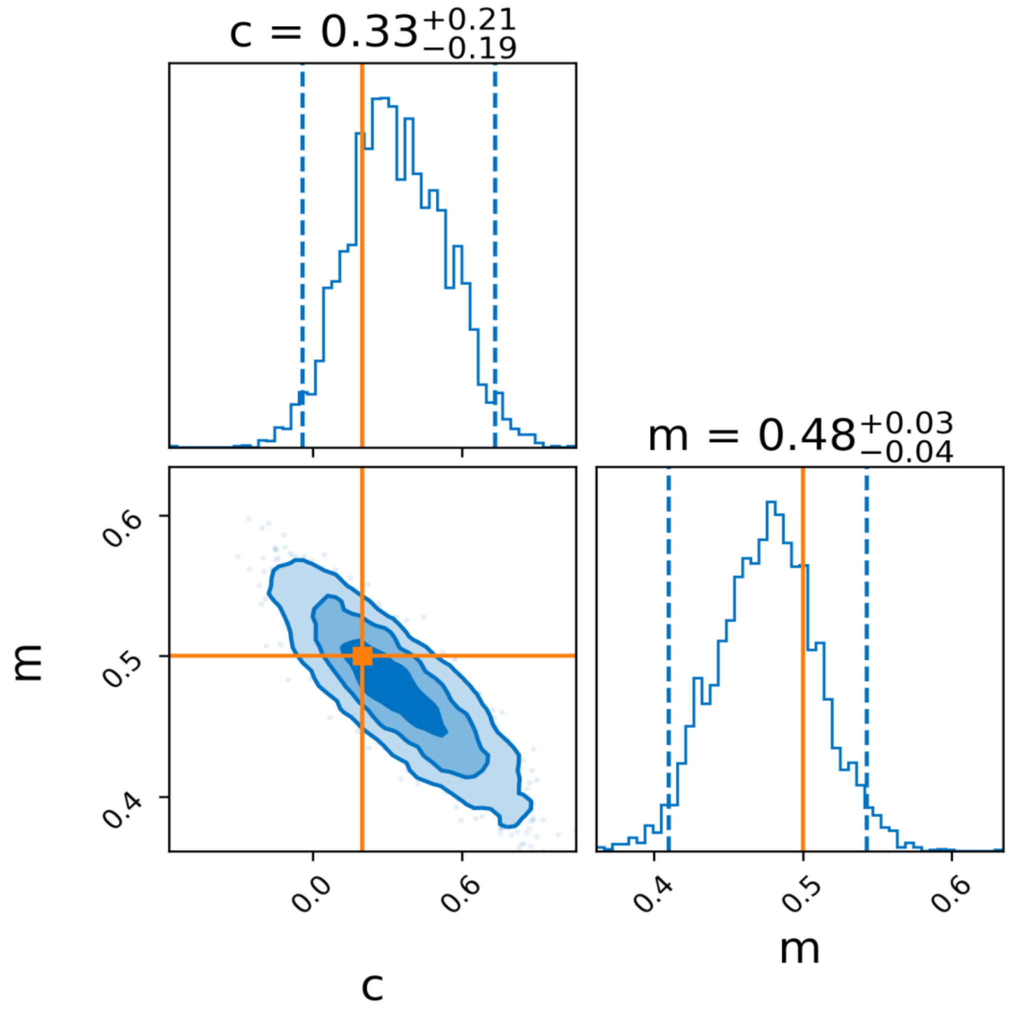
[Bayesian Linear Regression]

Bayesian vs frequentist: linear regression



Bayesian Linear Regression

Bayesian vs frequentist: linear regression



Bayesian Linear Regression